

We must solve this set of simultaneous equations. Solution can be achieved in many different ways; usually it is a good idea to express each variable as a function of only one other variable. b is already written simply as a function of c in (4); let us try expressing the other variables solely in terms of c . From (1):

$$d = 16 - c. \quad (6)$$

From (5):

$$a = \frac{d}{0.43} = 2.326 d. \quad (7)$$

Combining (6) and (7) gives an expression for a in terms of c only:

$$\begin{aligned} a &= 2.326 (16 - c) \\ a &= 37.22 - 2.326 c. \end{aligned} \quad (8)$$

Substituting (4) into (2) gives:

$$\begin{aligned} 34 + 3 (0.20 c) &= 1.66 c + 2 e \\ 34 &= 1.06 c + 2 e \\ e &= 17 - 0.53 c. \end{aligned} \quad (9)$$

Substituting (8), (6) and (9) into (3) gives:

$$\begin{aligned} 2 (37.22 - 2.326 c) &= 0.27 c + 2 (16 - c) + (17 - 0.53 c) \\ 25.44 &= 2.39 c \\ c &= 10.64. \end{aligned}$$

Using this result for c in (8), (4), (6) and (9) gives:

$$\begin{aligned} a &= 12.48 \\ b &= 2.13 \\ d &= 5.37 \\ e &= 11.36. \end{aligned}$$

Check that these coefficient values satisfy Eqs (1)–(5).

The complete reaction equation is:



Although elemental balances are useful, the presence of water in Eq. (4.4) causes some problems in practical application. Because water is usually present in great excess and changes in water concentration are inconvenient to measure or

experimentally verify, H and O balances can present difficulties. Instead, a useful principle is conservation of reducing power or available electrons, which can be applied to determine quantitative relationships between substrates and

products. An electron balance shows how available electrons from the substrate are distributed in reaction.

4.6.2 Electron Balances

Available electrons refers to the number of electrons available for transfer to oxygen on combustion of a substance to CO_2 , H_2O and nitrogen-containing compounds. The number of available electrons found in organic material is calculated from the valence of the various elements: 4 for C, 1 for H, -2 for O, 5 for P, and 6 for S. The number of available electrons for N depends on the reference state: -3 if ammonia is the reference, 0 for molecular nitrogen N_2 , and 5 for nitrate. The reference state for cell growth is usually chosen to be the same as the nitrogen source in the medium. In the following discussion it will be assumed for convenience that ammonia is used as nitrogen source; this can easily be changed if other nitrogen sources are employed [5].

Degree of reduction, γ , is defined as the number of equivalents of available electrons in that quantity of material containing 1 g atom carbon. Therefore, for substrate $\text{C}_w\text{H}_x\text{O}_y\text{N}_z$, the number of available electrons is $(4w + x - 2y - 3z)$. The degree of reduction for the substrate, γ_S , is therefore $(4w + x - 2y - 3z)/w$. Degrees of reduction relative to NH_3 and N_2 for several biological compounds are given in Table B.2 in Appendix B. Degree of reduction for CO_2 , H_2O and NH_3 is zero.

Electrons available for transfer to oxygen are conserved during metabolism. In a balanced growth equation, number of available electrons is conserved by virtue of the fact that the amounts of each chemical element are conserved. Applying this principle to Eq. (4.4) with ammonia as nitrogen source, the available-electron balance is:

$$w\gamma_S - 4a = c\gamma_B \quad (4.10)$$

where γ_S and γ_B are the degrees of reduction of substrate and biomass, respectively. Note that the available-electron balance is not independent of the complete set of elemental balances; if the stoichiometric equation is balanced in terms of each element including H and O, the electron balance is implicitly satisfied.

4.6.3 Biomass Yield

Typically, Eq. (4.10) is used with carbon and nitrogen balances Eqs (4.5) and (4.8) and a measured value of RQ for evaluation

of stoichiometric coefficients. However, one electron balance, two elemental balances and one measured quantity are still inadequate information for solution of five unknown coefficients; another experimental quantity is required. As cells grow there is, as a general approximation, a linear relationship between the amount of biomass produced and the amount of substrate consumed. This relationship is expressed quantitatively using the *biomass yield*, Y_{XS} :

$$Y_{XS} = \frac{\text{g cells produced}}{\text{g substrate consumed}} \quad (4.11)$$

A large number of factors influences biomass yield, including medium composition, nature of the carbon and nitrogen sources, pH and temperature. Biomass yield is greater in aerobic than in anaerobic cultures; choice of electron acceptor, e.g. O_2 , nitrate or sulphate, can also have a significant effect [5, 6].

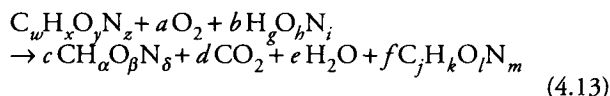
When Y_{XS} is constant throughout growth, its experimentally-determined value can be used to determine the stoichiometric coefficient c in Eq. (4.4). Eq. (4.11) expressed in terms of the stoichiometric Eq. (4.4) is:

$$Y_{XS} = \frac{c(\text{MW cells})}{(\text{MW substrate})} \quad (4.12)$$

where MW is molecular weight and 'MW cells' means the biomass formula-weight plus any residual ash. However, before applying measured values of Y_{XS} and Eq. (4.12) to evaluate c , we must be sure that the experimental culture system is well represented by the stoichiometric equation. For example, we must be sure that substrate is not used to synthesise extracellular products other than CO_2 and H_2O . One complication with real cultures is that some fraction of substrate consumed is always used for *maintenance activities* such as maintenance of membrane potential and internal pH, turnover of cellular components and cell motility. These metabolic functions require substrate but do not necessarily produce cell biomass, CO_2 and H_2O in the way described by Eq. (4.4). It is important to account for maintenance when experimental information is used to complete stoichiometric equations; maintenance requirements and the difference between observed and true yields are discussed further in Chapter 11. For the time being, we will assume that available values for biomass yield reflect substrate consumption for growth only.

4.6.4 Product Stoichiometry

Consider formation of an extracellular product $C_jH_kO_lN_m$ during growth. Eq. (4.4) can be extended to include product synthesis as follows:



where f is the stoichiometric coefficient for product. Product synthesis introduces one extra unknown stoichiometric coefficient to the equation; thus, an additional relationship between coefficients is required. This is usually provided as another experimentally-determined yield coefficient, the *product yield from substrate*, Y_{PS} :

$$Y_{PS} = \frac{\text{g product formed}}{\text{g substrate consumed}} = \frac{f(\text{MW product})}{(\text{MW substrate})}. \quad (4.14)$$

As mentioned above with regard to biomass yields, we must be sure that the experimental system used to measure Y_{PS} conforms to Eq. (4.13). Eq. (4.13) does not hold if product formation is not directly linked with growth; accordingly it cannot be applied for secondary-metabolite production such as penicillin fermentation, or for biotransformations such as steroid hydroxylation which involve only a small number of enzymes in cells. In these cases, independent reaction equations must be used to describe growth and product synthesis.

4.6.5 Theoretical Oxygen Demand

Oxygen demand is an important parameter in bioprocessing as oxygen is often the limiting substrate in aerobic fermentations. Oxygen demand is represented by the stoichiometric coefficient a in Eqs (4.4) and (4.13). Oxygen requirement is related directly to the electrons available for transfer to oxygen; the oxygen demand can therefore be derived from an appropriate electron balance. When product synthesis occurs as represented by Eq. (4.13), the electron balance is:

$$w\gamma_S - 4a = c\gamma_B + fj\gamma_P \quad (4.15)$$

where γ_P is the degree of reduction of the product. Rearranging gives:

$$a = 1/4 (w\gamma_S - c\gamma_B - fj\gamma_P). \quad (4.16)$$

Eq. (4.16) is a very useful equation. It means that if we know which organism (γ_B), substrate (w and γ_S) and product (j and γ_P) are involved in cell culture, and the yields of biomass (c) and product (f), we can quickly calculate the oxygen demand. Of course we could also determine a by solving for all the stoichiometric coefficients of Eq. (4.13) as described in Section 4.6.1. Eq. (4.16) allows more rapid evaluation and does not require that the quantities of NH_3 , CO_2 and H_2O involved in the reaction be known.

4.6.6 Maximum Possible Yield

From Eq. (4.15) the fractional allocation of available electrons in the substrate can be written as:

$$1 = \frac{4a}{w\gamma_S} + \frac{c\gamma_B}{w\gamma_S} + \frac{fj\gamma_P}{w\gamma_S}. \quad (4.17)$$

In Eq. (4.17), the first term on the right-hand side is the fraction of available electrons transferred from substrate to oxygen, the second term is the fraction of available electrons transferred to biomass, and the third term is the fraction of available electrons transferred to product. This relationship can be used to obtain upper bounds for the yields of biomass and product from substrate.

Let us define ζ_B as the fraction of available electrons in the substrate transferred to biomass:

$$\zeta_B = \frac{c\gamma_B}{w\gamma_S}. \quad (4.18)$$

In the absence of product formation, if all available electrons were used for biomass synthesis, ζ_B would equal unity. Under these conditions, the maximum value of the stoichiometric coefficient c is:

$$c_{\max} = \frac{w\gamma_S}{\gamma_B}. \quad (4.19)$$

$c_{\max}$ can be converted to a biomass yield with mass units using Eq. (4.12). Therefore, even if we do not know the stoichiometry of growth, we can quickly calculate an upper limit for biomass yield from the molecular formulae for substrate and product. If the composition of the cells is unknown, γ_B can be

taken as 4.2 corresponding to the average biomass formula $\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$. Maximum biomass yields for several substrates are listed in Table 4.4; maximum biomass yield can be expressed in terms of mass ($Y_{\text{XS,max}}$), or as number of C atoms in the biomass per substrate C atom consumed ($c_{\text{max}/w}$). These quantities are sometimes known as *thermodynamic maximum biomass yields*. Table 4.4 shows that substrates with high energy content, indicated by high γ_S values, give high maximum biomass yields.

Likewise, the maximum possible product yield in the

absence of biomass synthesis can be determined from Eq. (4.17):

$$f_{\text{max}} = \frac{w\gamma_S}{j\gamma_P} \quad (4.20)$$

Eq. (4.20) allows us to quickly calculate an upper limit for product yield from the molecular formulae for substrate and product.

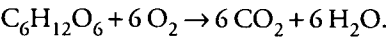
Table 4.4 Thermodynamic maximum biomass yields

(Adapted from L.E. Erickson, I.G. Minkevich and V.K. Eroshin, 1978, *Application of mass and energy balance regularities in fermentation*, Biotechnol. Bioeng. 20, 1595–1621)

Substrate	Formula	γ_{S}	Thermodynamic maximum yield corresponding to $\zeta_{\text{B}} = 1$	
			Carbon yield (c_{max}/w)	Mass yield $Y_{\text{XS,max}}$
Alkanes				
Methane	CH_4	8.0	1.9	2.9
Hexane (n)	C_6H_{14}	6.3	1.5	2.6
Hexadecane (n)	$\text{C}_{16}\text{H}_{34}$	6.1	1.5	2.5
Alcohols				
Methanol	CH_4O	6.0	1.4	1.1
Ethanol	$\text{C}_2\text{H}_6\text{O}$	6.0	1.4	1.5
Ethylene glycol	$\text{C}_2\text{H}_6\text{O}_2$	5.0	1.2	0.9
Glycerol	$\text{C}_3\text{H}_8\text{O}_3$	4.7	1.1	0.9
Carbohydrates				
Formaldehyde	CH_2O	4.0	0.95	0.8
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	4.0	0.95	0.8
Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	4.0	0.95	0.8
Starch	$(\text{C}_6\text{H}_{10}\text{O}_5)_x$	4.0	0.95	0.9
Organic acids				
Formic acid	CH_2O_2	2.0	0.5	0.3
Acetic acid	$\text{C}_2\text{H}_4\text{O}_2$	4.0	0.95	0.8
Propionic acid	$\text{C}_3\text{H}_6\text{O}_2$	4.7	1.1	1.1
Lactic acid	$\text{C}_3\text{H}_6\text{O}_3$	4.0	0.95	0.8
Fumaric acid	$\text{C}_4\text{H}_4\text{O}_4$	3.0	0.7	0.6
Oxalic acid	$\text{C}_2\text{H}_2\text{O}_4$	1.0	0.24	0.1

Example 4.8 Product yield and oxygen demand

The chemical reaction equation for respiration of glucose is:



Candida utilis cells convert glucose to CO_2 and H_2O during growth. The cell composition is $\text{CH}_{1.84}\text{O}_{0.55}\text{N}_{0.2}$ plus 5% ash. Yield of biomass from substrate is 0.5 g g^{-1} . Ammonia is used as nitrogen source.

- What is the oxygen demand with growth compared to that without?
- C. utilis* is also able to grow with ethanol as substrate, producing cells of the same composition as above. On a mass basis, how does the maximum possible biomass yield from ethanol compare with the maximum possible yield from glucose?

Solution:

Molecular weights: glucose = 180
 ethanol = 46

MW biomass is (25.44 + ash); since ash accounts for 5% of the total weight, 95% of the total = 25.44. Therefore, MW biomass = $25.44/0.95 = 26.78$. From Table B.2, γ_S for glucose is 4.00; γ_S for ethanol is 6.00. $\gamma_B = (4 \times 1 + 1 \times 1.84 - 2 \times 0.55 - 3 \times 0.2) = 4.14$. For glucose $w = 6$; for ethanol $w = 2$.

- $Y_{XS} = 0.5 \text{ g g}^{-1}$. Converting this mass yield to a molar yield:

$$Y_{XS} = \frac{0.5 \text{ g biomass}}{\text{g glucose}} \cdot \left| \frac{180 \text{ g glucose}}{1 \text{ gmol glucose}} \right| \cdot \left| \frac{1 \text{ gmol biomass}}{26.78 \text{ g biomass}} \right|$$

$$Y_{XS} = 3.36 \frac{\text{gmol biomass}}{\text{gmol glucose}} = c.$$

Oxygen demand is given by Eq. (4.16). In the absence of product formation:

$$a = 1/4 [6 (4.00) - 3.36 (4.14)] = 2.52.$$

Therefore, the oxygen demand for glucose respiration with growth is 2.5 gmol O_2 per gmol glucose consumed. By comparison with the chemical reaction equation for respiration, this is only about 42% that required in the absence of growth.

- Maximum possible biomass yield is given by Eq. (4.19). Using the data above, for glucose:

$$c_{\max} = \frac{6(4.00)}{4.14} = 5.80.$$

Converting this to a mass basis:

$$Y_{XS,\max} = \frac{5.80 \text{ g biomass}}{\text{gmol glucose}} \cdot \left| \frac{1 \text{ gmol glucose}}{180 \text{ g glucose}} \right| \cdot \left| \frac{26.78 \text{ g biomass}}{1 \text{ gmol biomass}} \right|$$

$$Y_{XS,\max} = 0.86 \frac{\text{g biomass}}{\text{g glucose}}.$$

For ethanol:

$$c_{\max} = \frac{2(6.00)}{4.14} = 2.90$$

and

$$Y_{XS,\max} = \frac{2.90 \text{ g mol biomass}}{\text{g mol ethanol}} \cdot \left| \frac{1 \text{ g mol ethanol}}{46 \text{ g ethanol}} \right| \cdot \left| \frac{26.78 \text{ g biomass}}{1 \text{ g mol biomass}} \right|$$

$$Y_{XS,\max} = 1.69 \frac{\text{g biomass}}{\text{g ethanol}}.$$

Therefore, on a mass basis, the maximum possible amount of biomass produced per gram ethanol consumed is roughly twice that per gram glucose consumed. This result is consistent with the data listed in Table 4.4.

Example 4.8 illustrates two important points. First, the chemical reaction equation for conversion of substrate without growth is a poor approximation of overall stoichiometry when cell growth occurs. When estimating yields and oxygen requirements for any process involving cell growth, the full stoichiometric equation including biomass should be used. Second, the chemical nature or oxidation state of the substrate has a major influence on product and biomass yield through the number of available electrons.

4.7 Summary of Chapter 4

At the end of Chapter 4 you should:

- (i) understand the terms: *system*, *surroundings*, *boundary* and *process* in thermodynamics;
- (ii) be able to identify *open* and *closed systems*, and *batch*, *semi-batch*, *fed-batch* and *continuous processes*;
- (iii) understand the difference between *steady state* and *equilibrium*;
- (iv) be able to write appropriate equations for conservation of mass for processes with and without reaction;
- (v) be able to solve simple mass-balance problems with and without reaction; and
- (vi) be able to apply stoichiometric principles for macroscopic analysis of cell growth and product formation.

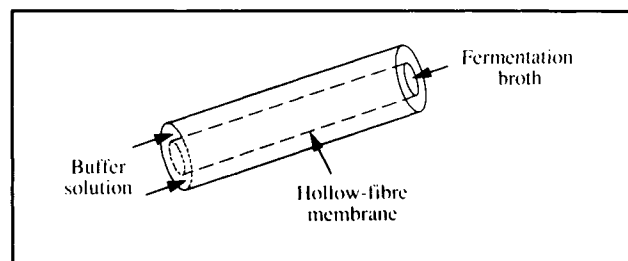
Problems

4.1 Cell concentration using membranes

A battery of cylindrical hollow-fibre membranes is operated at steady state to concentrate a bacterial suspension from a fermenter. 350 kg min⁻¹ fermenter broth is pumped through a stack of hollow-fibre membranes as shown in Figure 4P1.1. The broth contains 1% bacteria; the rest may be considered water. Buffer solution enters the annular space around the membrane tubes at a flow rate of 80 kg min⁻¹; because broth in the membrane tubes is under pressure, water is forced across

the membrane into the buffer. Cells in the broth are too large to pass through the membrane and pass out of the tubes as a concentrate.

Figure 4P1.1 Hollow-fibre membrane for concentration of cells.



The aim of the membrane system is to produce a cell suspension containing 6% biomass.

- (a) What is the flow rate from the annular space?
- (b) What is the flow rate of cell suspension from the membrane tubes?

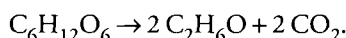
Assume that the cells are not active, i.e. they do not grow. Assume further that the membrane does not allow any molecules other than water to pass from annulus to inner cylinder, or vice versa.

4.2 Membrane reactor

A battery of cylindrical membranes similar to that shown in Figure 4P1.1 is used for an extractive bioconversion. Extractive bioconversion means that fermentation and extraction of product occur at the same time.

Yeast cells are immobilised within the membrane walls. A 10% glucose in water solution is passed through the annular space at a rate of 40 kg h⁻¹. An organic solvent, such as 2-ethyl-1,3-hexanediol, enters the inner tube at a rate of 40 kg h⁻¹.

Because the membrane is constructed of a polymer which repels organic solvents, the hexanediol cannot penetrate the membrane and the yeast is relatively unaffected by its toxicity. On the other hand, because glucose and water are virtually insoluble in 2-ethyl-1,3-hexanediol, these compounds do not enter the inner tube to an appreciable extent. Once immobilised in the membrane, the yeast cannot reproduce but convert glucose to ethanol according to the equation:



Ethanol is soluble in 2-ethyl-1,3-hexanediol; it diffuses into the inner tube and is carried out of the system. CO_2 gas exits from the membrane unit through an escape valve. In the aqueous stream leaving the annular space, the concentration of unconverted glucose is 0.2% and the concentration of ethanol is 0.5%. If the system operates at steady state:

- What is the concentration of ethanol in the hexanediol stream leaving the reactor?
- What is the mass flow rate of CO_2 ?

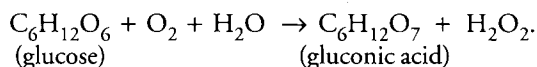
4.3 Ethanol distillation

Liquid from a brewery fermenter can be considered to contain 10% ethanol and 90% water. 50 000 kg h^{-1} of this fermentation product are pumped to a distillation column on the factory site. Under current operating conditions a distillate of 45% ethanol and 55% water is produced from the top of the column at a rate one-tenth that of the feed.

- What is the composition of the waste 'bottoms' from the still?
- What is the rate of alcohol loss in the bottoms?

4.4 Removal of glucose from dried egg

The enzyme, glucose oxidase, is used commercially to remove glucose from dehydrated egg to improve colour, flavour and shelf-life. The reaction is:



A continuous-flow reactor is set up using immobilised-enzyme beads which are retained inside the vessel. Dehydrated egg slurry containing 2% glucose, 20% water and the remainder unreactive egg solids, is available at a rate of 3 000 kg h^{-1} . Air is pumped through the reactor contents so that 18 kg oxygen are delivered per hour. The desired glucose level in the

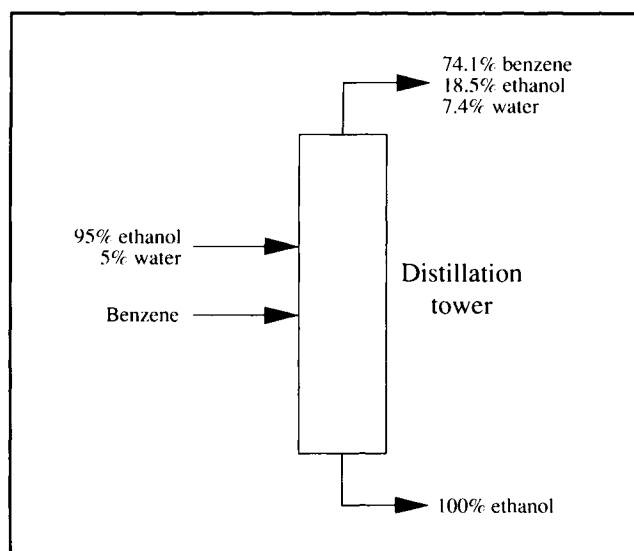
dehydrated egg product leaving the enzyme reactor is 0.2%. Determine:

- which is the limiting substrate;
- the percentage excess substrate;
- the composition of the reactor off-gas; and
- the composition of the final egg product.

4.5 Azeotropic distillation

Absolute or 100% ethanol is produced from a mixture of 95% ethanol and 5% water using the Keyes distillation process. A third component, benzene, is added to lower the volatility of the alcohol. Under these conditions, the overhead product is a constant-boiling mixture of 18.5% ethanol, 7.4% H_2O and 74.1% benzene. The process is outlined in Figure 4P5.1.

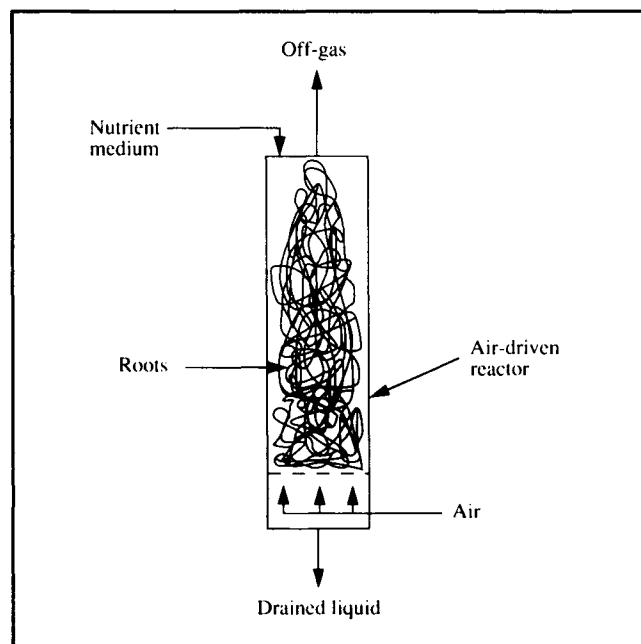
Figure 4P5.1 Flowsheet for Keyes distillation process.



Use the following data to calculate the volume of benzene which should be fed to the still in order to produce 250 litres absolute ethanol: ρ (100% alcohol) = 0.785 g cm^{-3} ; ρ (benzene) = 0.872 g cm^{-3} .

4.6 Culture of plant roots

Plant roots produce valuable chemicals *in vitro*. A batch culture of *Atropa belladonna* roots at 25°C is established in an air-driven reactor as shown in Figure 4P6.1. Because roots cannot be removed during operation of the reactor, it is proposed to monitor growth using mass balances.

Figure 4P6.1 Reactor for culture of plant roots.

1425 g nutrient medium containing 3% glucose and 1.75% NH_3 is fed into the reactor; the remainder of the medium can be considered water. Air at 25°C and 1 atm pressure is sparged into the fermenter at a rate of $22 \text{ cm}^3 \text{ min}^{-1}$. During a 10-day culture period, 47 litres O_2 and 15 litres CO_2 are collected in the off-gas. After 10 days, 1110 g liquid containing 0.063% glucose and 1.7% dissolved NH_3 is drained from the vessel. The ratio of fresh weight to dry weight for roots is known to be 14:1.

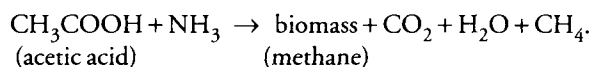
- What dry mass of roots is produced in 10 days?
- Write the reaction equation for growth, indicating the approximate chemical formula for the roots, $\text{CH}_\alpha\text{O}_\beta\text{N}_\delta$.
- What is the limiting substrate?
- What is the yield of roots from glucose?

4.7 Oxygen requirement for growth on glycerol

Klebsiella aerogenes is produced from glycerol in aerobic culture with ammonia as nitrogen source. The biomass contains 8% ash, 0.40 g biomass is produced for each g glycerol consumed, and no major metabolic products are formed. What is the oxygen requirement for this culture in mass terms?

4.8 Product yield in anaerobic digestion

Anaerobic digestion of volatile acids by methane bacteria is represented by the equation:



The composition of methane bacteria is approximated by the empirical formula $\text{CH}_{1.4}\text{O}_{0.40}\text{N}_{0.20}$. For each kg acetic acid consumed, 0.67 kg CO_2 is evolved. How does the yield of methane under these conditions compare with the maximum possible yield?

4.9 Stoichiometry of single-cell protein synthesis

- Cellulomonas* bacteria used as single-cell protein for human or animal food are produced from glucose under anaerobic conditions. All carbon in the substrate is converted into biomass; ammonia is used as nitrogen source. The molecular formula for the biomass is $\text{CH}_{1.56}\text{O}_{0.54}\text{N}_{0.16}$; the cells also contain 5% ash. How does the yield of biomass from substrate in mass and molar terms compare with the maximum possible biomass yield?
- Another system for manufacture of single-cell protein is *Methylophilus methylotrophus*. This organism is produced aerobically from methanol with ammonia as nitrogen source. The molecular formula for the biomass is $\text{CH}_{1.68}\text{O}_{0.36}\text{N}_{0.22}$; these cells contain 6% ash.
 - How does the maximum yield of biomass compare with (a) above? What is the main reason for the difference?
 - If the actual yield of biomass from methanol is 42% the thermodynamic maximum, what is the oxygen demand?

4.10 Ethanol production by yeast and bacteria

Both *Saccharomyces cerevisiae* yeast and *Zymomonas mobilis* bacteria produce ethanol from glucose under anaerobic conditions without external electron acceptors. The biomass yield from glucose is 0.11 g g^{-1} for yeast and 0.05 g g^{-1} for *Z. mobilis*. In both cases the nitrogen source is NH_3 . Both cell compositions are represented by the formula $\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$.

- What is the yield of ethanol from glucose in both cases?
- How do the yields calculated in (a) compare with the thermodynamic maximum?

4.11 Detecting unknown products

Yeast growing in continuous culture produce 0.37 g biomass per g glucose consumed; about 0.88 g O₂ is consumed per g cells formed. The nitrogen source is ammonia, and the biomass composition is CH_{1.79}O_{0.56}N_{0.17}. Are other products also synthesised?

4.12 Medium formulation

Pseudomonas 5401 is to be used for production of single-cell protein for animal feed. The substrate is fuel oil. The composition of *Pseudomonas* 5401 is CH_{1.83}O_{0.55}N_{0.25}. If the final cell concentration is 25 g l⁻¹, what minimum concentration of (NH₄)₂SO₄ must be provided in the medium if (NH₄)₂SO₄ is the sole nitrogen source?

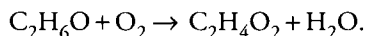
4.13 Oxygen demand for production of recombinant protein

Production of recombinant protein by a genetically-engineered strain of *Escherichia coli* is proportional to cell growth. Ammonia is used as nitrogen source for aerobic respiration of glucose. The recombinant protein has an overall formula CH_{1.55}O_{0.31}N_{0.25}. The yield of biomass from glucose is measured at 0.48 g g⁻¹; the yield of recombinant protein from glucose is about 20% that for cells.

- How much ammonia is required?
- What is the oxygen demand?
- If the biomass yield remains at 0.48 g g⁻¹, how much different are the ammonia and oxygen requirements for wild-type *E. coli* unable to synthesise recombinant protein?

4.14 Effect of growth on oxygen demand

The chemical reaction equation for conversion of ethanol (C₂H₆O) to acetic acid (C₂H₄O₂) is:



Acetic acid is produced from ethanol during growth of *Acetobacter aceti*, which has the composition CH_{1.8}O_{0.5}N_{0.2}. Biomass yield from substrate is 0.14 g g⁻¹; product yield from substrate is 0.92 g g⁻¹. Ammonia is used as nitrogen source. How does growth in this culture affect oxygen demand for acetic acid production?

References

- Felder, R.M. and R.W. Rousseau (1978) *Elementary Principles of Chemical Processes*, Chapter 5, John Wiley, New York.
- Himmelblau, D.M. (1974) *Basic Principles and Calculations in Chemical Engineering*, 3rd edn, Chapter 2, Prentice-Hall, New Jersey.
- Whitwell, J.C. and R.K. Toner (1969) *Conservation of Mass and Energy*, Chapter 4, Blaisdell, Waltham, Massachusetts.
- Cordier, J.-L., B.M. Butsch, B. Birou and U. von Stockar (1987) The relationship between elemental composition and heat of combustion of microbial biomass. *Appl. Microbiol. Biotechnol.* **25**, 305–312.
- Roels, J.A. (1983) *Energetics and Kinetics in Biotechnology*, Chapter 3, Elsevier Biomedical Press, Amsterdam.
- Atkinson, B. and F. Mavituna (1991) *Biochemical Engineering and Biotechnology Handbook*, 2nd edn, Chapter 4, Macmillan, Basingstoke.

Suggestions for Further Reading

Process Mass Balances (see also refs 1–3)

- Hougen, O.A., K.M. Watson and R.A. Ragatz (1954) *Chemical Process Principles: Material and Energy Balances*, 2nd edn, Chapter 7, John Wiley, New York.
- Shaheen, E.I. (1975) *Basic Practice of Chemical Engineering*, Chapter 4, Houghton Mifflin, Boston.

Metabolic Stoichiometry (see also ref 5)

- Erickson, L.E., I.G. Minkevich and V.K. Eroshin (1978) Application of mass and energy balance regularities in fermentation. *Biotechnol. Bioeng.* **20**, 1595–1621.
- Heijnen, J.J. and J.A. Roels (1981) A macroscopic model describing yield and maintenance relationships in aerobic fermentation processes. *Biotechnol. Bioeng.* **23**, 739–763.
- Nagai, S. (1979) Mass and energy balances for microbial growth kinetics. *Adv. Biochem. Eng.* **11**, 49–83.
- Roels, J.A. (1980) Application of macroscopic principles to microbial metabolism. *Biotechnol. Bioeng.* **22**, 2457–2514.

Energy Balances

Unlike many chemical processes, bioprocesses are not particularly energy intensive. Fermenters and enzyme reactors are operated at temperatures and pressures close to ambient; energy input for downstream processing is minimised to avoid damaging heat-labile products. Nevertheless, energy effects are important because biological catalysts are very sensitive to heat and changes in temperature. In large-scale processes, heat released during reaction can cause cell death or denaturation of enzymes if it is not quickly removed. For rational design of temperature-control facilities, energy flows in the system must be determined using energy balances. Energy effects are also important in other areas of bioprocessing such as steam sterilisation.

The law of conservation of energy means that an energy accounting system can be set up to determine the amount of steam or cooling water required to maintain optimum process temperatures. In this chapter, after the necessary thermodynamic concepts are explained, an energy-conservation equation applicable to biological processes is derived. The calculation techniques outlined in Chapter 4 are then extended for solution of simple energy-balance problems.

5.1 Basic Energy Concepts

Energy takes three forms:

- (i) kinetic energy, E_k ;
- (ii) potential energy, E_p ; and
- (iii) internal energy, U .

Kinetic energy is the energy possessed by a moving system because of its velocity. *Potential energy* is due to the position of the system in a gravitational or electromagnetic field, or due to the conformation of the system relative to an equilibrium position (e.g. compression of a spring). *Internal energy* is the sum of all molecular, atomic and sub-atomic energies of matter. Internal energy cannot be measured directly or known in absolute terms; we can only quantify change in internal energy.

Energy is transferred as either heat or work. *Heat* is energy which flows across system boundaries because of a temperature difference between the system and surroundings. *Work* is energy transferred as a result of any driving force other than temperature difference. There are two types of work: *shaft work* W_s , which is work done by a moving part within the system, e.g., an impeller mixing a fermentation broth, and *flow work* W_f . Flow work is the energy required to push matter into

the system. In a flow-through process, fluid at the inlet has work done on it by fluid just outside of the system, while fluid at the outlet does work on the fluid in front to push the flow along. Flow work is given by the expression:

$$W_f = pV \quad (5.1)$$

where p is pressure and V is volume. (Convince yourself that pV has the same dimensions as work and energy.)

5.1.1 Units

The SI unit for energy is the *joule* (J): $1 \text{ J} = 1 \text{ newton metre (N m)}$. Another unit is the *calorie* (cal), which is defined as the heat required to raise the temperature of 1 g pure water by 1°C at 1 atm pressure. The quantity of heat according to this definition depends somewhat on the temperature of the water; because there has been no universal agreement on a reference temperature, there are several slightly different calorie-units in use. The *international table calorie* (cal_{IT}) is fixed at 4.1868 J exactly. In imperial units, the British thermal unit (Btu) is common; this is defined as the amount of energy required to raise the temperature of 1 lb water by 1°F at 1 atm pressure. As with the calorie, a reference temperature is required for this definition; 60°F is common although other temperatures are sometimes used.

5.1.2 Intensive and Extensive Properties

Properties of matter fall into two categories: those whose magnitude depends on the quantity of matter present and those

whose magnitude does not. Temperature, density, and mole fraction are examples of properties which are independent of the size of the system; these quantities are called *intensive variables*. On the other hand, mass, volume and energy are *extensive variables* which change if mass is added to or removed from the system. Extensive variables can be converted to *specific* quantities by dividing by the mass of the system; for example, specific volume is volume divided by mass. Because specific properties are independent of the mass of the system, they are also intensive variables. In this chapter, for extensive properties denoted by an upper-case symbol, the specific property is given in lower-case notation. Therefore if U is internal energy, u denotes specific internal energy with units, e.g. kJ g^{-1} . Although, strictly speaking, the term 'specific' refers to the quantity per unit mass, we will use the same lower-case symbols for molar quantities, e.g. with units kJ gmol^{-1} .

5.1.3 Enthalpy

Enthalpy is a property used frequently in energy-balance calculations. It is defined as the combination of two energy terms:

$$H = U + pV \quad (5.2)$$

where H is enthalpy, U is internal energy, p is pressure and V is volume. Specific enthalpy h is therefore:

$$h = u + pv \quad (5.3)$$

where u is specific internal-energy and v is specific volume. Since internal energy cannot be measured or known in absolute terms, neither can enthalpy.

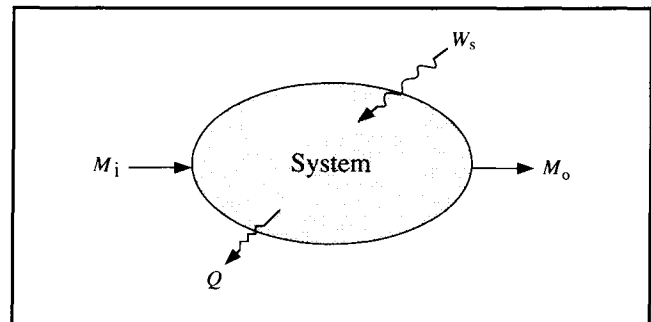
5.2 General Energy-Balance Equations

The principle underlying all energy-balance calculations is the law of conservation of energy, which states that energy can be neither created nor destroyed. Although this law does not apply to nuclear reactions, conservation of energy remains a valid principle for bioprocesses because nuclear rearrangements are not involved. In the following sections, we will derive the equations used for solution of energy-balance problems.

The law of conservation of energy can be written as:

$$\left\{ \begin{array}{l} \text{energy in through} \\ \text{system boundaries} \end{array} \right\} - \left\{ \begin{array}{l} \text{energy out through} \\ \text{system boundaries} \end{array} \right\} = \left\{ \begin{array}{l} \text{energy accumulated} \\ \text{within the system} \end{array} \right\} \quad (5.4)$$

Figure 5.1 Flow system for energy-balance calculations.



For practical application of this equation, consider the system depicted in Figure 5.1. Mass M_i enters the system while mass M_o leaves. Both these masses have energy associated with them in the form of internal, kinetic and potential energy; flow work is also being done. Energy leaves the system as heat Q ; shaft work W_s is done on the system by the surroundings. We will assume that the system is homogeneous without charge or surface-energy effects.

To apply Eq. (5.4), we must identify which forms of energy are involved in each term of the expression. If we group together the extensive properties and express them as specific variables multiplied by mass, Eq. (5.4) can be written:

$$M_i(u + e_k + e_p + pv)_i - M_o(u + e_k + e_p + pv)_o - Q + W_s = \Delta E \quad (5.5)$$

where subscripts i and o refer to inlet and outlet conditions, respectively, and ΔE represents the total change or accumulation of energy in the system. u is specific internal energy, e_k is specific kinetic energy, e_p is specific potential energy, p is pressure, and v is specific volume. All energies associated with masses crossing the system boundary are added together; the energy-transfer terms Q and W_s are considered separately. Shaft work appears explicitly in Eq. (5.5) as W_s ; flow work done by inlet and outlet streams is represented as pv multiplied by mass.

Energy flows represented by Q and W_s can be directed either into or out of the system; appropriate signs must be used to indicate the direction of flow. Because it is usual in bioprocesses that shaft work be done on the system by external sources, in this text we will adopt the convention that work is positive when energy flows *from* the surroundings *to* the system as shown in Figure 5.1. Conversely, work will be considered negative when the system supplies work energy to the surroundings. On the other hand, we will regard heat as positive when the surroundings receives energy from the system,

i.e. when the temperature of the system is higher than the surroundings. Therefore, when W_s and Q are positive quantities, W_s makes a positive contribution to the energy content of the system while Q causes a reduction. These effects are accounted for in Eq. (5.5) by the signs preceding Q and W_s . The opposite sign convention is sometimes used in thermodynamics texts. Choice of sign convention is arbitrary if used consistently.

Eq. (5.5) refers to a process with only one input and one output stream. A more general equation is Eq. (5.6), which can be used for any number of separate material flows:

$$\sum_{\text{input streams}} M(u + e_k + e_p + pv) - \sum_{\text{output streams}} M(u + e_k + e_p + pv) - Q + W_s = \Delta E. \quad (5.6)$$

The symbol Σ means summation; the internal, kinetic, potential and flow work energies associated with all output streams are added together and subtracted from the sum for all input streams. Eq. (5.6) is a basic form of the *first law of thermodynamics*, a simple mathematical expression of the law of conservation of energy. The equation can be shortened by substituting enthalpy h for $u + pv$ as defined by Eq. (5.3):

$$\sum_{\text{input streams}} M(h + e_k + e_p) - \sum_{\text{output streams}} M(h + e_k + e_p) - Q + W_s = \Delta E. \quad (5.7)$$

5.2.1 Special Cases

Eq. (5.7) can be simplified considerably if the following assumptions are made:

- (i) kinetic energy is negligible; and
- (ii) potential energy is negligible.

These assumptions are acceptable for bioprocesses, in which high-velocity motion and large changes in height or electromagnetic field do not generally occur. Thus, the energy-balance equation becomes:

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) - Q + W_s = \Delta E. \quad (5.8)$$

Eq. (5.8) can be simplified further in the following special cases:

- (i) *Steady-state flow process.* At steady state, all properties of the system are invariant. Therefore, there can be no accumulation or change in the energy of the system: $\Delta E = 0$.

The steady-state energy-balance equation is:

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) - Q + W_s = 0. \quad (5.9)$$

Eq. (5.9) can also be applied over the entire duration of batch and fed-batch processes if there is no energy accumulation; 'output streams' in this case refers to the harvesting of all mass in the system at the end of the process. Eq. (5.9) is used frequently in bioprocess energy balances.

- (ii) *Adiabatic process.* A process in which no heat is transferred to or from the system is termed *adiabatic*; if the system has an *adiabatic wall* it cannot receive or release heat to the surroundings. Under these conditions $Q = 0$ and Eq. (5.8) becomes:

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) + W_s = \Delta E. \quad (5.10)$$

Eqs (5.8)–(5.10) are energy-balance equations which allow us to predict, for example, how much heat must be removed from a fermenter to maintain optimum conditions, or the effect of evaporation on cooling requirements. To apply the equations we must know the specific enthalpy h of flow streams entering or leaving the system. Methods for calculating enthalpy are outlined in the following sections.

5.3 Enthalpy Calculation Procedures

Irrespective of how enthalpy changes occur, certain conventions are used in enthalpy calculations.

5.3.1 Reference States

Specific enthalpy h appears explicitly in energy-balance equations. What values of h do we use in these equations if enthalpy cannot be measured or known in absolute terms? Because energy balances are actually concerned with the *difference* in enthalpy between incoming and outgoing streams, we can overcome any difficulties by working always in terms of enthalpy change. In many energy-balance problems, changes in enthalpy are evaluated relative to reference states that must be defined at the beginning of the calculation.

Because H cannot be known absolutely, it is convenient to assign $H = 0$ to some reference state. For example, when 1 gmol carbon dioxide is heated at 1 atm pressure from 0°C to 25°C, the change in enthalpy of the gas can be calculated

(using methods explained later) as $\Delta H = 0.91$ kJ. If we assign $H = 0$ for CO_2 gas at 0°C , H at 25°C can be considered to be 0.91 kJ. This result does not mean that the absolute value of enthalpy at 25°C is 0.91 kJ; we can say only that the enthalpy at 25°C is 0.91 kJ relative to the enthalpy at 0°C .

We will use various reference states in energy-balance calculations to determine enthalpy change. Suppose for example we want to calculate the change in enthalpy as a system moves from State 1 to State 2. If the enthalpies of States 1 and 2 are known relative to the same reference condition H_{ref} , ΔH is calculated as follows:

$$\begin{aligned} \text{State 1} &\xrightarrow{\Delta H} \text{State 2} \\ \text{Enthalpy} = H_1 - H_{\text{ref}} &\quad \text{Enthalpy} = H_2 - H_{\text{ref}} \\ \Delta H = (H_2 - H_{\text{ref}}) - (H_1 - H_{\text{ref}}) &= H_2 - H_1. \end{aligned}$$

ΔH is therefore independent of the reference state because H_{ref} cancels out in the calculation.

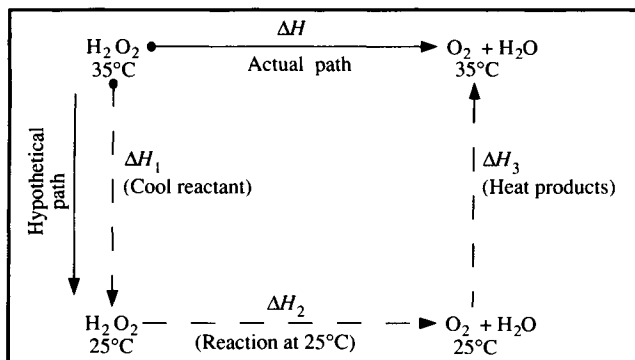
5.3.2 State Properties

Values of some variables depend only on the state of the system and not on how that state was reached. These variables are called *state properties* or *functions of state*; examples include temperature, pressure, density and composition. On the other hand, work is a *path function* since the amount of work done depends on the way in which the final state of the system is obtained from previous states.

Enthalpy is a state function. This property of enthalpy is very handy in energy-balance calculations; it means that change in enthalpy for a process can be calculated by taking a series of hypothetical steps or *process path* leading from the initial state and eventually reaching the final state. Change in enthalpy is calculated for each step; the total enthalpy change for the process is then equal to the sum of changes in the hypothetical path. This is true even though the process path used for calculation is not the same as that actually undergone by the system. As an example, consider the enthalpy change for the process shown in Figure 5.2 in which hydrogen peroxide is converted to oxygen and water by catalase enzyme. The enthalpy change for the direct process at 35°C can be calculated using an alternative pathway in which hydrogen peroxide is first cooled to 25°C , oxygen and water are formed by reaction at 25°C , and the products then heated to 35°C . Because the initial and final states for both actual and hypothetical paths are the same, the total enthalpy change is also identical:

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 \quad (5.11)$$

Figure 5.2 Hypothetical process path for calculation of enthalpy change.



The reason for using hypothetical rather than actual pathways to calculate enthalpy change will become apparent later in the chapter.

5.4 Enthalpy Change in Non-Reactive Processes

Change in enthalpy can occur as a result of:

- (i) temperature change;
- (ii) change of phase;
- (iii) mixing or solution; and
- (iv) reaction.

In the remainder of this section we will consider enthalpy changes associated with (i), (ii) and (iii). We will then consider how the results are used in energy-balance calculations. Processes involving reaction will be discussed in Sections 5.8–5.11.

5.4.1 Change in Temperature

Heat transferred to raise or lower the temperature of a material is called *sensible heat*; change in the enthalpy of a system due to variation in temperature is called *sensible heat change*. In energy-balance calculations, sensible heat change is determined using a property of matter called the *heat capacity at constant pressure*, or just *heat capacity*. We will use the symbol C_p for heat capacity; units for C_p are, e.g. $\text{J gmol}^{-1} \text{K}^{-1}$, $\text{cal g}^{-1} ^\circ\text{C}^{-1}$ and $\text{Btu lb}^{-1} ^\circ\text{F}^{-1}$. The term *specific heat* refers to heat capacity expressed on a per-unit-mass basis. Heat capacity must be known before enthalpy changes from heating or cooling can be determined. Tables B.3–B.6 in Appendix B list C_p values for several organic and inorganic compounds. Additional C_p data and information about estimating heat capacities can be found in references such as *Chemical*

Engineers' Handbook [1], *Handbook of Chemistry and Physics* [2] and *International Critical Tables* [3].

There are several methods for calculating enthalpy change using C_p values. When C_p is constant, the change in enthalpy of a substance due to change in temperature at constant pressure is:

$$\Delta H = MC_p \Delta T = MC_p (T_2 - T_1) \quad (5.12)$$

where M is either mass or moles of the substance depending on the dimensions of C_p , T_1 is the initial temperature and T_2 is the final temperature. The corresponding change in specific enthalpy is:

$$\Delta h = C_p \Delta T = C_p (T_2 - T_1). \quad (5.13)$$

Example 5.1 Sensible heat change with constant C_p

What is the enthalpy of 150 g formic acid at 70°C and 1 atm relative to 25°C and 1 atm?

Solution:

From Table B.5, C_p for formic acid in the temperature range of interest is $0.524 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$. Substituting into Eq (5.12):

$$\begin{aligned} \Delta H &= (150 \text{ g}) (0.524 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}) (70 - 25)^\circ\text{C} \\ \Delta H &= 3537.0 \text{ cal} \end{aligned}$$

or

$$\Delta H = 3.54 \text{ kcal.}$$

Relative to $H = 0$ at 25°C, the enthalpy of formic acid at 70°C is 3.54 kcal.

Heat capacities for most substances vary with temperature. This means that when we calculate enthalpy change due to change in temperature, the value of C_p itself varies over the range of ΔT . Heat capacities are often tabulated as polynomial functions of temperature, such as:

$$C_p = a + bT + cT^2 + dT^3. \quad (5.14)$$

Coefficients a , b , c and d for a number of substances are given in Table B.3 in Appendix B.

Sometimes we can assume that heat capacity is constant; this will give results for sensible heat change which approximate the true value. Because the temperature range of interest in bioprocessing is relatively small, assuming constant heat capacity for some materials does not introduce large errors. C_p data may not be available at all temperatures; heat capacities like most of those listed in Tables B.5 and B.6 are applicable only at a specified temperature or temperature range. As an example, in Table B.5 the heat capacity for liquid acetone between 24.2°C and 49.4°C is given as $0.538 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$ even though this value will vary within the temperature range. A useful rule of thumb for organic liquids near room temperature is that C_p increases by $0.001\text{--}0.002 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$

for each Celsius-degree temperature increase [4].

One method for calculating sensible heat change when C_p varies with temperature involves use of the *mean heat capacity*, C_{pm} . Table B.4 in Appendix B lists mean heat capacities for several common gases. These values are based on changes in enthalpy relative to a single reference temperature, $T_{\text{ref}} = 0^\circ\text{C}$. To determine the change in enthalpy for a change in temperature from T_1 to T_2 , read the values of C_{pm} at T_1 and T_2 and calculate:

$$\Delta H = M[(C_{pm})_{T_2}(T_2 - T_{\text{ref}}) - (C_{pm})_{T_1}(T_1 - T_{\text{ref}})]. \quad (5.15)$$

5.4.2 Change of Phase

Phase changes, such as vaporisation and melting, are accompanied by relatively large changes in internal energy and enthalpy as bonds between molecules are broken and reformed. Heat transferred to or from a system causing change of phase at constant temperature and pressure is known as *latent heat*. Types of latent heat are:

- (i) *latent heat of vaporisation* (Δh_v): heat required to vaporise a liquid;

- (ii) *latent heat of fusion* (Δh_f): heat required to melt a solid; and
- (iii) *latent heat of sublimation* (Δh_s): heat required to directly vaporise a solid.

Condensation of gas to liquid requires removal rather than addition of heat; the latent heat evolved in condensation is $-\Delta h_v$. Similarly, the latent heat evolved in freezing or solidification of liquid to solid is $-\Delta h_f$.

Latent heat is a property of substances and, like heat capacity, varies with temperature. Tabulated values of latent heats usually apply to substances at their normal boiling, melting or

sublimation point at 1 atm, and are called *standard heats of phase change*. Table B.7 in Appendix B lists latent heats for selected compounds; more values may be found in *Chemical Engineers' Handbook* [1] and *Handbook of Chemistry and Physics* [2].

The change in enthalpy resulting from phase change is calculated directly from the latent heat. For example, increase in enthalpy due to evaporation of liquid mass M at constant temperature is:

$$\Delta H = M \Delta h_v. \quad (5.16)$$

Example 5.2 Enthalpy of condensation

50 g benzaldehyde vapour is condensed at 179°C. What is the enthalpy of the liquid relative to the vapour?

Solution:

From Table B.7, the molecular weight of benzaldehyde is 106.12, the normal boiling point is 179.0°C, and the standard heat of vaporisation is 38.40 kJ gmol⁻¹. For condensation the latent heat is -38.40 kJ gmol⁻¹. The enthalpy change is:

$$\Delta H = 50 \text{ g} (-38.40 \text{ kJ gmol}^{-1}) \cdot \left| \frac{1 \text{ gmol}}{106.12 \text{ g}} \right| = -18.09 \text{ kJ}.$$

Therefore, the enthalpy of 50 g benzaldehyde liquid relative to the vapour at 179°C is -18.09 kJ. As heat is released during condensation, the enthalpy of the liquid is lower than the vapour.

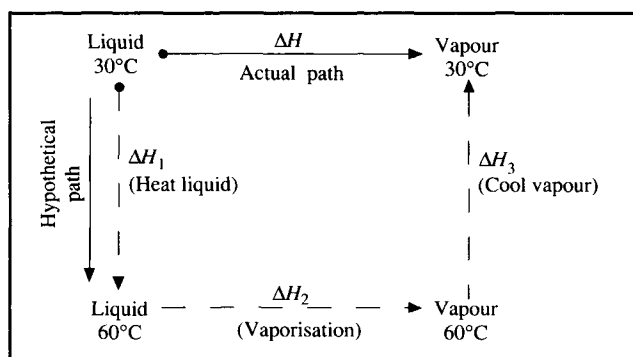
Phase changes often occur at temperatures other than the normal boiling, melting or sublimation point; for example, water can evaporate at temperatures higher or lower than 100°C. How can we determine ΔH when the latent heat at the actual temperature of the phase change is not listed in the tables? This problem is overcome by using a hypothetical process path as described in Section 5.3.2. Suppose a substance is vaporised isothermally at 30°C although tabulated values for standard heat of vaporisation refer to 60°C. As shown in Figure 5.3, we can consider a process whereby liquid is heated from 30°C to 60°C, vaporised at 60°C, and the vapour cooled to 30°C. The total enthalpy change for this process is the same as if vaporisation occurred directly at 30°C. ΔH_1 and ΔH_3 are sensible heat changes and can be calculated using heat-capacity values and the methods described in Section 5.4.1. ΔH_2 is the latent heat at standard conditions available from tables. Because enthalpy is a state property, ΔH for the actual path is the same as $\Delta H_1 + \Delta H_2 + \Delta H_3$.

5.4.3 Mixing and Solution

So far we have considered enthalpy changes for pure compounds. For an *ideal solution* or *ideal mixture* of several

compounds, the thermodynamic properties of the mixture are a simple sum of contributions from the individual components. However, when compounds are mixed or dissolved, bonds between molecules in the solvent and solute are broken and reformed. In *real solutions* a net absorption or release of energy accompanies these processes resulting in changes in the internal energy and enthalpy of the mixture. Dilution of sulphuric acid with water is a good example; in this case energy is

Figure 5.3 Process path for calculating latent-heat change at a temperature other than the normal boiling point.



released. For real solutions there is an additional energy term to consider in evaluating enthalpy: the *integral heat of mixing* or *integral heat of solution*, Δh_m . The integral heat of solution is defined as the change in enthalpy which occurs as one mole of solute is dissolved at constant temperature in a given quantity of solvent. The enthalpy of a non-ideal mixture of two compounds A and B is:

$$H_{\text{mixture}} = H_A + H_B + \Delta H_m \quad (5.17)$$

where ΔH_m is the heat of mixing.

Heat of mixing is a property of the solution components and is dependent on the temperature and concentration of the mixture. As a solution becomes more and more dilute, an asymptotic value of Δh_m is reached. This value is called the *integral heat of solution at infinite dilution*. When water is the primary component of solutions, Δh_m at infinite dilution can be used to calculate the enthalpy of the mixture. Δh_m values for selected aqueous solutions are listed in *Chemical Engineers' Handbook* [1], *Handbook of Chemistry and Physics* [2] and *Biochemical Engineering and Biotechnology Handbook* [5].

Example 5.3 Heat of solution

Malonic acid and water are initially at 25°C. If 15 g malonic acid is dissolved in 5 kg water, how much heat must be added for the solution to remain at 25°C? What is the solution enthalpy relative to the components?

Solution:

The molecular weight of malonic acid is 104. Because the solution is very dilute (< 0.3% w/w), we can use the integral heat of solution at infinite dilution. From handbooks, Δh_m at room temperature is 4.493 kcal gmol⁻¹. This value is positive; therefore the mixture enthalpy is greater than the components and heat is absorbed during solution. The heat required for the solution to remain at 25°C is:

$$\Delta H = 4.493 \text{ kcal gmol}^{-1} (15 \text{ g}) \cdot \left| \frac{1 \text{ gmol}}{104 \text{ g}} \right| = 0.648 \text{ kcal.}$$

Relative to $H = 0$ for water and malonic acid at 25°C, the enthalpy of the solution at 25°C is 0.648 kcal.

In biological systems, significant changes in enthalpy due to heats of mixing do not often occur. Most solutions in fermentation and enzyme processes are dilute aqueous mixtures; in energy-balance calculations these solutions are usually considered ideal without much loss of accuracy.

5.5 Steam Tables

Steam tables have been used for many years by engineers designing industrial processes and power stations. These tables list the thermodynamic properties of water, including specific volume, internal energy and enthalpy. As we are mainly concerned here with enthalpies, a list of enthalpy values for steam and water under various conditions has been extracted from the steam tables and given in Appendix C [6]. All enthalpy values must have a reference point; in the steam tables of Appendix C, $H = 0$ for liquid water at the triple point: 0.01°C and 0.6112 kPa pressure. (The triple point is an invariant condition of pressure and temperature at which ice, liquid water and water vapour are in equilibrium with each other.) Steam

tables from other sources may have different reference states. Steam tables eliminate the need for sensible-heat and latent-heat calculations for water and steam, and can be used directly in energy-balance calculations.

Tables C.1 and C.2 in Appendix C list enthalpy values for liquid water and *saturated steam*. When liquid and vapour are in equilibrium with each other, they are saturated; a gas saturated with water contains all the water it can hold at the prevailing temperature and pressure. For a pure substance such as water, once the temperature is specified, saturation occurs at only one pressure. For example, from Table C.2 saturated steam at 188°C has a pressure of 1200 kPa. Also from the table, the enthalpy of this steam relative to the triple point of water is 2782.7 kJ kg⁻¹; liquid water in equilibrium with the steam has an enthalpy of 798.4 kJ kg⁻¹. The latent heat of vaporisation under these conditions is the difference between liquid and vapour enthalpies; as indicated in the middle enthalpy column, Δh_v is 1984.3 kJ kg⁻¹. Table C.1 lists enthalpies of saturated water and steam by temperature; Table C.2 lists these enthalpies by pressure.

It is usual when using steam tables to ignore the effect of pressure on the enthalpy of liquid water. For example, the enthalpy of water at 40°C and 1 atm (101.3 kPa) is found by looking up the enthalpy of saturated water at 40°C in Table C.1, and assuming the value is independent of pressure. This assumption is valid at low pressure, i.e. less than about 50 atm. The enthalpy of liquid water at 40°C and 1 atm is therefore 167.5 kJ kg⁻¹.

Enthalpy values for *superheated steam* are given in Table C.3. If the temperature of saturated vapour is increased (or the pressure decreased at constant temperature), the vapour is said to be superheated. A superheated vapour cannot condense until it is returned to saturation conditions. The difference between the temperature of a superheated gas and its saturation temperature is called the *degrees of superheat* of the gas. In Table C.3, enthalpy is listed as a function of temperature at 15 different pressures from 10 kPa to 50 000 kPa; for example, superheated steam at 1000 kPa pressure and 250°C has an enthalpy of 2943 kJ kg⁻¹ relative to the triple point. Table C.3 also lists properties at saturation conditions; at 1000 kPa the saturation temperature is 179.9°C, the enthalpy of liquid water under these conditions is 762.6 kJ kg⁻¹, and the enthalpy of saturated vapour is 2776.2 kJ kg⁻¹. Thus, the degrees of superheat for superheated steam at 1000 kPa and 250°C can be calculated as (250 - 179.9) = 70.1 centigrade degrees. Water under pressure remains liquid even at relatively high temperatures. Enthalpy values for liquid water up to 350°C are found in the upper region of Table C.3 above the line extending to the critical pressure.

5.6 Procedure For Energy-Balance Calculations Without Reaction

Methods described in Section 5.4 for evaluating enthalpy can be used to solve energy-balance problems for systems in which reactions do not occur. Many of the points described in Section 4.3 for material balances also apply when setting out an energy balance.

- (i) A properly drawn and labelled *flow diagram* is essential to identify all inlet and outlet streams and their compositions. For energy balances, the temperatures, pressures

and phases of the material should also be indicated if appropriate.

- (ii) The *units* selected for the energy balance should be stated; these units are also used when labelling the flow diagram.
- (iii) As in mass balance problems, a *basis* for the calculation must be chosen and stated clearly.
- (iv) The *reference state* for $H = 0$ is determined. In the absence of reaction, reference states for each molecular species in the system can be arbitrarily assigned.
- (v) State all *assumptions* used in solution of the problem. Assumptions such as absence of leaks and steady-state operation for continuous processes are generally applicable.

Following on from (v), other assumptions commonly made for energy balances include the following:

- (a) The system is homogeneous or well mixed. Under these conditions, product streams including gases leave the system at the system temperature.
- (b) Heats of mixing are often neglected for mixtures containing compounds of similar molecular structure. Gas mixtures are always considered ideal.
- (c) Sometimes shaft work can be neglected even though the system is stirred by mechanical means. This assumption may not apply when vigorous agitation is used or when the liquid being stirred is very viscous. When shaft work is not negligible you will need to know how much mechanical energy is input through the stirrer.
- (d) Evaporation in liquid systems may be considered negligible if the components are not particularly volatile or if the operating temperature is relatively low.
- (e) Heat losses from the system to the surroundings are often ignored; this assumption is generally valid for large insulated vessels when the operating temperature is close to ambient.

5.7 Energy-Balance Worked Examples Without Reaction

As illustrated in the following examples, the format described in Chapter 4 for material balances can be used as a foundation for energy-balance calculations.

Example 5.4 Continuous water heater

Water at 25°C enters an open heating tank at a rate of 10 kg h⁻¹. Liquid water leaves the tank at 88°C at a rate of 9 kg h⁻¹; 1 kg h⁻¹ water vapour is lost from the system through evaporation. At steady state, what is the rate of heat input to the system?

Solution:

1. *Assemble*

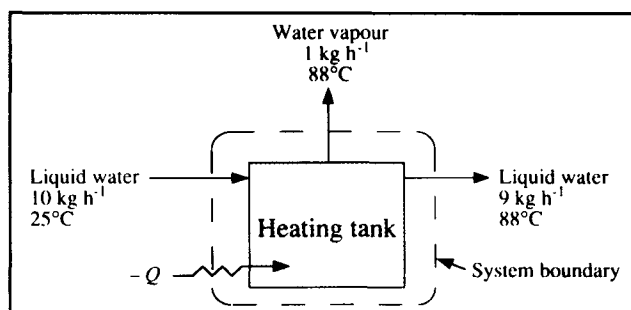
(i) *Select units for the problem.*

kg, h, kJ, °C.

(ii) *Draw the flowsheet showing all data and units.*

The flowsheet is shown in Figure 5E4.1.

Figure 5E4.1 Flowsheet for continuous water heater.



(iii) *Define the system boundary by drawing on the flowsheet.*

The system boundary is indicated in Figure 5E4.1.

2. *Analyse*

(i) *State any assumptions.*

- process is operating at steady state
- system does not leak
- system is homogeneous
- evaporation occurs at 88°C
- vapour is saturated
- shaft work is negligible
- no heat losses

(ii) *Select and state a basis.*

The calculation is based on 10 kg water entering the system, or 1 hour.

(iii) *Select and state a reference state.*

The reference state for water is the same as that used in the steam tables: 0.01°C and 0.6112 kPa.

(iv) *Collect any extra data needed.*

h (liquid water at 88°C) = 368.5 kJ kg⁻¹ (Table C.1)

h (saturated steam at 88°C) = 2656.9 kJ kg⁻¹ (Table C.1)

h (liquid water at 25°C) = 104.8 kJ kg⁻¹ (Table C.1).

(v) *Determine which compounds are involved in reaction.*

No reaction occurs.

(vi) *Write down the appropriate mass-balance equation.*

The mass balance is already complete.

(vii) *Write down the appropriate energy-balance equation.*

At steady state, Eq. (5.9) applies:

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) - Q + W_s = 0.$$

3. Calculate

(i) Identify the terms of the energy-balance equation.

For this problem $W_s = 0$. The energy-balance equation becomes:

$$(Mh)_{\text{liq in}} - (Mh)_{\text{liq out}} - (Mh)_{\text{vap out}} - Q = 0.$$

Substituting the information available:

$$(10 \text{ kg}) (104.8 \text{ kJ kg}^{-1}) - (9 \text{ kg}) (368.5 \text{ kJ kg}^{-1}) - (1 \text{ kg}) (2656.9 \text{ kJ kg}^{-1}) - Q = 0$$

$$Q = -4925.4 \text{ kJ}.$$

Q has a negative value. Thus, according to the sign convention outlined in Section 5.2, heat must be supplied to the system from the surroundings.

4. Finalise

Answer the specific questions asked in the problem; check the number of significant figures; state the answers clearly.

The rate of heat input is $4.93 \times 10^3 \text{ kJ h}^{-1}$.

Example 5.5 Cooling in downstream processing

In downstream processing of gluconic acid, concentrated fermentation broth containing 20% (w/w) gluconic acid is cooled in a heat exchanger prior to crystallisation. 2000 kg h^{-1} liquid leaving an evaporator at 90°C must be cooled to 6°C . Cooling is achieved by heat exchange with 2700 kg h^{-1} water initially at 2°C . If the final temperature of the cooling water is 50°C , what is the rate of heat loss from the gluconic acid solution to the surroundings? Assume the heat capacity of gluconic acid is $0.35 \text{ cal g}^{-1}^\circ\text{C}^{-1}$.

Solution:

1. Assemble

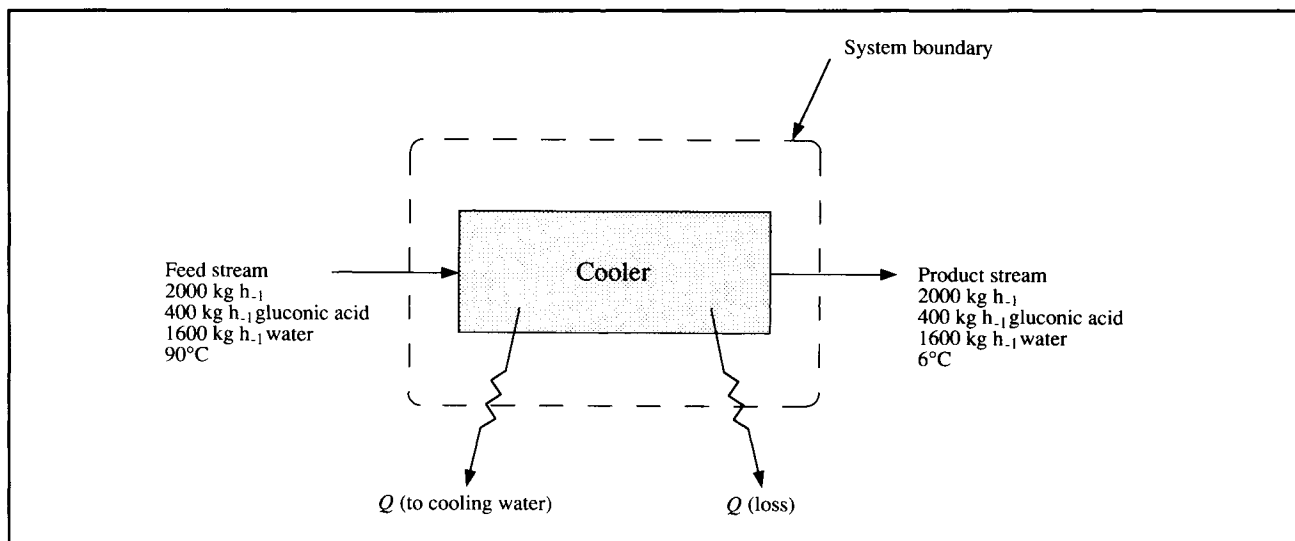
(i) Units.

kg, h, kJ, $^\circ\text{C}$.

(ii) Flowsheet.

The flowsheet is shown in Figure 5E5.1.

Figure 5E5.1 Flowsheet for cooling gluconic acid product stream.



(iii) *System boundary.*

The system boundary indicated in Figure 5E5.1 separates the gluconic acid solution from the cooling water

2. *Analyse*(i) *Assumptions.*

- steady state
- no leaks
- other components of the fermentation broth can be considered water
- gluconic acid and water form an ideal mixture
- no shaft work

(ii) *Basis.*

2000 kg feed stream, or 1 hour.

(iii) *Reference state.*

$H = 0$ for gluconic acid at 90°C

$H = 0$ for water at its triple point.

(iv) *Extra data.*

The heat capacity of gluconic acid is $0.35 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$; we will assume this C_p remains constant over the temperature range of interest. Converting units:

$$C_p (\text{gluconic acid}) = \frac{0.35 \text{ cal}}{\text{g } ^\circ\text{C}} \cdot \left| \frac{4.187 \text{ J}}{1 \text{ cal}} \right| \cdot \left| \frac{1 \text{ kJ}}{1000 \text{ J}} \right| \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right|$$

$$C_p (\text{gluconic acid}) = 1.47 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}.$$

$$h (\text{liquid water at } 90^\circ\text{C}) = 376.9 \text{ kJ kg}^{-1} \text{ (Table C.1)}$$

$$h (\text{liquid water at } 6^\circ\text{C}) = 25.2 \text{ kJ kg}^{-1} \text{ (Table C.1)}$$

$$h (\text{liquid water at } 2^\circ\text{C}) = 8.4 \text{ kJ kg}^{-1} \text{ (Table C.1)}$$

$$h (\text{liquid water at } 50^\circ\text{C}) = 209.3 \text{ kJ kg}^{-1} \text{ (Table C.1)}.$$

(v) *Compounds involved in reaction.*

No reaction occurs.

(vi) *Mass-balance equation.*

The mass-balance equation for total mass, gluconic acid and water is:

$$\text{mass in} = \text{mass out}.$$

The mass flow rates are as shown in Figure 5E5.1.

(vii) *Energy-balance equation.*

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) - Q + W_s = 0.$$

3. *Calculate*(i) *Identify terms in the energy-balance equation.*

$W_s = 0$. There are two heat flows out of the system: one to the cooling water (Q) and one representing loss to the surroundings (Q_{loss}). With symbols W = water and G = gluconic acid, the energy-balance equation is:

$$(Mh)_{W \text{ in}} + (Mh)_{G \text{ in}} - (Mh)_{W \text{ out}} - (Mh)_{G \text{ out}} - Q_{\text{loss}} - Q = 0.$$

$$(Mh)_{W \text{ in}} = (1600 \text{ kg}) (376.9 \text{ kJ kg}^{-1}) = 6.03 \times 10^5 \text{ kJ}.$$

$$(Mh)_{G \text{ in}} = 0 \text{ (reference state)}.$$

$$(Mh)_{W \text{ out}} = (1600 \text{ kg}) (1.47 \text{ kJ kg}^{-1}) = 4.03 \times 10^4 \text{ kJ}.$$

$(Mh)_{G \text{ out}}$ at 6°C is calculated as a sensible heat change from 90°C using Eq. (5.12):

$$(Mh)_{G \text{ out}} = MC_p (T_2 - T_1) = (400 \text{ kg}) (1.47 \text{ kJ kg}^{-1} \text{ °C}^{-1}) (6 - 90) \text{ °C}$$

$$\therefore (Mh)_{G \text{ out}} = -4.94 \times 10^4 \text{ kJ}.$$

The heat removed to the cooling water, Q_c is equal to the enthalpy change of the cooling water between 2°C and 50°C:

$$Q = (2700 \text{ kg}) (209.3 - 8.4) \text{ kJ kg}^{-1} = 5.42 \times 10^5 \text{ kJ}.$$

These results can now be substituted into the energy-balance equation:

$$(6.03 \times 10^5 \text{ kJ}) + (0 \text{ kJ}) - (4.03 \times 10^4 \text{ kJ}) - (-4.94 \times 10^4 \text{ kJ}) - Q_{\text{loss}} - 5.42 \times 10^5 \text{ kJ} = 0$$

$$\therefore Q_{\text{loss}} = 7.01 \times 10^4 \text{ kJ}.$$

4. Finalise

The rate of heat loss to the surroundings is $7.0 \times 10^4 \text{ kJ h}^{-1}$.

It is important to recognise that the final answers to energy-balance problems do not depend on the choice of reference states for the components. Although values of h depend on the reference states, as discussed in Section 5.3.1 this dependence disappears when the energy-balance equation is applied and the difference between input and output enthalpies determined. To prove the point, any of the examples in this chapter can be repeated using different reference conditions to obtain the same final answers.

5.8 Enthalpy Change Due to Reaction

Reactions in bioprocesses occur as a result of enzyme activity and cell metabolism. During reaction, relatively large changes in internal energy and enthalpy occur as bonds between atoms are rearranged. *Heat of reaction* ΔH_{rxn} is the energy released or absorbed during reaction, and is equal to the difference in enthalpy of reactants and products:

$$\Delta H_{\text{rxn}} = \sum_{\text{products}} Mh - \sum_{\text{reactants}} Mh \quad (5.18)$$

or

$$\Delta H_{\text{rxn}} = \sum_{\text{products}} nh - \sum_{\text{reactants}} nh \quad (5.19)$$

where Σ denotes the sum, M is mass, n is number of moles, and h is specific enthalpy expressed on either a per-mass or per-mole basis. Note that M and n represent the mass and moles

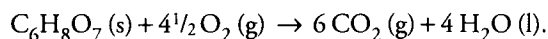
actually involved in the reaction, not the total amount present in the system. In an *exothermic reaction* the energy required to hold the atoms of product together is less than for the reactants; surplus energy is released as heat and ΔH_{rxn} is negative in value. On the other hand, energy is absorbed during *endothermic reactions*; the enthalpy of the products is greater than the reactants and ΔH_{rxn} is positive.

The specific heat of reaction Δh_{rxn} is a property of matter. The value of Δh_{rxn} depends on which reactants and products are involved in the reaction and the temperature and pressure. Because any molecule can participate in a large number of reactions, it is not feasible to tabulate all possible Δh_{rxn} values. Instead, Δh_{rxn} is calculated from the heats of combustion of individual compounds.

5.8.1 Heat of Combustion

Heat of combustion Δh_c is defined as the heat evolved during reaction of a substance with oxygen to yield certain oxidation products such as CO_2 gas, H_2O liquid and N_2 gas. The *standard heat of combustion* Δh_c° is the specific enthalpy change associated with this reaction at standard conditions, usually 25°C and 1 atm pressure. By convention, Δh_c° is zero for the products of oxidation, i.e. CO_2 gas, H_2O liquid, N_2 gas, etc.; standard heats of combustion for other compounds are always negative. Table B.8 in Appendix B lists selected values; heats of combustion for other materials can be found in *Chemical Engineers' Handbook* [1] and *Handbook of Chemistry and Physics* [2]. As an example, the standard heat of combustion for citric acid is given in Table B.8 as $-1962.0 \text{ kJ gmol}^{-1}$; this

refers to the heat evolved at 25°C and 1 atm in the following reaction :



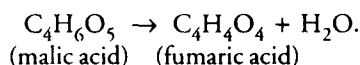
Standard heats of combustion are used to calculate the *standard heat of reaction* $\Delta H_{\text{rxn}}^\circ$ for reactions involving combustible reactants and combustion products:

$$\Delta H_{\text{rxn}}^\circ = \sum_{\text{reactants}} n \Delta h_c^\circ - \sum_{\text{products}} n \Delta h_c^\circ \quad (5.20)$$

where n is moles of reactant or product involved in the reaction, and Δh_c° is the standard heat of combustion per mole. The standard heat of reaction is the difference between the heats of combustion of reactants and products.

Example 5.6 Calculation of heat of reaction from heats of combustion

Fumaric acid is produced from malic acid using the enzyme, fumarase. Calculate the standard heat of reaction for the following enzyme transformation:



Solution:

$\Delta h_c^\circ = 0$ for liquid water. From Eq. (5.20):

$$\Delta H_{\text{rxn}}^\circ = (n \Delta h_c^\circ)_{\text{malic acid}} - (n \Delta h_c^\circ)_{\text{fumaric acid}}$$

Table B.8 lists the standard heats of combustion for these compounds:

$$(\Delta h_c^\circ)_{\text{malic acid}} = -1328.8 \text{ kJ gmol}^{-1}$$

$$(\Delta h_c^\circ)_{\text{fumaric acid}} = -1334.0 \text{ kJ gmol}^{-1}.$$

Therefore, using a basis of 1 gmol malic acid converted:

$$\Delta H_{\text{rxn}}^\circ = (1 \text{ gmol}) (-1328.8 \text{ kJ gmol}^{-1}) - (1 \text{ gmol}) (-1334.0 \text{ kJ gmol}^{-1})$$

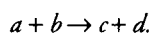
$$\Delta H_{\text{rxn}}^\circ = 5.2 \text{ kJ}.$$

As $\Delta H_{\text{rxn}}^\circ$ is positive, the reaction is endothermic and heat is absorbed.

5.8.2 Heat of Reaction at Non-Standard Conditions

Example 5.6 shows how to calculate the heat of reaction at standard conditions. However, most reactions do not occur at 25°C, and the standard heat of reaction calculated using Eq. (5.20) may not be the same as the actual heat of reaction at the reaction temperature.

Consider the following reaction between compounds A, B, C and D occurring at temperature T :



The standard heat of reaction at 25°C is known from tabulated heat of combustion data. $\Delta H_{\text{rxn}}^\circ$ at temperature T can be calculated using the alternative reaction pathway outlined in

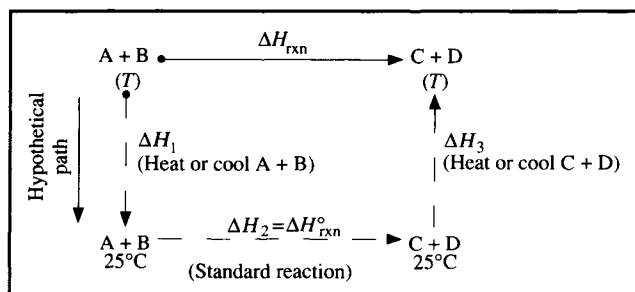
Figure 5.4, in which reaction occurs at 25°C and the reactants and products are heated or cooled between 25°C and T before and after the reaction. Because the initial and final states for the actual and hypothetical paths are the same, the total enthalpy change is also identical. Therefore:

$$\Delta H_{\text{rxn}} (\text{at } T) = \Delta H_1 + \Delta H_{\text{rxn}}^\circ + \Delta H_3 \quad (5.21)$$

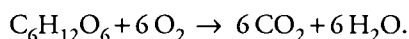
where ΔH_1 and ΔH_3 are changes in sensible heat and $\Delta H_{\text{rxn}}^\circ$ is the standard heat of reaction at 25°C. ΔH_1 and ΔH_3 are evaluated using heat capacities and the methods described in Section 5.4.1.

Depending on how much T deviates from 25°C and the magnitude of $\Delta H_{\text{rxn}}^\circ$, ΔH_{rxn} may not be much different from

Figure 5.4 Hypothetical process path for calculating heat of reaction at non-standard temperature.



$\Delta H_{\text{rxn}}^{\circ}$. For example, consider the reaction for respiration of glucose:



$\Delta H_{\text{rxn}}^{\circ}$ for this conversion is -2805.0 kJ; if the reaction occurs at 37°C , ΔH_{rxn} is -2801.7 kJ. Contributions from sensible heat amount to only 3.3 kJ, which is insignificant compared with the total magnitude of $\Delta H_{\text{rxn}}^{\circ}$ and can be ignored without much loss of accuracy. With reference to Figure 5.4, $\Delta H_1 = -4.8$ kJ for cooling 1 gmol glucose and 6 gmol oxygen from 37°C to 25°C ; $\Delta H_3 = 8.1$ kJ for heating the products back to 37°C . Having opposite signs, ΔH_1 and ΔH_3 act to cancel each other. This situation is typical of most reactions in bioprocessing where the actual temperature of reaction is not sufficiently different from 25°C to warrant concern about sensible heat changes. When heat of reaction is substantial compared with other types of enthalpy change, ΔH_{rxn} can be assumed equal to $\Delta H_{\text{rxn}}^{\circ}$ irrespective of reaction temperature.

A major exception to this general rule are single-enzyme conversions. Because most single-enzyme reactions involve only small molecular rearrangements, heats of reaction are relatively small. For instance, per mole of substrate, the fumase reaction of Example 5.6 involves a standard enthalpy change of only 5.2 kJ; other examples are 8.7 kJ gmol $^{-1}$ for the glucose isomerase reaction, -26.2 kJ gmol $^{-1}$ for hydrolysis of sucrose, and -29.4 kJ per gmol glucose for hydrolysis of starch. For conversions such as these, sensible energy changes of 5 to 10 kJ are clearly significant and should not be ignored. Calculated enthalpy changes for enzyme reactions are often imprecise; being the difference between two or more large numbers, the small $\Delta H_{\text{rxn}}^{\circ}$ for these conversions can carry considerable uncertainty depending on the accuracy of the heats of combustion data. When coupled with usual assumptions such as constant C_p and Δh_m within the temperature

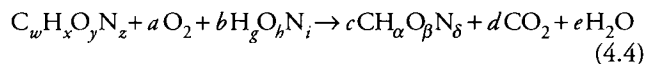
range of interest, this uncertainty means that estimates of heating and cooling requirements for enzyme reactors are sometimes quite rough.

5.9 Heat of Reaction For Processes With Biomass Production

Biochemical reactions in cells do not occur in isolation but are linked in a complex array of metabolic transformations. Catabolic and anabolic reactions take place at the same time, so that energy released in one reaction is used in other energy-requiring processes. Cells use chemical energy quite efficiently; however some is inevitably released as heat. How can we estimate the heat of reaction associated with cell metabolism and growth?

5.9.1 Thermodynamics of Microbial Growth

As described in Section 4.6.1, a macroscopic view of cell growth is represented by the equation:



where a , b , c , d and e are stoichiometric coefficients, $\text{C}_w\text{H}_x\text{O}_y\text{N}_z$ is the substrate, $\text{H}_g\text{O}_h\text{N}_i$ is the nitrogen source, and $\text{CH}_\alpha\text{O}_\beta\text{N}_\delta$ is dry biomass. Once the stoichiometric coefficients or yields are determined, Eq. (4.4) can be used as the reaction equation in energy-balance calculations. We need, however, to determine the heat of reaction for this conversion.

Heats of reaction for cell growth can be estimated using microbial stoichiometry and the concept of available electrons (Section 4.6.2). It has been found empirically that the energy content of organic compounds is related to degree of reduction as follows:

$$\Delta h_c^{\circ} = -q\gamma x_C \quad (5.22)$$

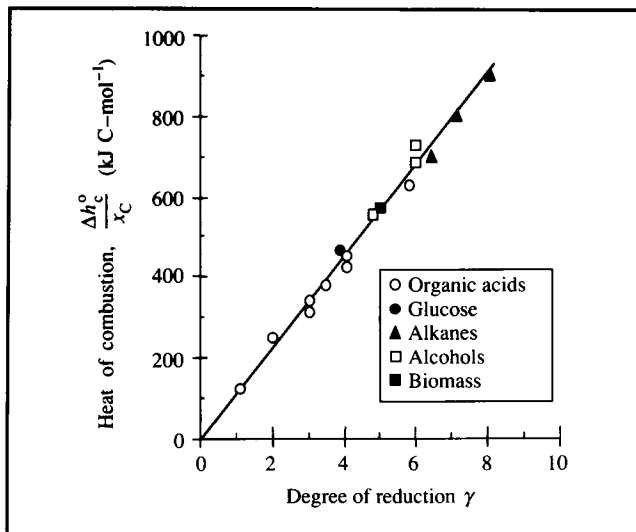
where Δh_c° is the molar heat of combustion at standard conditions, q is the heat evolved per mole of available electrons transferred to oxygen during combustion, γ is the degree of reduction of the compound defined with respect to N_2 , and x_C is the number of carbon atoms in the molecular formula. The coefficient q relating Δh_c° and γ is relatively constant for a large number of compounds. Patel and Erickson [7] assigned a value of 111 kJ gmol $^{-1}$ to q ; in another analysis, Roels [8] determined a value of 115 kJ gmol $^{-1}$. The correlation found

by Roels is based on analysis of several chemical and biochemical compounds including biomass; the results are shown in Figure 5.5.

5.9.2 Heat of Reaction With Oxygen as Electron Acceptor

The direct proportionality between heat of combustion and degree of reduction as shown in Figure 5.5 has important implications for determining the heat of reaction in aerobic culture. Degree of reduction is related directly to the amount of oxygen required for complete combustion of a substance; therefore, heat produced in reaction of compounds for which Figure 5.5 applies must be directly proportional to oxygen consumption. In aerobic cultures with oxygen the principal acceptor of electrons, because molecular oxygen O_2 accepts four electrons, if one mole O_2 is consumed during respiration, four moles of electrons must be transferred. Accepting the value of 115 kJ energy released per gmol electrons transferred, the amount of energy released from consumption of one gmol O_2 is therefore (4×115) kJ, or 460 kJ. This result, that the heat of reaction for aerobic metabolism is approximately -460 kJ per gmol O_2 consumed, is verified by the experimental data of Cooney, *et al.* [9] shown in Figure 3.2 (p. 33). The value is quite accurate for a wide range of conditions,

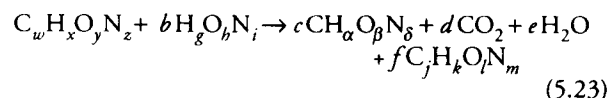
Figure 5.5 Relationship between degree of reduction and heat of combustion for various organic compounds. (From J.A. Roels, 1987, *Thermodynamics of growth*. In: J. Bu'Lock and B. Kristiansen, Eds, *Basic Biotechnology*, Academic Press, London.)



including fermentations involving product formation. Thus, once the amount of oxygen taken up during aerobic cell culture is known, the heat of reaction can be evaluated immediately.

5.9.3 Heat of Reaction With Oxygen Not the Principal Electron Acceptor

If a fermentation uses electron acceptors other than oxygen, for example in anaerobic culture, the simple relationship for heat of reaction derived in Section 5.9.2 does not apply. Heats of combustion must be used to estimate the heat of reaction for anaerobic conversions. Consider the following reaction equation for anaerobic growth with product formation:

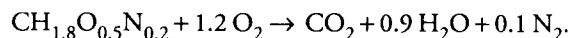


where $C_j H_k O_l N_m$ is an extracellular product and f is its stoichiometric coefficient. With ammonia as nitrogen source and heats of combustion of H_2O and CO_2 zero, from Eq. (5.20) the equation for standard heat of reaction is:

$$\Delta H_{rxn}^0 = (n \Delta h_c^0)_{\text{substrate}} + (n \Delta h_c^0)_{NH_3} - (n \Delta h_c^0)_{\text{biomass}} - (n \Delta h_c^0)_{\text{product}} \quad (5.24)$$

where n is number of moles and Δh_c^0 is the standard molar heat of combustion. Heats of combustion for substrate, NH_3 and product are available from tables; what is the heat of combustion of biomass?

As shown in Table 4.3 (p. 76), the elemental composition of biomass does not vary a great deal. If we assume an average biomass molecular formula of $CH_{1.8}O_{0.5}N_{0.2}$, the reaction equation for combustion of cells to CO_2 , H_2O and N_2 is:



From Table B.2, the degree of reduction of the biomass relative to N_2 is 4.80. Assuming an average of 5% ash associated with the biomass, the cell molecular weight is 25.9. Heat of combustion is obtained by applying Eq. (5.22):

$$\begin{aligned} (\Delta h_c^0)_{\text{biomass}} &= (-115 \text{ kJ gmol}^{-1}) (4.80) (1) \cdot \left| \frac{1 \text{ gmol}}{25.9 \text{ g}} \right| \\ &= -21.3 \text{ kJ g}^{-1}. \end{aligned} \quad (5.25)$$

Actual heats of combustion measured by Cordier *et al.* [10] for a range of microorganisms and culture conditions are listed in Table 5.1. The difference in Δh_c values for bacteria and yeast reflects their different elemental compositions. When the composition of a particular organism is unknown, the heat of combustion for bacteria can be taken as -23.2 kJ g^{-1} ; for yeast Δh_c is approximately -21.2 kJ g^{-1} . These experimentally determined values compare well with that calculated in Eq. (5.25). Once the heat of combustion of biomass is known, it can be used with the heats of combustion for the other products and substrates to determine the heat of reaction.

5.10 Energy-Balance Equation For Cell Culture

In fermentations, the heat of reaction so dominates the energy balance that small enthalpy effects due to sensible heat change and heats of mixing can generally be ignored. In this section we incorporate these observations into a simplified energy-balance equation for cell processes.

Consider Eq. (5.9) applied to a continuous fermenter. What are the major factors responsible for the enthalpy difference between input and output streams in fermentations? Because cell-culture media are usually dilute aqueous solutions with behaviour close to ideal, even though the composition of the broth may change as substrates are consumed and products formed, changes in heats of mixing of these solutes are generally negligible. Similarly, even though there may be a temperature difference between input and output streams, the overall change in enthalpy due to sensible heat is also small. Usually, heat of reaction, latent heat of phase change and shaft work are the only energy effects worth considering in fermentation energy balances. Evaporation is the most likely phase change in fermenter operation; if evaporation is controlled then latent heat effects can also be ignored. Per cubic metre of fermentation broth, metabolic reactions typically generate $5-20 \text{ kJ}$ heat per second for growth on carbohydrate, and up to 60 kJ s^{-1} for growth on hydrocarbon substrates. By way of comparison, in aerobic cultures sparged with dry air, evaporation of the fermentation broth removes only about

Table 5.1 Heats of combustion for bacteria and yeast

(From J.-L. Cordier, B.M. Butsch, B. Birou and U. von Stockar, 1987, *The relationship between elemental composition and heat of combustion of microbial biomass*. Appl. Microbiol. Biotechnol. 25, 305–312)

Organism	Substrate	$\Delta h_c \text{ (kJ g}^{-1}\text{)}$
Bacteria		
<i>Escherichia coli</i>	glucose	-23.04 ± 0.06
	glycerol	-22.83 ± 0.07
<i>Enterobacter cloacae</i>	glucose	-23.22 ± 0.14
	glycerol	-23.39 ± 0.12
<i>Methylophilus methylotrophus</i>	methanol	-23.82 ± 0.06
<i>Bacillus thuringiensis</i>	glucose	-22.08 ± 0.03
Yeast		
<i>Candida lipolytica</i>	glucose	-21.34 ± 0.16
<i>Candida boidinii</i>	glucose	-20.14 ± 0.18
	ethanol	-20.40 ± 0.14
	methanol	-21.52 ± 0.09
<i>Kluyveromyces fragilis</i>	lactose	-21.54 ± 0.07
	galactose	-21.78 ± 0.10
	glucose	-21.66 ± 0.19
	glucose*	-21.07 ± 0.07
		-21.30 ± 0.10
		-20.66 ± 0.26
		-21.22 ± 0.14

* Chemostat rather than batch culture: dilution rates were 0.036 h^{-1} , 0.061 h^{-1} , 0.158 h^{-1} and 0.227 h^{-1} , respectively.

$0.5 \text{ kJ s}^{-1} \text{ m}^{-3}$ as latent heat. Energy input due to shaft work varies between 0.5 and $5 \text{ kJ s}^{-1} \text{ m}^{-3}$ in large-scale vessels and $10\text{--}20 \text{ kJ s}^{-1} \text{ m}^{-3}$ in small vessels. Sensible heats and heats of mixing are generally several orders of magnitude smaller.

For cell processes, therefore, we can simplify energy-balance calculations by substituting expressions for heat of reaction and latent heat for the first two terms of Eq. (5.9). By the definition of Eq. (5.18), ΔH_{rxn} is the difference between product and reactant enthalpies. As the products are contained in the output flow and the reactants in the input, ΔH_{rxn} is approximately equal to the difference in enthalpy between input and output streams. If evaporation is also significant, the enthalpy of vapour leaving the system will be greater than that of liquid entering by $M_v \Delta h_v$, where M_v is the mass of liquid evaporated and Δh_v is the latent heat of vaporisation. The energy-balance equation can be modified as follows:

$$-\Delta H_{\text{rxn}} - M_v \Delta h_v - Q + W_s = 0. \quad (5.26)$$

ΔH_{rxn} has a negative sign in Eq. (5.26) because ΔH_{rxn} is equal to [enthalpy of products – enthalpy of reactants] whereas the energy-balance equation refers to [enthalpy of inlet streams – enthalpy of outlet streams]. Eq. (5.26) applies even if some proportion of the reactants remains unconverted or if there are

components in the system which do not react. At steady state, any material added to the system that does not participate in reaction must leave in the output stream; ignoring enthalpy effects due to change in temperature or solution and unless the material volatilises, the enthalpy of unreacted material in the output stream must be equal to its inlet enthalpy. As sensible heat effects are considered negligible, the difference between $\Delta H_{\text{rxn}}^\circ$ and ΔH_{rxn} at the reaction temperature can be ignored.

It must be emphasised that Eq. (5.26) is greatly simplified and, as discussed in Section 5.8.2, may not be applicable to single-enzyme conversions. It is, however, a very useful equation for fermentation processes.

5.11 Fermentation Energy-Balance Worked Examples

For processes involving cell growth and metabolism the enthalpy change accompanying reaction is relatively large. Energy balances for aerobic and anaerobic cultures can therefore be carried out using the modified energy-balance equation (5.26). Because this equation contains no enthalpy terms, it is not necessary to define reference states. Application of Eq. (5.26) to anaerobic fermentation is illustrated in Example 5.7.

Example 5.7 Continuous ethanol fermentation

Saccharomyces cerevisiae is grown anaerobically in continuous culture at 30°C . Glucose is used as carbon source; ammonia is the nitrogen source. A mixture of glycerol and ethanol is produced. At steady state, mass flows to and from the reactor at steady state are as follows:

glucose in	36.0 kg h^{-1}
NH_3 in	0.40 kg h^{-1}
cells out	2.81 kg h^{-1}
glycerol out	7.94 kg h^{-1}
ethanol out	11.9 kg h^{-1}
CO_2 out	13.6 kg h^{-1}
H_2O out	0.15 kg h^{-1}

Estimate the cooling requirements.

Solution:

1. *Assemble*

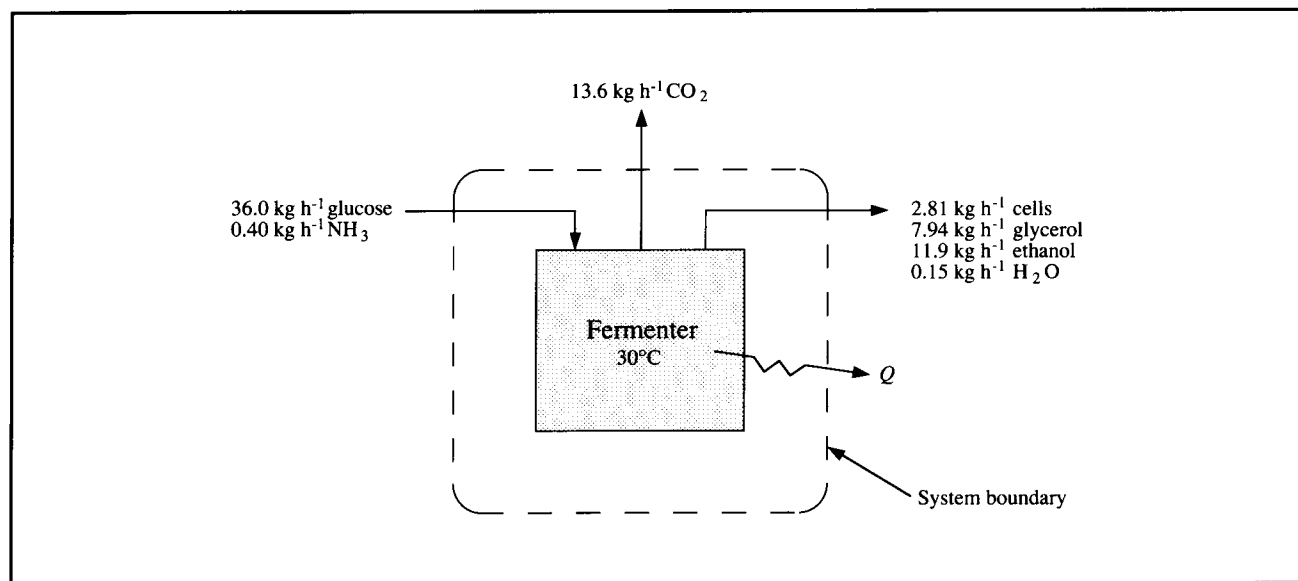
(i) *Units.*

kg, kJ, h, $^\circ\text{C}$.

(ii) *Flowsheet.*

The flowsheet for this process is shown in Figure 5E7.1.

Figure 5E7.1 Flowsheet for anaerobic yeast fermentation.



(iii) *System boundary.*

The system boundary is shown in Figure 5E7.1.

2. *Analyse*

(i) *Assumptions.*

- steady state
- no leaks
- system is homogeneous
- heat of combustion for yeast is -21.2 kJ g^{-1}
- ideal solutions
- negligible sensible heat change
- no shaft work
- no evaporation

(ii) *Basis.*

36.0 kg glucose, or 1 hour.

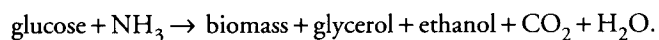
(iii) *Extra data.*

MW glucose	= 180
MW NH_3	= 17
MW glycerol	= 92
MW ethanol	= 46

Heats of combustion:

$$\begin{aligned}
 (\Delta h_c^\circ)_{\text{glucose}} &= -2805.0 \text{ kJ gmol}^{-1} \text{ (Table B.8)} \\
 (\Delta h_c^\circ)_{\text{NH}_3} &= -382.6 \text{ kJ gmol}^{-1} \text{ (Table B.8)} \\
 (\Delta h_c^\circ)_{\text{glycerol}} &= -1655.4 \text{ kJ gmol}^{-1} \text{ (Table B.8)} \\
 (\Delta h_c^\circ)_{\text{ethanol}} &= -1366.8 \text{ kJ gmol}^{-1} \text{ (Table B.8)}.
 \end{aligned}$$

(iv) *Reaction.*



All components are involved in reaction.

(v) *Mass-balance equation.*

The mass balance is already complete.

(vi) *Energy-balance equation.*

For cell metabolism, the modified energy-balance equation is Eq. (5.26):

$$-\Delta H_{\text{rxn}} - M_v \Delta h_v - Q + W_s = 0.$$

3. *Calculate*

(i) *Identify terms in the energy-balance equation.*

$W_s = 0$; $M_v = 0$. Therefore:

$$-\Delta H_{\text{rxn}} - Q = 0.$$

Evaluate the heat of reaction using Eq. (5.20). As the heat of combustion of H_2O and CO_2 is zero, the heat of reaction is:

$$\Delta H_{\text{rxn}} = (n \Delta h_c^\circ)_G + (n \Delta h_c^\circ)_A - (n \Delta h_c^\circ)_B - (n \Delta h_c^\circ)_{\text{Gly}} - (n \Delta h_c^\circ)_E$$

where G = glucose, A = ammonia, B = cells, Gly = glycerol and E = ethanol. Converting to a mass basis:

$$\Delta H_{\text{rxn}} = (M \Delta h_c^\circ)_G + (M \Delta h_c^\circ)_A - (M \Delta h_c^\circ)_B - (M \Delta h_c^\circ)_{\text{Gly}} - (M \Delta h_c^\circ)_E$$

where Δh_c° is expressed per unit mass. Converting the Δh_c° data to kJ kg^{-1} :

$$(\Delta h_c^\circ)_G = -2805.0 \frac{\text{kJ}}{\text{gmol}} \cdot \left| \frac{1 \text{ gmol}}{180 \text{ g}} \right| \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| = -1.558 \times 10^4 \text{ kJ kg}^{-1}$$

$$(\Delta h_c^\circ)_A = -382.6 \frac{\text{kJ}}{\text{gmol}} \cdot \left| \frac{1 \text{ gmol}}{17 \text{ g}} \right| \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| = -2.251 \times 10^4 \text{ kJ kg}^{-1}$$

$$(\Delta h_c^\circ)_B = -21.2 \frac{\text{kJ}}{\text{g}} \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| = -2.120 \times 10^4 \text{ kJ kg}^{-1}$$

$$(\Delta h_c^\circ)_{\text{Gly}} = -1655.4 \frac{\text{kJ}}{\text{gmol}} \cdot \left| \frac{1 \text{ gmol}}{92 \text{ g}} \right| \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| = -1.799 \times 10^4 \text{ kJ kg}^{-1}$$

$$(\Delta h_c^\circ)_E = -1366.8 \frac{\text{kJ}}{\text{gmol}} \cdot \left| \frac{1 \text{ gmol}}{46 \text{ g}} \right| \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| = -2.971 \times 10^4 \text{ kJ kg}^{-1}$$

Therefore:

$$\begin{aligned} \Delta H_{\text{rxn}} &= (36.0 \text{ kg}) (-1.558 \times 10^4 \text{ kJ kg}^{-1}) + (0.4 \text{ kg}) (-2.251 \times 10^4 \text{ kJ kg}^{-1}) - (2.81 \text{ kg}) (-2.120 \times 10^4 \text{ kJ kg}^{-1}) \\ &\quad - (7.94 \text{ kg}) (-1.799 \times 10^4 \text{ kJ kg}^{-1}) - (11.9 \text{ kg}) (-2.971 \times 10^4 \text{ kJ kg}^{-1}) \\ &= -1.392 \times 10^4 \text{ kJ}. \end{aligned}$$

Substituting this result into the energy-balance equation:

$$Q = 1.392 \times 10^4 \text{ kJ}.$$

Q is positive indicating that heat must be removed from the system.

4. Finalise

1.4×10^4 kJ heat must be removed from the fermenter per hour.

In Example 5.7, the water used as solvent for components of the nutrient medium was ignored. This water was effectively a tie component, moving through the system unchanged and not contributing to the energy balance. Cooling requirements could be determined directly from the heat of reaction.

For aerobic cultures, we can relate the heat of reaction directly to oxygen consumption, providing a short-cut

method for determining ΔH_{rxn} . Heats of combustion are not required in these calculations. Also, as long as the amount of oxygen consumed is known, the mass balance for the problem need not be completed. The procedure for energy-balance problems involving aerobic fermentation is illustrated in Example 5.8.

Example 5.8 Citric acid production

Citric acid is manufactured using submerged culture of *Aspergillus niger* in a batch reactor operated at 30°C. Over a period of two days, 2500 kg glucose and 860 kg oxygen are consumed to produce 1500 kg citric acid, 500 kg biomass and other products. Ammonia is used as nitrogen source. Power input to the system by mechanical agitation of the broth is about 15 kW; approximately 100 kg water is evaporated over the culture period. Estimate the cooling requirements.

Solution:

1. Assemble

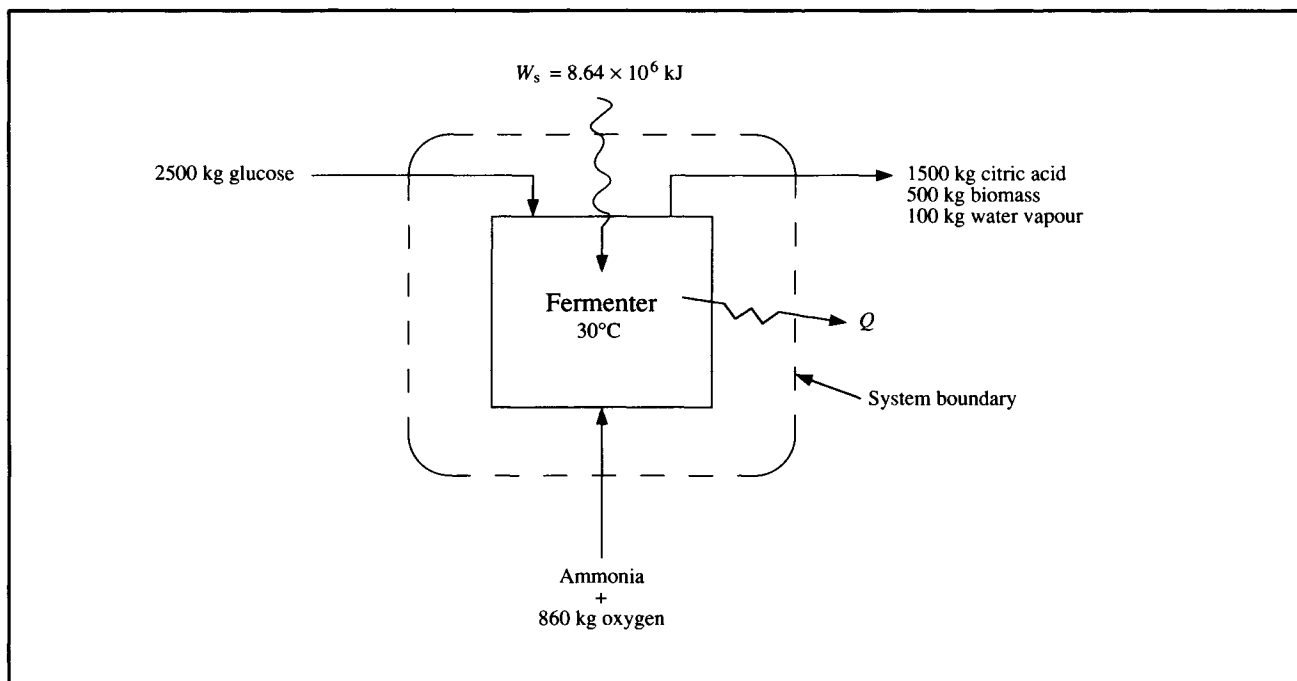
(i) *Units.*

kg, kJ, h, °C.

(ii) *Flowsheet.*

The flowsheet is shown in Figure 5E8.1.

Figure 5E8.1 Flowsheet for microbial production of citric acid.



(iii) *System boundary.*

The system boundary is shown in Figure 5E8.1.

2. *Analyse*

(i) *Assumptions.*

- system is homogeneous
- no leaks
- ideal solutions
- negligible sensible heat
- heat of reaction at 30°C is $-460 \text{ kJ gmol}^{-1} \text{ O}_2$ consumed

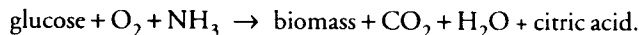
(ii) *Basis.*

1500 kg citric acid produced, or 2 days.

(iii) *Extra data.*

Δh_v water at 30°C = $2430.7 \text{ kJ kg}^{-1}$ (Table C.1)

(iv) *Reaction.*



All components are involved in reaction.

(v) *Mass balance.*

The mass balance need not be completed as the sensible energy associated with inlet and outlet streams is negligible.

(vi) *Energy balance.*

The aim of the integral energy balance for batch culture is to calculate the amount of heat which must be removed to produce zero accumulation of energy in the system. Eq. (5.26) is appropriate:

$$-\Delta H_{\text{rxn}} - M_v \Delta h_v - Q + W_s = 0$$

where each term refers to the two-day culture period.

3. *Calculate*

(i) *Identify terms in the energy-balance equation.*

ΔH_{rxn} is related to the amount of oxygen consumed:

$$\begin{aligned} \Delta H_{\text{rxn}} &= (-460 \text{ kJ gmol}^{-1}) (860 \text{ kg}) \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| \cdot \left| \frac{1 \text{ gmol}}{32 \text{ g}} \right| \\ &= -1.24 \times 10^7 \text{ kJ}. \end{aligned}$$

Heat lost through evaporation is:

$$M_v \Delta h_v = (100 \text{ kg}) (2430.7 \text{ kJ kg}^{-1}) = 2.43 \times 10^5 \text{ kJ}.$$

Power input by mechanical agitation is 15 kW or 15 kJ s^{-1} . Over a period of 2 days:

$$W_s = (15 \text{ kJ s}^{-1}) (2 \text{ days}) \cdot \left| \frac{3600 \text{ s}}{1 \text{ h}} \right| \cdot \left| \frac{24 \text{ h}}{1 \text{ day}} \right| = 2.59 \times 10^6 \text{ kJ}.$$

These results can now be substituted into the energy-balance equation.

$$-(-1.24 \times 10^7 \text{ kJ}) - (2.43 \times 10^5 \text{ kJ}) - Q + (2.59 \times 10^6 \text{ kJ}) = 0.$$

$$Q = 1.47 \times 10^7 \text{ kJ}.$$

Q is positive, indicating that heat must be removed from the system. Note the relative magnitudes of the energy contributions from heat of reaction, shaft work and evaporation; the effects of evaporation can often be ignored.

4. Finalise

$1.5 \times 10^7 \text{ kJ}$ heat must be removed from the fermenter per 1500 kg citric acid produced.

5.12 Summary of Chapter 5

At the end of Chapter 5 you should:

- (i) know which forms of energy are common in bioprocesses;
- (ii) know the *general energy balance* in words and as a mathematical equation, and the simplifications that can be made for bioprocesses;
- (iii) be familiar with *heat capacity* tables and be able to calculate *sensible heat changes*;
- (iv) be able to calculate *latent heat changes*;
- (v) understand *heats of mixing* for non-ideal solutions;
- (vi) be able to use *steam tables*;
- (vii) be able to determine *standard heats of reaction* from *heats of combustion*;
- (viii) know how to determine heats of reaction for aerobic and anaerobic cell cultures; and
- (ix) be able to carry out energy-balance calculations for biological systems with and without reaction.

Problems

5.1 Sensible energy change

Calculate the enthalpy change associated with the following processes:

- (a) *m*-cresol is heated from 25°C to 100°C;
- (b) ethylene glycol is cooled from 20°C to 10°C;
- (c) succinic acid is heated from 15°C to 120°C; and
- (d) air is cooled from 150°C to 65°C.

5.2 Heat of vaporisation

Nitrogen is sometimes bubbled into fermenters to maintain anaerobic conditions. It does not react, and leaves in the fermenter off-gas. However it can strip water from the fermenter, so that water vapour also leaves in the off-gas. In a continuous fermenter operated at 33°C, 20 g h⁻¹ water is evaporated in this way. How much heat must be put into the system to compensate for evaporative cooling?

5.3 Steam tables

Use the steam tables to find:

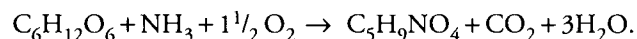
- (a) the heat of vaporisation of water at 85°C;
- (b) the enthalpy of liquid water at 35°C relative to $H = 0$ at 10°C;
- (c) the enthalpy of saturated water vapour at 40°C relative to $H = 0$ at the triple point; and
- (d) the enthalpy of superheated steam at 2.5 atm absolute pressure and 275°C relative to $H = 0$ at the triple point.

5.4 Pre-heating nutrient medium

Steam is used to heat nutrient medium in a continuous-flow process. Saturated steam at 150°C enters a coil on the outside of the heating vessel and is completely condensed. Medium enters the vessel at 15°C and leaves at 44°C. Heat losses from the jacket to the surroundings are estimated as 0.22 kW. If the flow rate of medium is 3250 kg h⁻¹ and the heat capacity is $C_p = 0.9 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$, how much steam is required?

5.5 Production of glutamic acid

Immobilised cells of a genetically-improved strain of *Brevibacterium lactofermentum* are used to convert glucose to glutamic acid for production of MSG (monosodium glutamate). The immobilised cells are unable to grow, but metabolise glucose according to the equation:



A feed stream of 4% glucose in water enters a 25 000-litre reactor at 25°C at a flow rate of 2000 kg h⁻¹. A gaseous mixture of 12% NH₃ in air is sparged into the reactor at 1 atm and 15°C at a flow rate of 4 vvm (1 vvm means 1 vessel volume per minute). The product stream from the reactor contains residual sugar at a concentration of 0.5%.

- (a) Estimate the cooling requirements.
- (b) How important is cooling in this fermentation? For

example, assuming the reaction rate remains constant irrespective of temperature, if cooling were not provided and the reactor operated adiabatically, what would be the temperature? (In fact, the rate of conversion will decline rapidly at high temperatures due to cell death and enzyme deactivation.)

5.6 Bacterial production of alginate

Azotobacter vinelandii is investigated for production of alginate from sucrose. In a continuous fermenter at 28°C with ammonia as nitrogen source, the yield of alginate was found to be 4 g g⁻¹ oxygen consumed. It is planned to produce alginate at a rate of 5 kg h⁻¹. Since the viscosity of alginate in aqueous solution is considerable, energy input due to mixing the broth cannot be neglected. The fermenter is equipped with a flat-bladed disc turbine; at a satisfactory mixing speed and air flow-rate, power requirements are estimated at 1.5 kW. Estimate the cooling requirements.

5.7 Acid fermentation

Propionibacterium species are tested for commercial-scale production of propionic acid. Propionic and other acids are synthesised in anaerobic culture using sucrose as substrate and ammonia as nitrogen source. Overall yields from sucrose are as follows:

propionic acid	40% (w/w)
acetic acid	20% (w/w)
butyric acid	5% (w/w)
lactic acid	3.4% (w/w)
biomass	12% (w/w)

Bacteria are inoculated into a vessel containing sucrose and ammonia; a total of 30 kg sucrose is consumed over a period of 10 days. What are the cooling requirements?

5.8 Ethanol fermentation

A crude fermenter is set up in a shed in the backyard of a suburban house. Under anaerobic conditions with ammonia as nitrogen source, about 0.45 g ethanol are formed per g glucose consumed. At steady state, the production rate of ethanol averages 0.4 kg h⁻¹.

The owner of this enterprise decides to reduce her electricity bill by using the heat released during the fermentation to warm water as an adjunct to the household hot-water system. 2.5 litres h⁻¹ cold water at 10°C is fed into a jacket surrounding the fermenter. To what temperature is the water heated?

Heat losses from the system are negligible. Use a biomass composition of CH_{1.75}O_{0.58}N_{0.18} plus 8% ash.

5.9 Production of bakers' yeast

Bakers' yeast is produced in a 50 000-litre fermenter under aerobic conditions. The carbon substrate is sucrose; ammonia is provided as nitrogen source. The average biomass composition is CH_{1.83}O_{0.55}N_{0.17} with 5% ash. Under efficient growth conditions, biomass is the only major product; the biomass yield from sucrose is 0.5 g g⁻¹. If the specific growth-rate is 0.45 h⁻¹, estimate the rate of heat removal required to maintain constant temperature in the fermenter when the yeast concentration is 10 g l⁻¹.

References

1. Perry, R.H., D.W. Green and J.O. Maloney (Eds) (1984) *Chemical Engineers' Handbook*, 6th edn, McGraw-Hill, New York.
2. *Handbook of Chemistry and Physics*, CRC Press, Boca Raton.
3. *International Critical Tables* (1926) McGraw-Hill, New York.
4. Perry, R.H. and C.H. Chilton (Eds) (1973) *Chemical Engineers' Handbook*, 5th edn, McGraw-Hill, Tokyo.
5. Atkinson, B. and F. Mavituna (1991) *Biochemical Engineering and Biotechnology Handbook*, 2nd edn, Macmillan, Basingstoke.
6. Haywood, R.W. (1972) *Thermodynamic Tables in SI (Metric) Units*, 2nd edn, Cambridge University Press, Cambridge.
7. Patel, S.A. and L.E. Erickson (1981) Estimation of heats of combustion of biomass from elemental analysis using available electron concepts. *Biotechnol. Bioeng.* 23, 2051–2067.
8. Roels, J.A. (1983) *Energetics and Kinetics in Biotechnology*, Chapter 3, Elsevier Biomedical Press, Amsterdam.
9. Cooney, C.L., D.I.C. Wang and R.I. Mateles (1968) Measurement of heat evolution and correlation with oxygen consumption during microbial growth. *Biotechnol. Bioeng.* 11, 269–281.
10. Cordier, J.-L., B.M. Butsch, B. Birou and U. von Stockar (1987) The relationship between elemental composition and heat of combustion of microbial biomass. *Appl. Microbiol. Biotechnol.* 25, 305–312.

Suggestions for Further Reading

Process Energy Balances

- Felder, R.M. and R.W. Rousseau (1978) *Elementary Principles of Chemical Processes*, Chapters 8–10, John Wiley, New York.
- Himmelblau, D.M. (1974) *Basic Principles and Calculations in Chemical Engineering*, 3rd edn, Chapter 4, Prentice-Hall, New Jersey.
- Shaheen, E.I. (1975) *Basic Practice of Chemical Engineering*, Chapter 5, Houghton Mifflin, Boston.
- Whitwell, J.C. and R.K. Toner (1969) *Conservation of Mass and Energy*, Chapters 6–9, Blaisdell, Waltham, Massachusetts.

Metabolic Thermodynamics

- Erickson, L.E. (1980) Biomass elemental composition and energy content. *Biotechnol. Bioeng.* 22, 451–456.
- Marison, I. and U. von Stockar (1987) A calorimetric investigation of the aerobic cultivation of *Kluyveromyces fragilis* on various substrates. *Enzyme Microbiol. Technol.* 9, 33–43.
- Minkevich, I.G. and V.K. Eroshin (1973) Productivity and heat generation of fermentation under oxygen limitation. *Folia Microbiol.* 18, 376–385.
- Nagai, S. (1979) Mass and energy balances for microbial growth kinetics. *Adv. Biochem. Eng.* 11, 49–83.
- Roels, J.A. (1980) Application of macroscopic principles to microbial metabolism. *Biotechnol. Bioeng.* 22, 2457–2514.
- Roels, J.A. (1987) Thermodynamics of growth. In: J. Bu'Lock and B. Kristiansen (Eds), *Basic Biotechnology*, pp. 57–74, Academic Press, London.

Unsteady-State Material and Energy Balances

An unsteady-state or transient process is one which causes system properties to vary with time. Batch and semi-batch systems are inherently transient; continuous systems are unsteady during start-up and shut-down. Changing from one set of process conditions to another also creates an unsteady state, as does any fluctuation in input or control variables.

The principles of mass and energy balances developed in Chapters 4 and 5 can be applied to unsteady-state processes. Balance equations are used to determine the rate of change of system parameters; solution of these equations generally requires application of calculus. Questions such as what is the concentration of product in the reactor as a function of time?, and how long will it take to reach a particular temperature after flow of steam is started? can be answered using unsteady-state mass and energy balances. In this chapter we will consider some simple unsteady-state problems.

6.1 Unsteady-State Material-Balance Equations

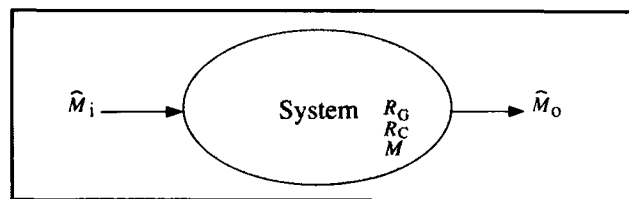
When mass of a system is not constant, we generally need to know how the mass varies as a function of time. To evaluate the *rate of change* of mass in the system, let us first return to the general mass-balance equation introduced in Chapter 4 (see p. 52).

$$\left\{ \begin{array}{l} \text{mass in} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} - \left\{ \begin{array}{l} \text{mass out} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} + \left\{ \begin{array}{l} \text{mass} \\ \text{generated} \\ \text{within} \\ \text{system} \end{array} \right\} - \left\{ \begin{array}{l} \text{mass} \\ \text{consumed} \\ \text{within} \\ \text{system} \end{array} \right\} = \left\{ \begin{array}{l} \text{mass} \\ \text{accumulated} \\ \text{within} \\ \text{system} \end{array} \right\}, \quad (4.1)$$

Consider the flow system of Figure 6.1 in which reactions are taking place. Species A is involved in the process; M is the mass of A in the system. Using the 'hat' symbol $\hat{}$ to denote rate, let $\hat{M}_i$ be the mass flow rate of A entering the system, and $\hat{M}_o$ the mass flow rate of A leaving. R_G is the mass rate of generation of species A by chemical reaction; R_C is the mass rate of consumption by reaction. The dimensions of $\hat{M}_i$, $\hat{M}_o$, R_G and R_C are MT^{-1} , with units such as $g\ s^{-1}$, $kg\ h^{-1}$, $lb\ min^{-1}$, etc.

All of the variables: $\hat{M}_i$, $\hat{M}_o$, R_G and R_C , may vary with time. However, let us focus on an infinitesimally-small

Figure 6.1 Flow system for an unsteady-state mass balance.



interval of time Δt between times t and $t + \Delta t$. Even though the system variables may be changing, if Δt is sufficiently small we can treat the flow rates $\hat{M}$ and rates of reaction R s as if they were constant during this period. Under these circumstances, the terms of the general balance equation (4.1) may be written as follows.

- (i) *Mass in.* During period Δt , the mass of species A transported into the system is $\hat{M}_i \Delta t$. Note that the dimensions of $\hat{M}_i \Delta t$ are M , with units g , kg , lb , etc.
- (ii) *Mass out.* Similarly, the mass of species A transported out during time Δt is $\hat{M}_o \Delta t$.
- (iii) *Generation.* The mass of A generated during Δt is $R_G \Delta t$.
- (iv) *Consumption.* The mass of A consumed during Δt is $R_C \Delta t$.
- (v) *Accumulation.* Let ΔM be the mass of A accumulated in the system during Δt . ΔM may be either positive (accumulation) or negative (depletion).

Entering these terms into the general mass balance Eq. (4.1) with the accumulation term on the left-hand side:

$$\Delta M = \hat{M}_i \Delta t - \hat{M}_o \Delta t + R_G \Delta t - R_C \Delta t. \quad (6.1)$$

We can divide both sides of Eq. (6.1) by Δt to give:

$$\frac{\Delta M}{\Delta t} = \hat{M}_i - \hat{M}_o + R_G - R_C.$$

(6.2)

$$\frac{dM}{dt} = \hat{M}_i - \hat{M}_o + R_G - R_C.$$

(6.5)

Eq. (6.2) applies when Δt is infinitesimally small. If we take the limit as Δt approaches zero, i.e. as t and $t + \Delta t$ become virtually the same, Eq. (6.2) represents the system at an instant rather than over an interval of time. Mathematical techniques for handling this type of situation are embodied in the rules of calculus. In calculus, the *derivative* of y with respect to x , $\frac{dy}{dx}$, is defined as:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

(6.3)

where

$$\lim_{\Delta x \rightarrow 0}$$

represents the limit as Δx approaches zero. As Eq. (6.2) is valid for $\Delta t \rightarrow 0$, we can write:

$$\frac{dM}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta M}{\Delta t} = \hat{M}_i - \hat{M}_o + R_G - R_C.$$

(6.4)....

The derivative $\frac{dM}{dt}$ represents the rate of change of mass with time measured at a particular instant. We have thus derived a differential equation for rate of change of M as a function of system variables $\hat{M}_i$, $\hat{M}_o$, R_G and R_C :

At steady state there can be no change in mass of the system, so the rate of change, $\frac{dM}{dt}$, must be zero. At steady state, therefore, Eq. (6.5) reduces to the familiar steady-state mass-balance equation (see p. 53):

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}.$$

(4.2)

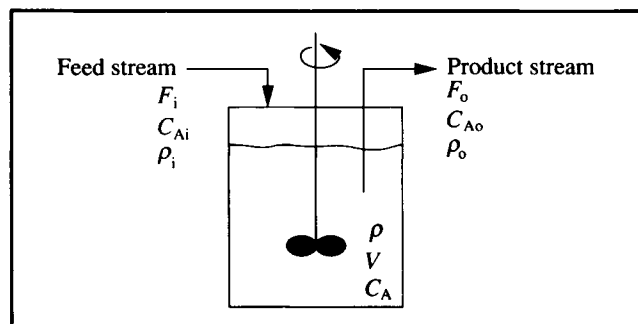
Unsteady-state mass-balance calculations begin with derivation of a differential equation to describe the process. Eq. (6.5) was developed on a mass basis and contains parameters such as mass flow rate $\hat{M}$ and mass rate of reaction R . Another common form of the unsteady-state mass balance is based on volume. The reason for this variation is that reaction rates are usually expressed on a per-volume basis. For example, the rate of a first-order reaction is expressed in terms of the concentration of reactant:

$$r_C = k_1 C_A$$

where r_C is the *volumetric rate of consumption of A by reaction* with units, e.g. $\text{g cm}^{-3} \text{s}^{-1}$, k_1 is the first-order reaction-rate constant, and C_A is the concentration of reactant A. This and other reaction-rate equations are described in more detail in Chapter 11. When rate expressions are used in mass- and energy-balance problems, the relationship between mass and volume must enter the analysis. This is illustrated in Example 6.1.

Example 6.1 Unsteady-state material balance for a CSTR

A continuous stirred-tank reactor is operated as shown in Figure 6E1.1. The volume of liquid in the tank is V . Feed enters with volumetric flow rate F_i ; product leaves with flow rate F_o . The concentration of reactant A in the feed is C_{Ai} ; the concentration of A in the exit stream is C_{Ao} . The density of the feed stream is ρ_i ; the density of the product stream is ρ_o . The tank is well mixed. The concentration of A in the tank is C_A and the density of liquid in the tank is ρ . In the reactor, compound A undergoes reaction and is transformed into compound B. The volumetric rate of consumption of A by reaction is given by the expression $r_C = k_1 C_A$.

Figure 6E1.1 Continuous stirred-tank reactor.

Using unsteady-state balances, derive differential equations for:

- (a) total mass; and
- (b) mass of component A.

Solution:

The general unsteady-state mass-balance equation is Eq. (6.5):

$$\frac{dM}{dt} = \hat{M}_i - \hat{M}_o + R_G - R_C.$$

- (a) For the balance on total mass R_G and R_C are zero; total mass cannot be generated or consumed by chemical reaction. From the definition of density (Section 2.4.1), total mass can be expressed as the product of volume and density. Similarly, mass flow rate can be expressed as the product of volumetric flow rate and density.

$$\begin{aligned} \text{Total mass in the tank: } M &= \rho V; \text{ therefore } \frac{dM}{dt} = \frac{d(\rho V)}{dt} \\ \text{Mass flow rate in: } \hat{M}_i &= F_i \rho_i \\ \text{Mass flow rate out: } \hat{M}_o &= F_o \rho_o. \end{aligned}$$

Substituting these terms into Eq. (6.5):

$$\frac{d(\rho V)}{dt} = F_i \rho_i - F_o \rho_o.$$

(6.6)

Eq. (6.6) is a differential equation representing the unsteady-state total-mass balance.

- (b) A is not generated in the reaction; therefore $R_G = 0$. The other terms of Eq. (6.5) can be expressed as follows.

$$\begin{aligned} \text{Mass of A in the tank: } M &= V C_A; \text{ therefore } \frac{dM}{dt} = \frac{d(V C_A)}{dt} \\ \text{Mass flow rate of A in: } \hat{M}_i &= F_i C_{Ai} \\ \text{Mass flow rate of A out: } \hat{M}_o &= F_o C_{Ao} \\ \text{Rate of consumption of A: } R_C &= k_1 C_A V. \end{aligned}$$

Substituting into Eq. (6.5):

$$\frac{d(VC_A)}{dt} = F_i C_{Ai} - F_o C_{Ao} - k_1 C_A V \quad (6.7)$$

Eq. (6.7) is the differential equation representing the unsteady-state mass balance on A.

6.2 Unsteady-State Energy-Balance Equations

In Chapter 5, the law of conservation of energy was represented by the equation (see p. 87):

$$\left\{ \begin{array}{l} \text{energy in through} \\ \text{system boundaries} \end{array} \right\} - \left\{ \begin{array}{l} \text{energy out through} \\ \text{system boundaries} \end{array} \right\} = \left\{ \begin{array}{l} \text{energy accumulated} \\ \text{within the system} \end{array} \right\}. \quad (5.4)$$

Consider the system shown in Figure 6.2. E is the total energy in the system, $\hat{W}_s$ is the rate at which shaft work is done on the system, $\hat{Q}$ is the rate of heat removal from the system, and $\hat{M}_i$ and $\hat{M}_o$ are mass flow rates to and from the system. All these parameters may vary with time.

Ignoring kinetic and potential energies as discussed in Section 5.2.1, the energy-balance equation can be applied over an infinitesimally-small interval of time Δt , during which we can treat $\hat{W}_s$, $\hat{Q}$, $\hat{M}_i$ and $\hat{M}_o$ as if they were constant. Under these conditions, the terms of the general energy-balance equation are:

- (i) *Input.* During time interval Δt , the amount of energy entering the system is $\hat{M}_i h_i \Delta t + \hat{W}_s \Delta t$, where h_i is the specific enthalpy of the incoming flow stream.
- (ii) *Output.* Similarly, the amount of energy leaving the system is $\hat{M}_o h_o \Delta t + \hat{Q} \Delta t$.
- (iii) *Accumulation.* Let ΔE be the energy accumulated in the system during time Δt . ΔE may be either positive (accumulation) or negative (depletion). In the absence of kinetic and potential energies, E represents the enthalpy of the system.

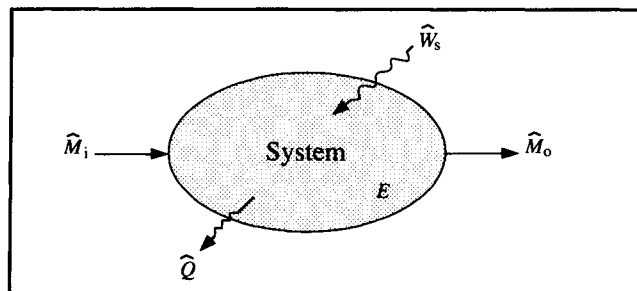
Entering these terms into Eq. (5.4) with the accumulation term first:

$$\Delta E = \hat{M}_i h_i \Delta t - \hat{M}_o h_o \Delta t - \hat{Q} \Delta t + \hat{W}_s \Delta t. \quad (6.8)$$

We can divide both sides of Eq. (6.8) by Δt :

$$\frac{\Delta E}{\Delta t} = \hat{M}_i h_i - \hat{M}_o h_o - \hat{Q} + \hat{W}_s. \quad (6.9)$$

Figure 6.2 Flow system for an unsteady-state energy balance.



Eq. (6.9) is valid for small Δt . The equation for rate of change of energy at a particular instant in time is determined by taking the limit of Eq. (6.9) as Δt approaches zero:

$$\frac{dE}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta E}{\Delta t} = \hat{M}_i h_i - \hat{M}_o h_o - \hat{Q} + \hat{W}_s. \quad (6.10)$$

Eq. (6.10) is the unsteady-state energy-balance equation for a system with only one inlet and one outlet stream. If there are several such streams, all mass flow rates and enthalpies must be added together:

$$\frac{dE}{dt} = \sum_{\text{input streams}} (\hat{M}h) - \sum_{\text{output streams}} (\hat{M}h) - \hat{Q} + \hat{W}_s. \quad (6.11)$$

Eq. (6.11) can be simplified for fermentation processes using the same arguments as those presented in Section 5.10. If $\Delta \hat{H}_{\text{rxn}}$ is the rate at which heat is absorbed or liberated by reaction, and $\hat{M}_v$ is the mass flow rate of evaporated liquid leaving the system, for fermentation processes in which sensible heat changes and heats of mixing can be ignored, the following unsteady-state energy-balance equation applies:

$$\frac{dE}{dt} = -\Delta \hat{H}_{\text{rxn}} - \hat{M}_v \Delta h_v - \hat{Q} + \hat{W}_s. \quad (6.12)$$

For exothermic reactions $\Delta \hat{H}_{\text{rxn}}$ is negative; for endothermic reactions $\Delta \hat{H}_{\text{rxn}}$ is positive.

6.3 Solving Differential Equations

As shown in Sections 6.1 and 6.2, unsteady-state mass and energy balances are represented using differential equations. Once the differential equation for a particular system has been found, the equation must be solved to obtain an expression for mass M or energy E as a function of time. Differential equations are solved by integration. Some simple rules for differentiation and integration are outlined in Appendix D. Of course, there are many more rules of calculus than those included in Appendix D; however those shown are sufficient for handling the unsteady-state problems in this chapter. Further details can be found in any elementary calculus textbook, or in mathematics handbooks written especially for biological scientists, e.g. [1–3].

Before we proceed with solution techniques for unsteady-state mass and energy balances, there are several general points to consider.

- (i) *A differential equation can be solved directly only if it contains no more than two variables.* For mass- and energy-balance problems, the differential equation must have the form:

$$\frac{dM}{dt} = f(M, t) \quad \text{or} \quad \frac{dE}{dt} = f(E, t)$$

where $f(M, t)$ represents some function of M and t , and $f(E, t)$ represents some function of E and t . The function may contain constants, but no other variables besides M and t should appear in the expression for dM/dt , and no other variables besides E and t should appear in the expression for dE/dt . Before you attempt to solve these differential equations, check first that all parameters except M and t , or E and t , are constants.

- (ii) *Solution of differential equations requires knowledge of boundary conditions.* Boundary conditions contain extra information about the system. The number of boundary conditions required depends on the *order* of the differential equation, which is equal to the order of the highest differential coefficient in the equation. For example, if the equation contains a second derivative, e.g. d^2x/dt^2 , the equation is second order. All equations developed in this chapter have been first order; they involve only first order derivatives of the form dx/dt . One boundary condition is required to solve a first-order differential equation; two

boundary conditions are required for a second-order differential equation, and so on. Boundary conditions which apply at the beginning of the process when $t = 0$ are called *initial conditions*.

- (iii) *Not all differential equations can be solved algebraically*, even if the equation contains only two variables and the boundary conditions are available. Solution of some differential equations requires application of numerical techniques, preferably using a computer. In this chapter we will be concerned mostly with simple equations that can be solved using elementary calculus.

The easiest way of solving differential equations is to *separate variables* so that each variable appears on only one side of the equation. For example, consider the simple differential equation:

$$\frac{dx}{dt} = a(b - x) \tag{6.13}$$

where a and b are constants. First we must check that the equation contains only two variables x and t , and that all other parameters in the equation are constants. Once this is verified, the equation is separated so that x and t each appear on only one side of the equation. In the case of Eq. (6.13), this is done by dividing each side of the equation by $(b - x)$, and multiplying each side by dt :

$$\frac{1}{(b - x)} dx = a dt. \tag{6.14}$$

The equation is now ready for integration:

$$\int \frac{1}{(b - x)} dx = \int a dt. \tag{6.15}$$

Using integration rules (D-28) and (D-24) from Appendix D:

$$-\ln(b - x) = at + K. \tag{6.16}$$

Note that the constants of integration from both sides of the equation have been condensed into one constant K ; this is valid because a constant $\pm$ a constant = a constant.

6.4 Solving Unsteady-State Mass Balances

Solution of unsteady-state mass balances is sometimes difficult unless certain simplifications are made. Because the aim here is to illustrate application of unsteady-state balances without becoming too involved in integral calculus, the problems presented in this section will be relatively simple. For the majority of problems in this chapter analytical solution is possible.

The following restrictions are common in unsteady-state mass-balance problems.

- (i) The system is *well mixed* so that properties of the system do not vary with position. If properties within the system

are the same at all points, this includes the point from which any product stream is drawn. Accordingly, when the system is well mixed, properties of the outlet stream are the same as those within the system.

- (ii) Expressions for reaction rate involve the concentration of only one reactive species. The mass-balance equation for this species can then be derived and solved; if other species appear in the kinetic expression this introduces extra variables into the differential equation making solution more complex.

The following example illustrates solution of an unsteady-state mass balance without reaction.

Example 6.2 Dilution of salt solution

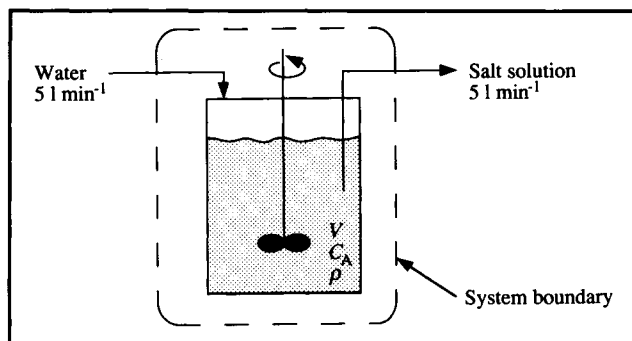
1.5 kg salt is dissolved in water to make 100 litres. Pure water runs into a tank containing this solution at a rate of 5 l min^{-1} ; salt solution overflows at the same rate. The tank is well mixed. How much salt is in the tank at the end of 15 min? Assume the density of salt solution is constant and equal to that of water.

Solution:

- (i) *Flowsheet and system boundary.*

These are shown in Figure 6E2.1.

Figure 6E2.1 Well-mixed tank for dilution of salt solution.



- (ii) *Define variables.*

C_A = concentration of salt in the tank; V = volume of solution in the tank; ρ = density of salt solution and water.

- (iii) *Assumptions.*

- no leaks
- tank is well mixed
- density of salt solution is the same as water

- (iv) *Boundary conditions.*

At the beginning of the process, the salt concentration in the tank is 1.5 kg in 100 l, or 0.015 kg l^{-1} . If we call this initial salt concentration C_{A0} , the initial condition is:

$$\text{at } t = 0 \quad C_A = C_{A0} = 0.015 \text{ kg l}^{-1}.$$

(1)

We also know that the initial volume of liquid in the tank is 100 l. Another initial condition is:

$$\text{at } t=0 \quad V = V_0 = 100 \text{ l.} \quad (2)$$

(v) *Total mass balance.*

The unsteady-state balance equation for total mass was derived in Example 6.1:

$$\frac{d(\rho V)}{dt} = F_i \rho_i - F_o \rho_o. \quad (6.6)$$

In this problem we are told that the volumetric flow rates of inlet and outlet streams are equal; therefore $F_i = F_o$. In addition, the density of the system is constant so that $\rho_i = \rho_o = \rho$. Under these conditions, the terms on the right-hand side of Eq. (6.6) cancel. On the left-hand side, because ρ is constant it can be taken outside of the differential. Therefore:

$$\rho \frac{dV}{dt} = 0$$

or

$$\frac{dV}{dt} = 0.$$

If the derivative of V with respect to t is zero, V must be a constant:

$$V = K$$

where K is the constant of integration. This result means that the volume of the tank is constant and independent of time. Initial condition (2) tells us that $V = 100 \text{ l}$ at $t = 0$; therefore V must equal 100 l at all times. Consequently, the constant of integration K is equal to 100 l , and the volume of liquid in the tank does not vary from 100 l .

(vi) *Mass balance for salt.*

The unsteady mass-balance equation for component A such as salt was derived in Example 6.1:

$$\frac{d(V C_A)}{dt} = F_i C_{Ai} - F_o C_{Ao} - k_1 C_A V. \quad (6.7)$$

In the present problem there is no reaction, so k_1 is zero. Also, $F_i = F_o = F = 5 \text{ l min}^{-1}$. Because the tank is well mixed, the concentration of salt in the outlet stream is equal to that inside the tank, i.e. $C_{Ao} = C_A$. In addition, since the inlet stream does not contain salt, $C_{Ai} = 0$. From the balance on total mass we know that V is constant and can be placed outside of the differential. Taking these factors into consideration, Eq. (6.7) becomes:

$$V \frac{dC_A}{dt} = -F C_A.$$

This differential equation contains only two variables C_A and t ; F and V are constants. The variables are easy to separate by dividing both sides by $V C_A$ and multiplying by dt :

$$\frac{dC_A}{C_A} = \frac{-F}{V} dt.$$

The equation is now ready to integrate:

$$\int \frac{dC_A}{C_A} = \int \frac{-F}{V} dt.$$

Using integration rules (D-27) and (D-24) from Appendix D and combining the constants of integration:

$$\ln C_A = \frac{-F}{V} t + K \quad (3)$$

We have yet to determine the value of K . From initial condition (1), at $t = 0$, $C_A = C_{A0}$. Substituting this information into (3):

$$\ln C_{A0} = K$$

We have thus determined K . Substituting this value for K back into (3):

$$\ln C_A = \frac{-F}{V} t + \ln C_{A0} \quad (4)$$

This is the solution to the mass balance; it gives an expression for concentration of salt in the tank as a function of time. Notice that if we had forgotten to add the constant of integration, the answer would not contain the term $\ln C_{A0}$. The equation would then say that at $t = 0$, $\ln C_A = 0$; i.e. $C_A = 1$. We know this is not true; instead, at $t = 0$, $C_A = 0.015 \text{ kg l}^{-1}$, so the result without the boundary condition is incorrect. It is important to apply boundary conditions every time you integrate.

The solution equation is usually rearranged to give an exponential expression. This is achieved by subtracting $\ln C_{A0}$ from both sides of (4):

$$\ln C_A - \ln C_{A0} = \frac{-F}{V} t$$

and noting from Eq. (D-9) that $(\ln C_A - \ln C_{A0})$ is the same as $\ln C_A / C_{A0}$:

$$\ln \frac{C_A}{C_{A0}} = \frac{-F}{V} t.$$

Taking the anti-logarithm of both sides:

$$\frac{C_A}{C_{A0}} = e^{(-F/V)t}$$

or

$$C_A = C_{A0} e^{(-F/V)t}.$$

We can check that this is the correct solution by taking the derivative of both sides with respect to t and making sure that the original differential equation is recovered.

For $F = 5 \text{ l min}^{-1}$, $V = 100 \text{ l}$ and $C_{A0} = 0.015 \text{ kg l}^{-1}$, at $t = 15 \text{ min}$:

$$C_A = (0.015 \text{ kg l}^{-1}) e^{(-5 \text{ l min}^{-1}/100 \text{ l})(15 \text{ min})} = 7.09 \times 10^{-3} \text{ kg l}^{-1}.$$

The salt concentration after 15 min is $7.09 \times 10^{-3} \text{ kg l}^{-1}$. Therefore:

$$\text{Mass of salt} = C_A V = (7.09 \times 10^{-3} \text{ kg l}^{-1}) 100 \text{ l} = 0.71 \text{ kg}.$$

(vii) *Finalise.*

After 15 min, the mass of salt in the tank is 0.71 kg.

In Example 6.2 the density of the system was assumed constant. This simplified the mathematics of the problem so that ρ could be taken outside the differential and cancelled from the total mass balance. The assumption of constant density is justified for dilute solutions because the density does not differ greatly from that of the solvent. The result of the total mass

balance makes intuitive sense: for a tank with equal flow rates in and out and constant density, the volume of liquid inside the tank should remain constant.

The effect of reaction on the unsteady-state mass balance is illustrated in Example 6.3.

Example 6.3 Flow reactor

Rework Example 6.2 to include reaction. Assume that a reaction in the tank consumes salt, at a rate given by the first-order equation:

$$r = k_1 C_A$$

where k_1 is the first-order reaction constant and C_A is the concentration of salt in the tank. Derive an expression for C_A as a function of time. If $k_1 = 0.02 \text{ min}^{-1}$, how long does it take for the concentration of salt to fall to a value $1/20$ the initial level?

Solution:

The flowsheet, boundary conditions and assumptions for this problem are the same as in Example 6.2. The total mass balance is also the same; total mass in the system is unaffected by reaction.

(i) *Mass balance for salt.*

From Example 6.1, the unsteady mass-balance equation for salt is:

$$\frac{d(VC_A)}{dt} = F_i C_{Ai} - F_o C_{Ao} - k_1 C_A V. \quad (6.7)$$

In this problem $F_i = F_o = F$, $C_{Ai} = 0$, and V is constant. Because the tank is well mixed $C_{Ao} = C_A$. Therefore, Eq. (6.7) becomes:

$$V \frac{dC_A}{dt} = -FC_A - k_1 C_A V.$$

This equation contains only two variables C_A and t ; F , V and k_1 are constants. Separate variables by dividing both sides by VC_A and multiplying by dt :

$$\frac{dC_A}{C_A} = \left(\frac{-F}{V} - k_1 \right) dt.$$

Integrating both sides gives:

$$\ln C_A = \left(\frac{-F}{V} - k_1 \right) t + K.$$

where K is the constant of integration. K is determined from initial condition (1) in Example 6.2: at $t = 0$, $C_A = C_{A0}$. Substituting these values gives:

$$\ln C_{A0} = K.$$

Substituting this value for K back into the answer:

$$\ln C_A = \left(\frac{-F}{V} - k_1 \right) t + \ln C_{A0}$$

or

$$\ln \frac{C_A}{C_{A0}} = \left(\frac{-F}{V} - k_1 \right) t.$$

For $F = 5 \text{ l min}^{-1}$, $V = 100 \text{ l}$, $k_1 = 0.02 \text{ min}^{-1}$ and $C_A/C_{A0} = 1/20$, this equation becomes:

$$\ln \left(\frac{1}{20} \right) = \left(\frac{-5 \text{ l min}^{-1}}{100 \text{ l}} - 0.02 \text{ min}^{-1} \right) t$$

or

$$-3.00 = (0.07 \text{ min}^{-1}) t.$$

Solving for t :

$$t = 42.8 \text{ min.}$$

(ii) *Finalise.*

The concentration of salt in the tank reaches $1/20$ its initial level after 43 min.

6.5 Solving Unsteady-State Energy Balances

Solution of unsteady-state energy-balance problems can be mathematically quite complex. In this chapter only problems with the following characteristics will be treated for ease of mathematical handling.

- (i) The system has at most one input and one output stream; furthermore, these streams have the same mass flow rate.

Under these conditions, total mass of the system is constant.

- (ii) The system is well mixed with uniform temperature and composition. Properties of the outlet stream are therefore the same as within the system.
- (iii) No chemical reactions or phase changes occur.
- (iv) Mixtures and solutions are ideal.
- (v) Heat capacities of the system contents and inlet and outlet streams are independent of composition and temperature, and therefore invariant with time.

(vi) Internal energy U and enthalpy H are independent of pressure.

The principles and equations for unsteady-state energy balances are entirely valid when these conditions are not met; the

only difference is that solution of the differential equation is greatly simplified for systems with the above characteristics. The procedure for solution of unsteady-state energy balances is illustrated in Example 6.4.

Example 6.4 Solvent heater

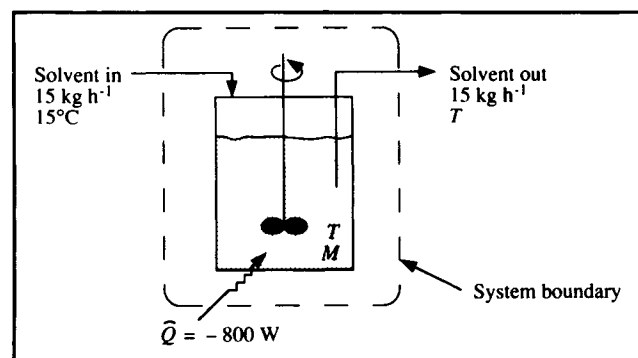
An electric heating-coil is immersed in a stirred tank. Solvent at 15°C with heat capacity $2.1 \text{ kJ kg}^{-1} ^\circ\text{C}^{-1}$ is fed into the tank at a rate of 15 kg h^{-1} . Heated solvent is discharged at the same flow rate. The tank is filled initially with 125 kg cold solvent at 10°C . The rate of heating by the electric coil is 800 W . Calculate the time required for the temperature of the solvent to reach 60°C .

Solution:

(i) *Flowsheet and system boundary.*

These are shown in Figure 6E4.1.

Figure 6E4.1 Continuous process for heating solvent.



(ii) *Define variables.*

If the tank is well mixed, the temperature of the outlet stream is the same as inside the tank. Let T be the temperature in the tank; T_i is the temperature of the incoming stream, M is the mass of solvent in the tank, and $\dot{M}$ is the mass flow rate of solvent to and from the tank.

(iii) *Assumptions.*

- no leaks
- tank is well mixed
- negligible shaft work
- heat capacity is independent of temperature
- no evaporation
- heat losses to the environment are negligible

(iv) *Reference state.*

$H = 0$ for solvent at 10°C , i.e. $T_{\text{ref}} = 10^\circ\text{C}$.

(v) *Boundary conditions.*

The initial condition is:

$$\text{at } t = 0 \quad T = T_0 = 10^\circ\text{C}.$$

(1)

(vi) *Mass balance.*

Since the mass flow rates of solvent to and from the tank are equal, the mass M of solvent inside the tank is constant and equal to the initial value, 125 kg. The mass balance is therefore complete.

(vii) *Energy balance.*

The unsteady-state energy-balance equation for two flow streams is given by Eq. (6.10):

$$\frac{dE}{dt} = \hat{M}_i h_i - \hat{M}_o h_o - \hat{Q} + \hat{W}_s.$$

In the absence of phase change, reaction and heats of mixing, the enthalpies of input and output streams can be determined from sensible heats only. Similarly, any change in the energy content of the system must be due to change in temperature and sensible heat. With solvent enthalpy defined as zero at T_{ref} , change in E can be calculated from the difference between the system temperature and T_{ref} . Therefore, the terms of the energy-balance equation are:

$$\text{Accumulation: } \frac{dE}{dt} = \frac{d}{dt} (MC_p \Delta T) = \frac{d}{dt} (MC_p [T - T_{\text{ref}}])$$

$$\text{As } M, C_p \text{ and } T_{\text{ref}} \text{ are constants, } \frac{dE}{dt} = MC_p \frac{dT}{dt}$$

$$\text{Flow input: } \hat{M}_i h_i = \hat{M}_i C_p (T_i - T_{\text{ref}})$$

$$\text{Flow output: } \hat{M}_o h_o = \hat{M}_o C_p (T - T_{\text{ref}})$$

$$\text{Shaft work: } \hat{W}_s = 0.$$

Substituting these expressions into the energy-balance equation gives:

$$MC_p \frac{dT}{dt} = \hat{M}_i C_p (T_i - T_{\text{ref}}) - \hat{M}_o C_p (T - T_{\text{ref}}) - \hat{Q}.$$

$M = 125 \text{ kg}$, $C_p = 2.1 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}$, $\hat{M}_i = \hat{M}_o = 15 \text{ kg h}^{-1}$, $T_i = 15^\circ\text{C}$ and $T_{\text{ref}} = 10^\circ\text{C}$. Converting data for $\hat{Q}$ into consistent units:

$$\begin{aligned} \hat{Q} &= -800 \text{ W} \cdot \left| \frac{1 \text{ J s}^{-1}}{1 \text{ W}} \right| \cdot \left| \frac{1 \text{ kJ}}{1000 \text{ J}} \right| \cdot \left| \frac{3600 \text{ s}}{1 \text{ h}} \right| \\ &= -2.88 \times 10^3 \text{ kJ h}^{-1}. \end{aligned}$$

$\hat{Q}$ is negative because heat flows into the system. Substituting these values into the energy-balance equation:

$$\begin{aligned} (125 \text{ kg}) (2.1 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}) \frac{dT}{dt} &= (15 \text{ kg h}^{-1}) (2.1 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}) (15^\circ\text{C} - 10^\circ\text{C}) \\ &\quad - (15 \text{ kg h}^{-1}) (2.1 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}) (T - 10^\circ\text{C}) - (-2.88 \times 10^3 \text{ kJ h}^{-1}). \end{aligned}$$

Grouping terms gives a differential equation for rate of temperature change:

$$\frac{dT}{dt} = 12.77 - 0.12 T$$

where T has units $^{\circ}\text{C}$ and t has units h. Separating variables and integrating:

$$\int \frac{dT}{12.77 - 0.12T} = \int dt.$$

Using the integration rule (D-28) from Appendix D:

$$\frac{-1}{0.12} \ln(12.77 - 0.12T) = t + K.$$

The initial condition is: at $t = 0$, $T = 15^{\circ}\text{C}$. Therefore, $K = -19.96$ and the solution is:

$$19.96 - \frac{1}{0.12} \ln(12.77 - 0.12T) = t.$$

From this equation, at $T = 60^{\circ}\text{C}$, $t = 5.65$ h.

(viii) *Finalise.*

It takes 5.7 h for the temperature to reach 60°C .

6.6 Summary of Chapter 6

At the end of Chapter 6 you should:

- know what types of process require unsteady-state analysis;
- be able to derive appropriate *differential equations* for unsteady-state mass and energy balances;
- understand the need for *boundary conditions* to solve differential equations representing actual processes; and
- be able to solve simple unsteady-state mass and energy balances to obtain equations for system parameters as a function of time.

Problems

6.1 Dilution of sewage

In a sewage-treatment plant, a large concrete tank initially contains 440 000 litres liquid and 10 000 kg fine suspended solids. To flush this material out of the tank, water is pumped into the vessel at a rate of 40 000 litres h^{-1} . Liquid containing solids leaves at the same rate. Estimate the concentration of suspended solids in the tank at the end of 5 h.

6.2 Production of fish-protein concentrate

Whole gutted fish are dried to make a protein paste. In a batch drier, the rate at which water is removed from the fish is

roughly proportional to the moisture content. If a batch of gutted fish loses half its initial moisture content in the first 20 min, how long will the drier take to remove 95% of the water?

6.3 Contamination of vegetable oil

Vegetable oil is used in a food-processing factory for preparing instant breadcrumbs. A stirred tank is used to hold the oil; during operation of the breadcrumb process, oil is pumped from the tank at a rate of 4.8 l h^{-1} . At 8 p.m. during the night shift, the tank is mistakenly connected to a drum of cod-liver oil which is then pumped into the tank. The volume of vegetable oil in the tank at 8 p.m. is 60 l.

- If the flow rate of cod-liver oil into the tank is 7.5 l h^{-1} and the tank has a maximum capacity of 100 l, will the tank overflow before the factory manager arrives at 9 a.m.? Assume that the density of both oils is the same.
- If cod-liver oil is pumped into the tank at a rate of 4.8 l h^{-1} instead of 7.5 l h^{-1} , what is the composition of oil in the tank at midnight?

6.4 Batch growth of bacteria

During exponential phase in batch culture, the growth rate of a culture is proportional to the concentration of cells present. When *Streptococcus lactis* bacteria are cultured in milk, the concentration of cells doubles in 45 min. If this rate of growth

is maintained for 12 h, what is the final concentration of cells relative to the inoculum level?

6.5 Radioactive decay

A radioactive isotope decays at a rate proportional to the amount of isotope present. If the concentration of isotope is C (mg l^{-1}), its rate of decay is:

$$r_C = k_1 C.$$

- A solution of radioactive isotope is prepared at concentration C_0 . Show that the half-life of the isotope, i.e. the time required for the isotope concentration to reach half of its original value, is equal to $\ln 2 / k_1$.
- A solution of the isotope ^{32}P is used to radioactively label DNA for hybridisation studies. The half-life of ^{32}P is 14.3 days. According to institutional safety requirements, the solution cannot be discarded until the activity is 1% its present value. How long will this take?

6.6 Continuous fermentation

A well-mixed fermenter of volume V contains cells initially at concentration x_0 . A sterile feed enters the fermenter with volumetric flow rate F ; fermentation broth leaves at the same rate. The concentration of substrate in the feed is s_1 . The equation for rate of cell growth is:

$$r_X = k_1 x$$

and the expression for rate of substrate consumption is:

$$r_S = k_2 x$$

where k_1 and k_2 are rate constants with dimensions T^{-1} , r_X and r_S have dimensions $\text{L}^{-3}\text{MT}^{-1}$, and x is the concentration of cells in the fermenter.

- Derive a differential equation for the unsteady-state mass balance of cells.
- From this equation, what must be the relationship between F , k_1 and the volume of liquid in the fermenter at steady state?
- Solve the differential equation to obtain an expression for cell concentration in the fermenter as a function of time.
- Use the following data to calculate how long it takes for the cell concentration in the fermenter to reach 4.0 g l^{-1} :

$$F = 2200 \text{ l h}^{-1}$$

$$\begin{aligned} V &= 10\,000 \text{ l} \\ x_0 &= 0.5 \text{ g l}^{-1} \\ k_1 &= 0.33 \text{ h}^{-1}. \end{aligned}$$

V is the volume of liquid in the fermenter.

- Set up a differential equation for the mass balance of substrate. Substitute the result for x from (c) to obtain a differential equation in which the only variables are substrate concentration and time. (Do you think you would be able to solve this equation algebraically?)
- At steady state, what must be the relationship between s and x ?

6.7 Fed-batch fermentation

A feed stream containing glucose enters a fed-batch fermenter at constant flow rate. The initial volume of liquid in the fermenter is V_0 . Cells in the fermenter consume glucose at a rate given by:

$$r_S = k_1 s$$

where k_1 is the rate constant (h^{-1}) and s is the concentration of glucose in the fermenter (g l^{-1}).

- Assuming constant density, derive an equation for the total mass balance. What is the expression relating volume and time?
- Derive the differential equation for rate of change of substrate concentration.

6.8 Plug-flow reactor

When fluid flows through a pipe or channel with sufficiently large Reynolds number, it approximates *plug flow*. Plug flow means that there is no variation of axial velocity over the flow cross-section. When reaction occurs in a plug-flow tubular reactor (PFTR), as reactant is consumed its concentration changes down the length of the tube.

- Derive the differential equation for change in reactant concentration with distance at steady state.
- What are the boundary conditions?
- If the reaction is first order, solve the differential equation to determine an expression for concentration as a function of distance from the front of the tube.
- How does this expression compare with that for a well-mixed batch reactor?

Hint: Referring to Figure 6P8.1, consider the accumulation of reactant within a section of the reactor between z and $z + \Delta z$.

In this case, the volumetric flow rate of liquid in and out of the section is Au . Use the following symbols:

A is the reactor cross-sectional area

u is the fluid velocity

z is distance along the tube from the entrance

L is the total length of the reactor

C_{Ai} is the concentration of reactant in the feed stream

C_A is the concentration of reactant in the reactor; C_A is a function of z

r_C is the volumetric rate of consumption of reactant.

6.9 Boiling water

A beaker containing 2 litres water at 18°C is placed on a laboratory hot-plate. The water begins to boil in 11 min.

- Neglecting evaporation, write the energy balance for the process.
- The hot-plate delivers heat at a constant rate. Assuming that the heat capacity of water is constant, what is that rate?

6.10 Heating glycerol solution

An adiabatic stirred tank is used to heat 100 kg of a 45% glycerol solution in water. An electrical coil delivers 2.5 kW of power to the tank; 88% of the energy delivered by the coil goes into heating the vessel contents. The glycerol solution is initially at 15°C.

- Write a differential equation for the energy balance.
- Integrate the equation to obtain an expression for temperature as a function of time.
- Assuming glycerol and water form an ideal solution, how long will the solution take to reach 90°C?

6.11 Heating molasses

Diluted molasses is heated in a well-stirred steel tank by saturated steam at 40 psia condensing in a jacket on the outside of

the tank. The outer walls of the jacket are insulated. 1020 kg h⁻¹ molasses solution at 20°C enters the tank, and 1020 kg h⁻¹ of heated molasses leaves. The rate of heat transfer from the steam through the jacket and to the molasses is given by the equation:

$$\hat{Q} = UA(T_{\text{steam}} - T_{\text{molasses}})$$

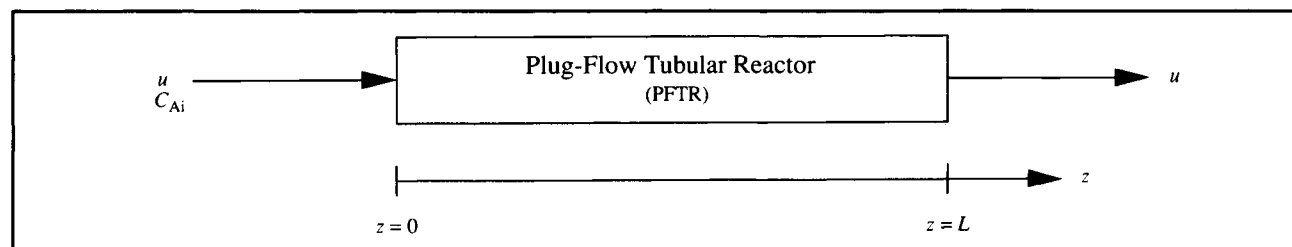
where $\hat{Q}$ is the rate of heat transfer, U is the overall heat-transfer coefficient, A is the surface area for heat transfer, and T is the temperature. For this system the value of U is 190 kcal m⁻² h⁻¹ °C⁻¹; C_p for the molasses solution is 0.85 kcal kg⁻¹ °C⁻¹. The initial mass of molasses solution in the tank is 5000 kg; the initial temperature is 20°C. The surface area for heat transfer between the steam and tank is 1.5 m².

- Derive the differential equation describing the rate of change of temperature in the tank.
- Solve the differential equation to obtain an equation relating temperature and time.
- Plot the temperature of molasses leaving the tank as a function of time.
- What is the maximum temperature that can be achieved in the tank?
- Estimate the time required for this system to reach steady state.
- How long does it take for the outlet molasses temperature to rise from 20°C to 40°C?

6.12 Pre-heating culture medium

A glass fermenter used for culture of hybridoma cells contains nutrient medium at 15°C. The fermenter is wrapped in an electrical heating mantle which delivers heat at a rate of 450 W. Before inoculation, the medium and vessel must be at 36°C. The medium is well mixed during heating. Use the following information to determine the time required for medium pre-heating.

Figure 6P8.1 Plug-flow tubular reactor (PFTR).



Glass fermenter vessel: mass = 12.75 kg; $C_p = 0.20 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$
 Nutrient medium: mass = 7.50 kg; $C_p = 0.92 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$.

6.13 Water heater

A tank contains 1000 kg water at 24°C . It is planned to heat this water using saturated steam at 130°C in a coil inside the tank. The rate of heat transfer from the steam is given by the equation:

$$\hat{Q} = UA(T_{\text{steam}} - T_{\text{molasses}})$$

where $\hat{Q}$ is the rate of heat transfer, U is the overall heat-transfer coefficient, A is the surface area for heat transfer, and T is the temperature. The heat-transfer area provided by the coil is 0.3 m^2 ; the heat-transfer coefficient is $220 \text{ kcal m}^{-2} \text{ h}^{-1} \text{ } ^\circ\text{C}^{-1}$. Condensate leaves the coil saturated.

- (a) The tank has a surface area of 0.9 m^2 exposed to the ambient air. The tank exchanges heat through this exposed surface at a rate given by an equation similar to that above. For heat transfer to or from the surrounding air the heat-transfer coefficient is $25 \text{ kcal m}^{-2} \text{ h}^{-1} \text{ } ^\circ\text{C}^{-1}$. If the air temperature is 20°C , calculate the time required to heat the water to 80°C .

- (b) What time is saved if the tank is insulated?

Assume the heat capacity of water is constant, and neglect the heat capacity of the tank walls.

References

1. Cornish-Bowden, A. (1981) *Basic Mathematics for Biochemists*, Chapman and Hall, London.
2. Newby, J.C. (1980) *Mathematics for the Biological Sciences*, Oxford University Press, Oxford.
3. Arya, J.C. and R.W. Lardner (1979) *Mathematics for the Biological Sciences*, Prentice-Hall, New Jersey.

Suggestions for Further Reading

- Felder, R.M. and R.W. Rousseau (1978) *Elementary Principles of Chemical Processes*, Chapter 11, John Wiley, New York.
- Himmelblau, D.M. (1974) *Basic Principles and Calculations in Chemical Engineering*, 3rd edn, Chapter 6, Prentice-Hall, New Jersey.
- Shaheen, E.I. (1975) *Basic Practice of Chemical Engineering*, Chapter 4, Houghton Mifflin, Boston, Massachusetts.
- Whitwell, J.C. and R.K. Toner (1969) *Conservation of Mass and Energy*, Chapter 9, Blaisdell, Waltham, Massachusetts.

This Page Intentionally Left Blank

Part 3

Physical

Processes

This Page Intentionally Left Blank

Fluid Flow and Mixing

Fluid mechanics is an important area of engineering science. The nature of flow in pipes, pumps and reactors depends on the power input to the system and the physical characteristics of the fluid. In fermenters, fluid properties affect process energy requirements and the effectiveness of mixing, which can have a dramatic influence on productivity and the success of equipment scale-up. As we shall see in the following chapters, transport of heat and mass is often coupled with fluid flow. To understand the mechanisms of these important transport processes, we must first examine the behaviour of fluid near surfaces and interfaces. Fluids in bioprocessing often contain suspended solids, consist of more than one phase, and have non-Newtonian properties. All of these features complicate analysis of flow behaviour and present many challenges in bioprocess design.

Fluid dynamics accounts for a substantial fraction of the chemical engineering literature. Accordingly, complete treatment of the subject is beyond the scope of this book. Here, we content ourselves with study of those aspects of flow behaviour particularly relevant to fermentation fluids. Further information can be found in the references at the end of the chapter.

which is incompressible and has zero viscosity. The term *inviscid* applies to fluids with zero viscosity. All *real* fluids have finite viscosity and are therefore called *viscid* or *viscous* fluids.

Fluids can be classified further as *Newtonian* or *non-Newtonian*. This distinction is explained in detail in Sections 7.3 and 7.5.

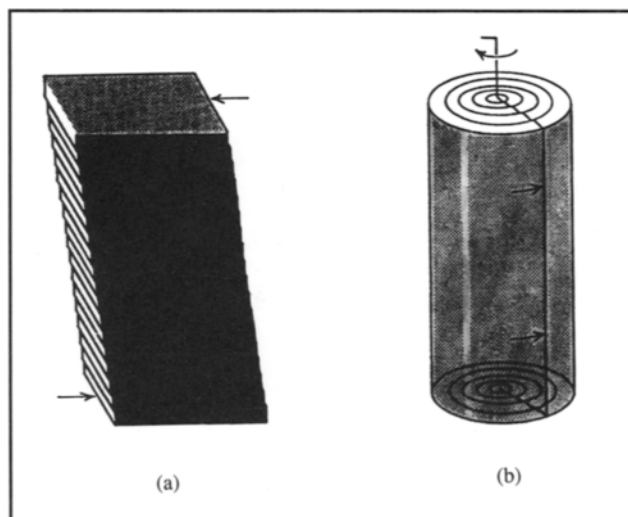
7.1 Classification of Fluids

A fluid is a substance which undergoes continuous deformation when subjected to a shearing force. A simple shearing force is one which causes thin parallel plates to slide over each other, as in a pack of cards. Shear can also occur in other geometries; the effect of shear force in planar and rotational systems is illustrated in Figure 7.1. Shear forces in these examples cause *deformation*, which is a change in the relative positions of parts of a body. A shear force must be applied to produce fluid flow.

According to the above definition, fluids can be either gases or liquids. Two physical properties, viscosity and density, are used to classify fluids. If the density of a fluid changes with pressure, the fluid is *compressible*. Gases are generally classed as compressible fluids. The density of liquids is practically independent of pressure; liquids are *incompressible* fluids. Sometimes the distinction between compressible and incompressible fluid is not well defined; for example, a gas may be treated as incompressible if variations of pressure and temperature are small.

Fluids are also classified on the basis of viscosity. Viscosity is the property of fluids responsible for internal friction during flow. An *ideal* or *perfect* fluid is a hypothetical liquid or gas

Figure 7.1 Laminar deformation due to (a) planar shear and (b) rotational shear. (From J.R. van Wazer, J.W. Lyons, K.Y. Kim and R.E. Colwell, 1963, *Viscosity and Flow Measurement*, Interscience, New York.)



7.2 Fluids in Motion

Bioprocesses involve fluids in motion in vessels and pipes. General characteristics of fluid flow are described in the following sections.

7.2.1 Streamlines

When a fluid flows through a pipe or over a solid object, the velocity of the fluid varies depending on position. One way of representing variation in velocity is *streamlines*, which follow the flow path. Constant velocity is shown by equidistant spacing of parallel streamlines as shown in Figure 7.2(a). The velocity profile for slow-moving fluid flowing over a submerged object is shown in Figure 7.2(b); reduced spacing between the streamlines indicates that the velocity at the top and bottom of the object is greater than at the front and back.

Streamlines show only the net effect of fluid motion; although streamlines suggest smooth continuous flow, fluid molecules may actually be moving in an erratic fashion. The slower the flow the more closely the streamlines represent actual motion. Slow fluid flow is therefore called *streamline* or *laminar* flow. In fast motion, fluid particles frequently cross and recross the streamlines. This motion is called *turbulent* flow and is characterised by formation of *eddies*.

7.2.2 Reynolds Number

Transition from laminar to turbulent flow depends not only on the velocity of the fluid, but also on its viscosity and density and the geometry of the flow conduit. A parameter used to characterise fluid flow is the *Reynolds number*. For full flow in pipes with circular cross-section, Reynolds number Re is defined as:

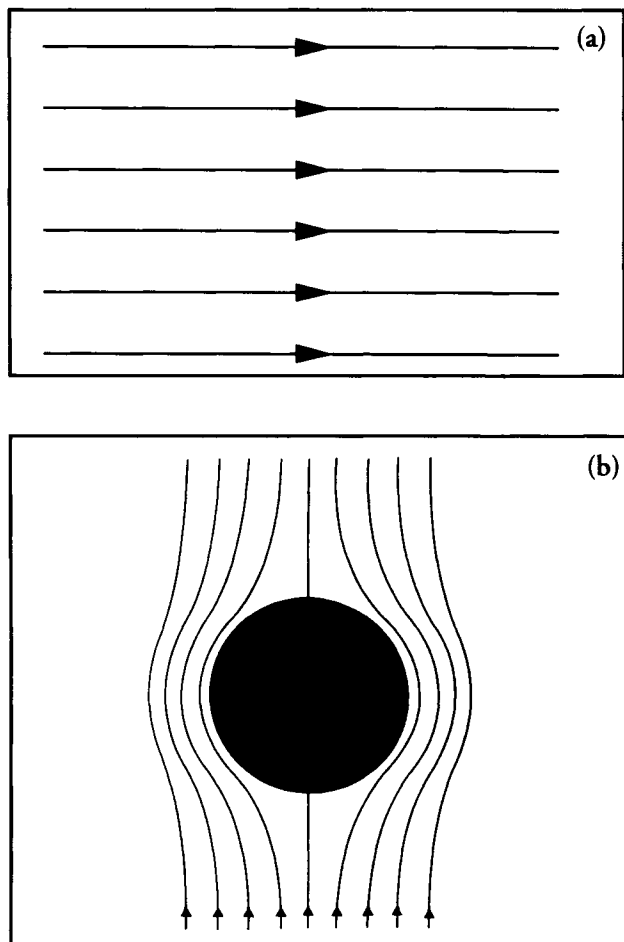
$$Re = \frac{D u \rho}{\mu} \quad (7.1)$$

where D is pipe diameter, u is average linear velocity of the fluid, ρ is fluid density, and μ is fluid viscosity. For stirred vessels there is another definition of Reynolds number:

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} \quad (7.2)$$

where Re_i is the *impeller Reynolds number*, N_i is stirrer speed, D_i is impeller diameter, ρ is fluid density and μ is fluid viscosity.

Figure 7.2 Streamlines for (a) constant fluid velocity; (b) steady flow over a submerged object.



The Reynolds number is a dimensionless variable; the units and dimensions of the parameters in Eqs (7.1) and (7.2) cancel completely.

Reynolds number is named after Osborne Reynolds, who published in 1883 a classical series of papers on the nature of flow in pipes. One of the most significant outcomes of Reynolds' experiments is that there is a *critical Reynolds number* which marks the upper boundary for laminar flow in pipes. In smooth pipes, laminar flow is encountered at Reynolds numbers less than 2100. Under normal conditions, flow is turbulent at Re above about 4000. Between 2100 and 4000 is the *transition region* where flow may be either laminar or turbulent depending on conditions at the entrance of the pipe and other variables. Flow in stirred tanks may also be laminar or turbulent as a function of the impeller Reynolds number.

The value of Re_i marking the transition between these flow regimes depends on the geometry of the impeller and tank; for several commonly-used mixing systems, laminar flow is found at $Re_i \leq 10$.

7.2.3 Hydrodynamic Boundary Layers

In most practical applications, fluid flow occurs in the presence of a stationary solid surface, such as the walls of a pipe or tank. That part of the fluid where flow is affected by the solid is called the *boundary layer*. As an example, consider flow of fluid parallel to the flat plate shown in Figure 7.3. Contact between the moving fluid and the plate causes formation of a boundary layer, u , is represented by arrows; u is zero at the surface of the plate. Viscous drag forces are transmitted upwards through the fluid from the stationary layer at the surface. The fluid layer just above the surface moves at a slow but finite velocity; layers further above move at increasing velocity as the drag forces associated with the stationary layer decrease. At the edge of the boundary layer, fluid is unaffected by the presence of the plate and the velocity is close to that of the bulk flow, u_B . The magnitude of u at various points in the boundary layer is indicated in Figure 7.3(b) by the length of the arrows in the direction of flow. The line connecting the heads of the velocity arrows shows the *velocity profile* in the fluid. A *velocity gradient*, i.e. a change in velocity with distance from the plate, is thus established in a direction perpendicular to the direction of flow. The velocity gradient forms as the drag force resulting from retardation of fluid at the surface is transmitted through the fluid.

As indicated by the arrows in Figure 7.3(a), the bulk fluid velocity in front of the plate is uniform and of magnitude u_B . The extent of the boundary layer is indicated by the broken

line. Above the boundary layer, fluid motion is the same as if the plate were not there. The boundary layer grows in thickness from the leading edge until it develops its full size. Final thickness of the boundary layer depends on the Reynolds number for bulk flow.

When fluid flows over a stationary object, a thin film of fluid in contact with the surface adheres to it to prevent slippage over the surface. Fluid velocity at the surface of the plate in Figure 7.3 is therefore zero. When part of a flowing fluid has been brought to rest, the flow of adjacent fluid layers is slowed down by the action of *viscous drag*. This phenomenon is illustrated in Figure 7.3(b). Velocity of fluid within the boundary layer, u , is represented by arrows; u is zero at the surface of the plate. Viscous drag forces are transmitted upwards through the fluid from the stationary layer at the surface. The fluid layer just above the surface moves at a slow but finite velocity; layers further above move at increasing velocity as the drag forces associated with the stationary layer decrease. At the edge of the boundary layer, fluid is unaffected by the presence of the plate and the velocity is close to that of the bulk flow, u_B . The magnitude of u at various points in the boundary layer is indicated in Figure 7.3(b) by the length of the arrows in the direction of flow. The line connecting the heads of the velocity arrows shows the *velocity profile* in the fluid. A *velocity gradient*, i.e. a change in velocity with distance from the plate, is thus established in a direction perpendicular to the direction of flow. The velocity gradient forms as the drag force resulting from retardation of fluid at the surface is transmitted through the fluid.

Formation of boundary layers is important not only in determining characteristics of fluid flow, but also for transfer of heat and mass between phases. These topics are discussed further in Chapters 8 and 9.

7.2.4 Boundary-Layer Separation

What happens when contact is broken between a fluid and a solid immersed in the flow path? As an example, consider a flat plate aligned perpendicular to the direction of fluid flow, as shown in Figure 7.4. Fluid impinges on the surface of the plate, and forms a boundary layer as it flows either up or down the object. When fluid reaches the top or bottom of the plate its momentum prevents it from making the sharp turn around the edge. As a result, fluid separates from the plate and proceeds outwards into the bulk fluid. Directly behind the plate is a zone of highly decelerating fluid in which large eddies or *vortices* are formed. This zone is called the *wake*. Eddies in the wake are kept in rotational motion by the force of bordering currents.

Figure 7.3 Fluid boundary layer for flow over a flat plate. (a) The boundary layer forms at the leading edge. (b) Compared with velocity u_B in the bulk fluid, velocity in the boundary layer is zero at the plate surface but increases with distance from the plate to reach u_B near the outer limit of the boundary layer.

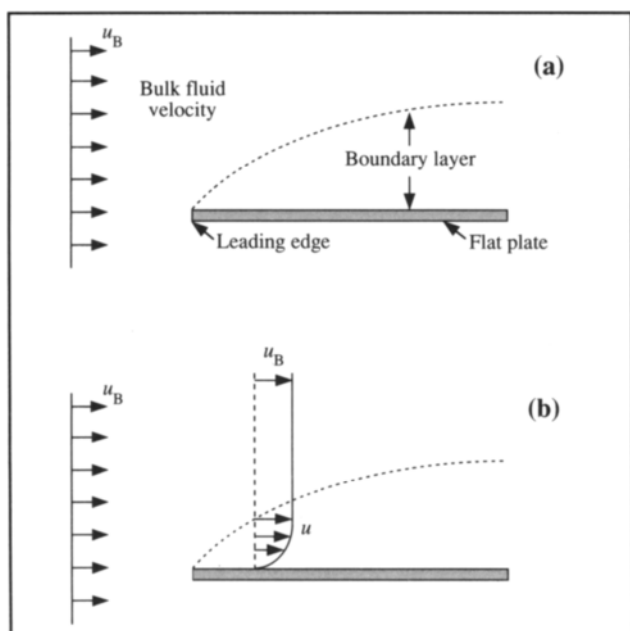
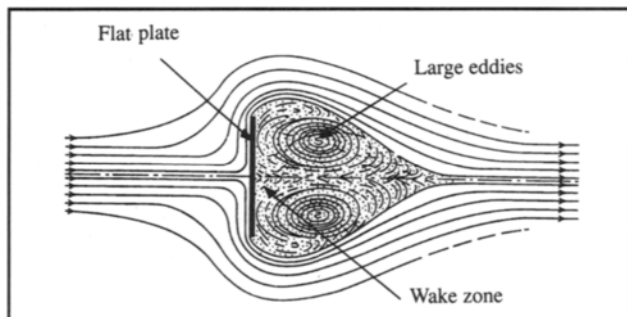


Figure 7.4 Flow around a flat plate aligned perpendicular to the direction of flow. (From W.L. McCabe and J.C. Smith, 1976, *Unit Operations of Chemical Engineering*, 3rd edn, McGraw-Hill, Tokyo.)



Boundary-layer separation such as that shown in Figure 7.4 occurs whenever an abrupt change in either magnitude or direction of fluid velocity is too great for the fluid to keep to a solid surface. It occurs in sudden contractions, expansions or bends in the flow channel, or when an object is placed across the flow path. Considerable energy is associated with the wake; this energy is taken from the bulk flow. Formation of wakes should be minimised if large pressure losses in the fluid are to be avoided; however, for some purposes such as promotion of mixing and heat transfer, boundary-layer separation may be desirable.

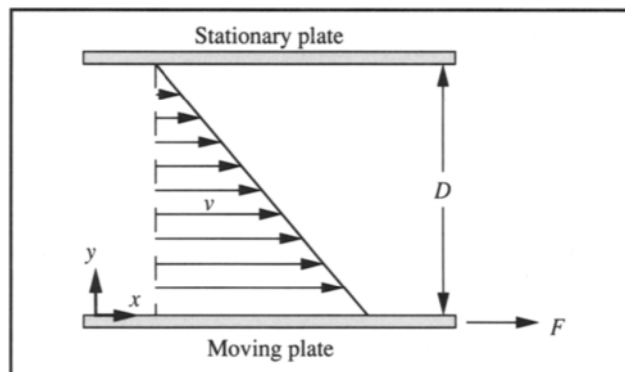
7.3 Viscosity

Viscosity is the most important property affecting flow behaviour of a fluid; viscosity is related to the fluid's resistance to motion. Viscosity has a marked effect on pumping, mixing, mass transfer, heat transfer and aeration of fluids; these in turn exert a major influence on bioprocess design and economics. Viscosity of fermentation fluids is affected by the presence of cells, substrates, products and air.

Viscosity is an important aspect of *rheology*, the science of deformation and flow. Viscosity is determined by relating the velocity gradient in fluids to the shear force causing flow to occur. This relationship can be explained by considering the development of laminar flow between parallel plates, as shown in Figure 7.5. The plates are a relatively short distance apart and, initially, the fluid between them is stationary. The lower plate is then moved steadily to the right with shear force F , while the upper plate remains fixed.

A thin film of fluid adheres to the surface of each plate. Therefore as the lower plate moves, fluid moves with it, while at the surface of the stationary plate the fluid velocity is zero. Due to viscous drag, fluid just above the moving plate is set

Figure 7.5 Velocity profile for Couette flow between parallel plates.



into motion, but with reduced speed. Layers further above also move; however, as we get closer to the top plate, the fluid is affected by viscous drag from the stationary film attached to the upper plate surface. As a consequence, fluid velocity between the plates decreases from that of the moving plate at $y = 0$, to zero at $y = D$. The velocity at different levels between the plates is indicated in Figure 7.5 by the arrows marked v . Laminar flow due to a moving surface as shown in Figure 7.5 is called *Couette flow*.

When steady Couette flow is attained in simple fluids, the velocity profile is as indicated in Figure 7.5; the slope of the line connecting all the velocity arrows is constant and proportional to the shear force F responsible for motion of the plate. The slope of the line connecting the velocity arrows is the velocity gradient, dv/dy . When the magnitude of the velocity gradient is directly proportional to F , we can write:

$$\frac{dv}{dy} \propto F. \quad (7.3)$$

If we define τ as the *shear stress*, equal to the shear force per unit area of plate:

$$\tau = \frac{F}{A} \quad (7.4)$$

it follows from Eq. (7.3) that:

$$\tau \propto \frac{dv}{dy} \quad (7.5)$$

This proportionality is represented by the equation:

$$\tau = -\mu \frac{dv}{dy} \quad (7.6)$$

where μ is the proportionality constant. Eq. (7.6) is called *Newton's law of viscosity*, and μ is the viscosity. The minus sign is necessary in Eq. (7.6) because the velocity gradient is always negative if the direction of F , and therefore τ , is considered positive. $-dv/dy$ is called the *shear rate*, and is usually denoted by the symbol $\dot{\gamma}$.

Viscosity as defined in Eq. (7.6) is sometimes called *dynamic viscosity*. As τ has dimensions $L^{-1}MT^{-2}$ and $\dot{\gamma}$ has dimensions T^{-1} , μ must therefore have dimensions $L^{-1}MT^{-1}$. The SI unit of viscosity is the Pascal second (Pa s), which is equal to 1 N s m^{-2} or $1 \text{ kg m}^{-1} \text{ s}^{-1}$. Other units include centipoise, cP. Direct conversion factors for viscosity units are given in Table A.9 in Appendix A. The viscosity of water at 20°C is approximately 1 cP or 10^{-3} Pa s. A modified form of viscosity is the *kinematic viscosity*, defined as μ/ρ where ρ is fluid density; kinematic viscosity is usually given the symbol ν .

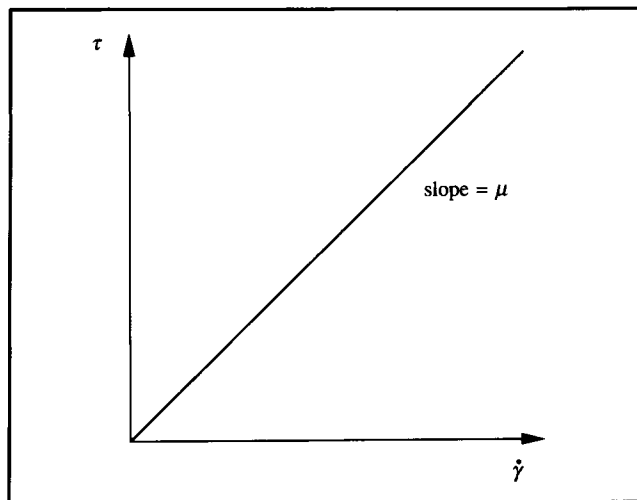
Fluids which obey Eq. (7.6) with constant μ are known as *Newtonian fluids*. The *flow curve* or *rheogram* for a Newtonian fluid is shown in Figure 7.6; the slope of a plot of τ versus $\dot{\gamma}$ is constant and equal to μ . The viscosity of Newtonian fluids remains constant despite changes in shear stress (force applied) or shear rate (velocity gradient). This does not imply that the viscosity is invariant; viscosity depends on many parameters such as temperature, pressure and fluid composition. However, under a given set of these conditions, viscosity of Newtonian fluids is independent of shear stress and shear rate. On the other hand, the ratio between shear stress and shear rate is not constant for *non-Newtonian* fluids, but depends on the shear force exerted on the fluid. Accordingly, μ in Eq. (7.6) is not a constant, and the velocity profile during Couette flow is not as simple as that shown in Figure 7.5.

7.4 Momentum Transfer

Viscous drag forces responsible for the velocity gradient in Figure 7.5 are the instrument of *momentum transfer* in fluids. At $y = 0$ the fluid acquires momentum in the x -direction due to motion of the lower plate. This fluid imparts some of its momentum to the adjacent layer of fluid above the plate, causing it also to move in the x -direction. Momentum in the x -direction is thus transmitted through the fluid in the y -direction.

Momentum transfer in fluids is represented by Eq. (7.6).

Figure 7.6 Flow curve for a Newtonian fluid.



To interpret this equation in terms of momentum transfer, shear stress τ is considered as the *flux* of x -momentum in the y -direction. The validity of this definition can be verified by checking the dimensions of momentum flux and shear stress.

Momentum is given by the expression Mv where M is mass and v is velocity; momentum has dimensions LMT^{-1} . *Flux means rate per unit area*; therefore momentum flux has dimensions $L^{-1}MT^{-2}$, which are also the dimensions of τ . So with τ representing momentum flux, according to Eq. (7.6), flux of momentum is directly proportional to the velocity gradient dv/dy . The negative sign in Eq. (7.6) means that momentum is transferred from regions of high velocity to regions of low velocity, i.e. in a direction opposite to the direction of increasing velocity. The magnitude of the velocity gradient dv/dy determines the magnitude of the momentum flux; dv/dy thus acts as the 'driving force' for momentum transfer.

Interpretation of fluid flow as momentum transfer perpendicular to the direction of flow may seem peculiar at first. The reason it is mentioned here is that there are many parallels between momentum transfer, heat transfer and mass transfer in terms of mechanism and equations. The analogy between these physical processes will be discussed further in Chapters 8 and 9.

7.5 Non-Newtonian Fluids

Most slurries, suspensions and dispersions are non-Newtonian, as are homogeneous solutions of long-chain polymers and other large molecules. Many fermentation

processes involve materials which exhibit non-Newtonian behaviour, such as starches, extracellular polysaccharides, and culture broths containing cell suspensions or pellets. Examples of non-Newtonian fluids are listed in Table 7.1.

Classification of non-Newtonian fluids depends on the relationship between the shear stress imposed on the fluid and the shear rate developed. Common types of non-Newtonian fluid include *pseudoplastic*, *dilatant*, *Bingham plastic* and *Casson plastic*; flow curves for these materials are shown in Figure 7.7. In each case, the ratio between shear stress and shear rate is not constant; nevertheless, this ratio for non-Newtonian fluids is often called the *apparent viscosity*, μ_a . Apparent viscosity is not a physical property of the fluid in the same way as Newtonian viscosity; it is dependent on the shear force exerted on the fluid. It is therefore meaningless to specify the apparent viscosity of a non-Newtonian fluid without noting the shear stress or shear rate at which it was measured.

7.5.1 Two-Parameter Models

Pseudoplastic and dilatant fluids obey the *Ostwald–de Waele* or *power law*:

$$\tau = K \dot{\gamma}^n \tag{7.7}$$

where τ is shear stress, K is the *consistency index*, $\dot{\gamma}$ is shear rate,

and n is the *flow behaviour index*. The parameters K and n characterise the rheology of power-law fluids. The flow behaviour index n is dimensionless; the dimensions of K , $L^{-1}MT^{n-2}$, depend on n . As indicated in Figure 7.7, when $n < 1$ the fluid exhibits pseudoplastic behaviour; when $n > 1$ the fluid is dilatant. $n = 1$ corresponds to Newtonian behaviour. For power-law fluids, apparent viscosity μ_a is expressed as:

$$\mu_a = \frac{\tau}{\dot{\gamma}} = K \dot{\gamma}^{n-1}. \tag{7.8}$$

For pseudoplastic fluids $n < 1$ and the apparent viscosity decreases with increasing shear rate; these fluids are said to exhibit *shear thinning*. On the other hand, apparent viscosity increases with shear rate for dilatant or *shear thickening* fluids.

Also included in Figure 7.7 are flow curves for plastic flow. Some fluids do not produce motion until some finite *yield stress* has been applied. For *Bingham plastic* fluids:

$$\tau = \tau_0 + K_p \dot{\gamma} \tag{7.9}$$

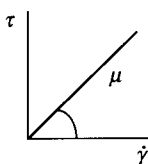
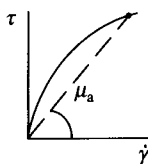
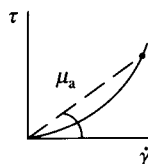
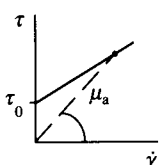
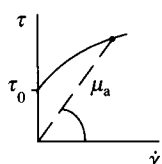
where τ_0 is the yield stress. Once the yield stress is exceeded and flow initiated, Bingham plastics behave like Newtonian fluids; a constant ratio K_p exists between change in shear stress

Table 7.1 Common non-Newtonian fluids

(Adapted from B. Atkinson and F. Mavituna, 1991, Biochemical Engineering and Biotechnology Handbook, 2nd edn, Macmillan, Basingstoke)

Fluid type	Examples
Newtonian	All gases, water, dispersions of gas in water, low-molecular-weight liquids, aqueous solutions of low-molecular-weight compounds
Non-Newtonian	
Pseudoplastic	Rubber solutions, adhesives, polymer solutions, some greases, starch suspensions, cellulose acetate, mayonnaise, some soap and detergent slurries, some paper pulps, paints, wallpaper paste, biological fluids
Dilatant	Some cornflour and sugar solutions, starch, quicksand, wet beach sand, iron powder dispersed in low-viscosity liquids, wet cement aggregates
Bingham plastic	Some plastic melts, margarine, cooking fats, some greases, toothpaste, some soap and detergent slurries, some paper pulps
Casson plastic	Blood, tomato sauce, orange juice, melted chocolate, printing ink

Figure 7.7 Classification of fluids according to their rheological behaviour. (From B. Atkinson and F. Mavituna, 1991, *Biochemical Engineering and Biotechnology Handbook*, 2nd edn, Macmillan, Basingstoke.)

Fluid	Flow curve	Equation	Apparent viscosity μ_a
Newtonian		$\tau = \mu \dot{\gamma}$	Constant. $\mu_a = \mu$
Pseudoplastic (power law)		$\tau = K \dot{\gamma}^n$ $n < 1$	Decreases with increasing shear rate. $\mu_a = K \dot{\gamma}^{n-1}$
Dilatant (power law)		$\tau = K \dot{\gamma}^n$ $n > 1$	Increases with increasing shear rate. $\mu_a = K \dot{\gamma}^{n-1}$
Bingham plastic		$\tau = \tau_0 + K_p \dot{\gamma}$	Decreases with increasing shear rate when yield stress τ_0 is exceeded. $\mu_a = \frac{\tau_0}{\dot{\gamma}} + K_p$
Casson plastic		$\tau^{1/2} = \tau_0^{1/2} + K_p \dot{\gamma}^{1/2}$	Decreases with increasing shear rate when yield stress τ_0 is exceeded. $\mu_a = \left[\left(\frac{\tau_0}{\dot{\gamma}} \right)^{1/2} + K_p \right]^2$

and change in shear rate. Another common plastic behaviour is described by the *Casson* equation:

$$\tau^{1/2} = \tau_0^{1/2} + K_p \dot{\gamma}^{1/2}. \quad (7.10)$$

Once the yield stress is exceeded, the behaviour of Casson fluids is pseudoplastic. Several other equations describing non-Newtonian flow have also been developed [1].

7.5.2 Time-Dependent Viscosity

When a shear force is exerted on some fluids, the apparent viscosity either increases or decreases with duration of the force. If apparent viscosity increases with time, the fluid is said to be *rheopectic*; rheopectic fluids are relatively rare in occurrence. If apparent viscosity decreases with time the fluid is *thixotropic*. Thixotropic behaviour is not uncommon in cultures containing fungal mycelia or extracellular microbial polysaccharides,

and appears to be related to reversible 'structure' effects associated with the orientation of cells and macromolecules in the fluid. Rheological properties vary during application of the shear force because it takes time for equilibrium to be established between structure breakdown and re-development.

7.5.3 Viscoelasticity

Viscoelastic fluids, such as some polymer solutions, exhibit an elastic response to changes in shear stress. When shear forces are removed from a moving viscoelastic fluid, the direction of flow may be reversed due to elastic forces developed during flow. Most viscoelastic fluids are also pseudoplastic and may exhibit other rheological characteristics such as yield stress. Mathematical analysis of viscoelasticity is therefore quite complex.

7.6 Viscosity Measurement

Many different instruments or *viscometers* are available for measurement of rheological properties. Space does not permit a detailed discussion of viscosity measurement in this text; further information can be found elsewhere [1–5]. Specifications for commercial viscometers are also available [2, 3, 6].

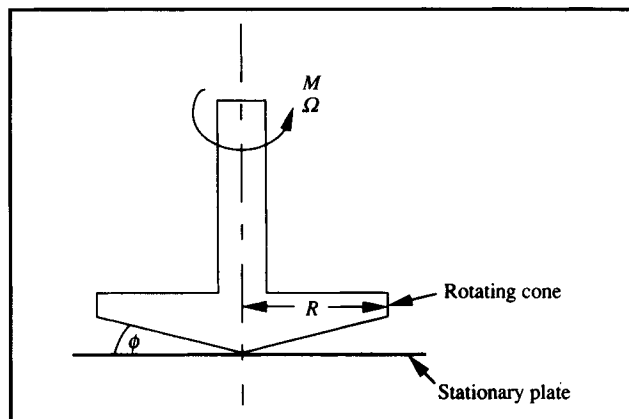
The objective of any viscosity measurement system is to create a controlled flow situation where easily measured parameters can be related to the shear stress τ and shear rate $\dot{\gamma}$. Usually the fluid is set in rotational motion and the parameters measured are torque M and angular velocity Ω . These quantities are used to calculate τ and $\dot{\gamma}$ using approximate formulae which depend on the geometry of the apparatus. Once obtained, τ and $\dot{\gamma}$ are applied for evaluation of viscosity in Newtonian fluids, or viscosity parameters such as K , n , and τ_0 for non-Newtonian fluids. Equations for particular viscometers can be found in other texts [2, 3, 6]. Most modern viscometers use microprocessors to provide automatic read-out of parameters such as shear stress, shear rate and apparent viscosity.

Three types of viscometer commonly used in bioprocessing applications are the cone-and-plate viscometer, the coaxial-cylinder rotary viscometer, and the impeller viscometer.

7.6.1 Cone-and-Plate Viscometer

The cone-and-plate viscometer consists of a flat horizontal plate and an inverted cone, the apex of which is near contact with the plate as shown in Figure 7.8. The angle ϕ between the plate and cone is very small, usually less than 3° , and the fluid to be measured is located in this small gap. Large cone angles are not used for routine work for a variety of reasons, the most important being that analysis of the results for non-Newtonian fluids would be complex or impossible. The cone is rotated in the fluid, and the angular velocity Ω and torque M measured. It

Figure 7.8 Cone-and-plate viscometer.

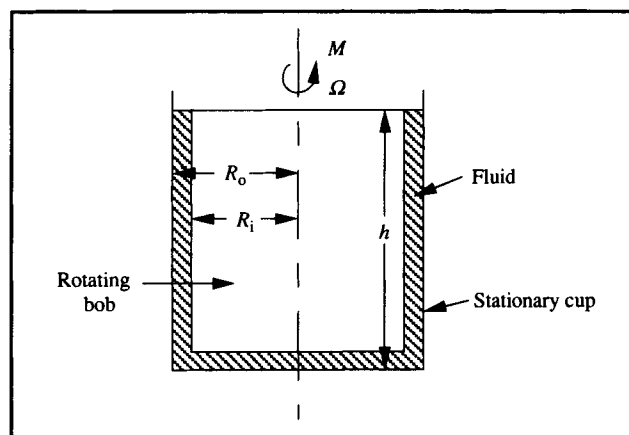


is generally assumed that the fluid undergoes streamline flow in concentric circles about the axis of rotation of the cone. This assumption is not always valid; however for ϕ less than about 3° , the error is small. Temperature can be controlled by circulating water from a constant-temperature bath beneath the plate; this is effective provided the speed of rotation is not too high. Limitations of the cone-and-plate method for measurement of flow properties, including corrections for edge and temperature effects and turbulence, are discussed elsewhere [3].

7.6.2 Coaxial-Cylinder Rotary Viscometer

The coaxial-cylinder viscometer is a popular rotational device for measuring rheological properties. A typical coaxial-cylinder instrument is shown in Figure 7.9. This device is designed to shear fluid located in the annulus between two

Figure 7.9 Coaxial-cylinder viscometer.



concentric cylinders, one of which is held stationary while the other rotates. A cylindrical bob of radius R_i is suspended in sample fluid held in a stationary cylindrical cup of radius R_o . Liquid covers the bob to a height h from the bottom of the outer cup. As the inner cylinder rotates, the angular velocity Ω and torque M are measured. In some designs the outer cylinder rather than the inner bob rotates; in any case the motion is relative with magnitude Ω .

Coaxial-cylinder viscometers are used with Newtonian or non-Newtonian fluids. When flow is non-Newtonian, shear rate is not related simply to rotational speed and geometric factors and the calculations can be somewhat complicated. Limitations of the coaxial-cylinder method, including corrections for end effects, slippage, temperature variation and turbulence, are discussed elsewhere [2, 3, 6].

7.6.3 Impeller Viscometer

Because of difficulties (discussed in Section 7.6.4) associated with standard rotational viscometers, modified apparatus employing turbine and other impellers have been developed for rheological study of fermentation fluids [7, 8]. Instead of the rotating inner cylinder of Figure 7.9, a small impeller on a stirring shaft is used to shear the fluid sample. As the impeller rotates slowly in the fluid, accurate measurements of torque M and rotational speed N_i are made. For a turbine impeller under laminar-flow conditions, the following relationships apply [8]:

$$\dot{\gamma} = k N_i \quad (7.11)$$

and

$$\tau = \frac{2\pi M k}{64 D_i^3} \quad (7.12)$$

where D_i is the impeller diameter and k is a constant which depends on the geometry of the impeller (see Section 7.13). The relationship of Eq. (7.11) is experimentally derived; for turbine impellers k is approximately 10. The exact value of k for a particular apparatus is evaluated using liquid with a known viscosity–shear rate relationship.

Because Eqs (7.11) and (7.12) are valid only for laminar flow, viscosity measurements using the impeller method must be carried out under laminar flow conditions. Accordingly, if a turbine impeller is used, Re_i should not be greater than about 10. As Re_i is directly proportional to N_i which, from Eq. (7.11), determines the value of $\dot{\gamma}$, the necessity for laminar flow limits the range of shear rates that may be investigated. This

range can be extended if anchor or helical agitators are used instead of the conventional disc turbine (see Figure 7.15 for illustrations of these impellers). Laminar flow is maintained at higher Re_i with anchor and helical impellers; the value of k in Eq. (7.11) is also greater so that higher shear rates can be tested. As Eq. (7.12) is valid only for turbine impellers, the relationship between τ and M must be modified if alternative impellers are used. Application of anchor and helical impellers for viscosity measurement is described in the literature [9, 10].

Because the flow patterns in stirred fluids are relatively complex, analysis of data from impeller viscometers is not absolutely rigorous from a rheological point of view. However, the procedure is based on well-proven and widely-accepted empirical correlations and is considered the most reliable technique for mycelial broths. As discussed below, the method eliminates many of the operating problems associated with conventional viscometers for study of fermentation fluids.

7.6.4 Use of Viscometers With Fermentation Broths

Measurement of rheological properties is difficult when the fluid contains suspended solids such as cells. Viscosity of fermentation broths often appears time-dependent due to artifacts associated with the measuring device. With viscometers such as the cone-and-plate and coaxial cylinder, the following problems can arise:

- (i) the suspension is effectively centrifuged in the viscometer so that a region with lower cell density is formed near the rotating surface;
- (ii) solids settle out of suspension during measurement;
- (iii) large particles about the same size as the gap in the coaxial viscometer, or about the same size as the cone angle in the cone-and-plate, interfere with accurate measurement;
- (iv) the measurement will depend somewhat on the orientation of particles in the flow field;
- (v) some types of particle will begin to flocculate or deflocculate when the shear field is applied; and
- (vi) particles can be destroyed during measurement.

The first problem is particularly troublesome because it is hard to detect and can give viscosity results which are too small by a factor of up to 100. For suspensions containing solids, the impeller method offers significant advantages compared with other measurement procedures. Stirring by the impeller prevents sedimentation, promotes uniform distribution of solids through the fluid, and reduces time-dependent changes in suspension composition. The method has proved very useful for rheological measurements on microbial suspensions [5].

Table 7.2 Rheological properties of microbial and plant-cell suspensions

(Adapted from M. Charles, 1978, *Technical aspects of the rheological properties of microbial cultures*. Adv. Biochem. Eng. 8, 1–62)

Culture	Shear rate (s ⁻¹)	Viscometer	Comments	Reference
<i>Saccharomyces cerevisiae</i> (pressed cake diluted with water)	2–100	rotating spindle	Newtonian below 10% solids ($\mu < 4\text{--}5$ cP); pseudoplastic above 10% solids	[11]
<i>Aspergillus niger</i> (washed cells in buffer)	0–21.6	rotating spindle (guard removed)	pseudoplastic	[12]
<i>Penicillium chrysogenum</i> (whole broth)	1–15	turbine impeller	Casson plastic	[8]
<i>Penicillium chrysogenum</i> (whole broth)	not given	coaxial cylinder	Bingham plastic	[13]
<i>Penicillium chrysogenum</i> (whole broth)	not given	coaxial cylinder	pseudoplastic; K and n vary with CO ₂ content of inlet gas	[14]
<i>Endomyces</i> sp. (whole broth)	not given	coaxial cylinder	pseudoplastic; K and n vary over course of batch culture	[15]
<i>Streptomyces noursei</i> (whole broth)	4–28	rotating spindle (guard removed)	Newtonian in batch culture; viscosity 40 cP after 96 h	[16]
<i>Streptomyces aureofaciens</i> (whole broth)	2–58	rotating spindle/ coaxial cylinder	initially Bingham plastic due to high starch concentration in medium; changes to Newtonian as starch is broken down; increasingly pseudoplastic as mycelium concentration increases	[17]
<i>Aureobasidium pullulans</i> (whole broth)	10.2–1020	coaxial cylinder	Newtonian at the beginning of culture; increasingly pseudoplastic as concentration of product (exopolysaccharide) increases	[18]
<i>Xanthomonas campestris</i>	0.0035–100	cone-and-plate	pseudoplastic; K increases continually; n levels off when xanthan concentration reaches 0.5%; cell mass (max 0.6%) has relatively little effect on viscosity	[4]

contd.

Culture	Shear rate (s^{-1})	Viscometer	Comments	Reference
<i>Cellulomonas uda</i> (whole broth)	0.8–100	anchor impeller	shredded newspaper used as substrate; broth pseudoplastic with constant n until end of cellulose degradation; Newtonian thereafter	[10]
<i>Nicotiana tabacum</i> (whole broth)	not given	rotating spindle	pseudoplastic	[19]
<i>Datura stramonium</i> (whole broth)	0–1000	rotating spindle/ parallel-plate	pseudoplastic and viscoelastic, with yield stress	[20]

7.7 Rheological Properties of Fermentation Broths

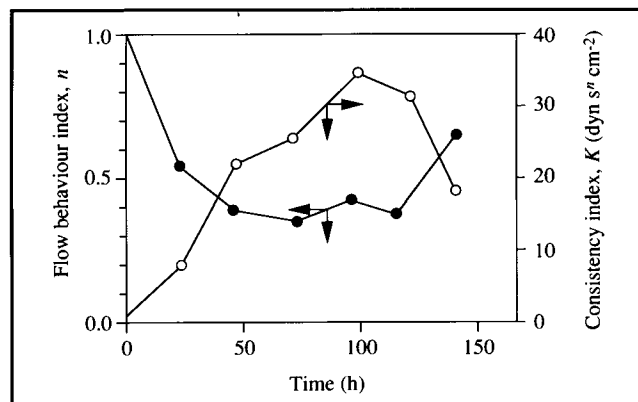
Rheological data have been reported for a range of fermentation fluids. This information has been obtained using various viscometers and measurement techniques; however, operating problems such as particle settling and broth centrifugation have been ignored in many cases. Most mycelial suspensions have been modelled as pseudoplastic fluids or, if there is a yield stress, Bingham or Casson plastic. On the other hand, the rheology of dilute broths and cultures of yeast and non-chain-forming bacteria is usually Newtonian. Rheological properties of some microbial and plant-cell suspensions are listed in Table 7.2. In most cases, the results are valid over only a limited range of shear conditions which is largely dictated by the choice of viscometer. When the fermentation produces extracellular polymers such as in microbial production of pullulan and xanthan, the rheological characteristics of the broth depend strongly on the properties and concentration of these materials.

7.8 Factors Affecting Broth Viscosity

The rheology of fermentation broths often changes throughout batch culture. For broths obeying the power law, the flow behaviour index n and consistency index K can vary substantially depending on culture time. As an example, Figure 7.10 shows changes in n and K during batch culture of *Endomyces* [15]; the culture starts off Newtonian ($n = 1$) but quickly becomes pseudoplastic ($n < 1$). K rises steadily throughout most of the batch period; this gives a direct indication of the increase in apparent viscosity since, as indicated in Eq. (7.8), μ_a is directly proportional to K .

Changes in rheology of fermentation broths are caused by variation of one or more of the following properties:

Figure 7.10 Variation of rheological parameters in *Endomyces* fermentation. (From H. Taguchi and S. Miyamoto, 1966, Power requirement in non-Newtonian fermentation broth. *Biotechnol. Bioeng.* 8, 43–54.)



- (i) cell concentration;
- (ii) cell morphology, including size, shape and mass;
- (iii) flexibility and deformability of cells;
- (iv) osmotic pressure of the suspending fluid;
- (v) concentration of polymeric substrate;
- (vi) concentration of polymeric product; and
- (vii) rate of shear.

Some of these parameters are considered below.

7.8.1 Cell Concentration

The viscosity of a suspension of spheres in Newtonian liquid can be predicted using the *Vand equation*:

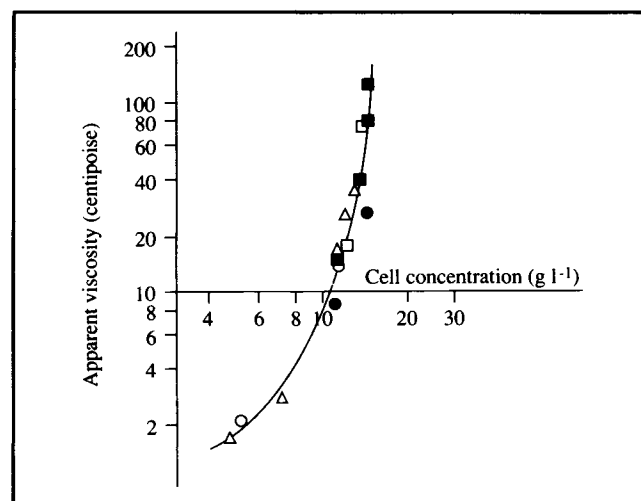
$$\mu = \mu_L (1 + 2.5\psi + 7.25\psi^2) \quad (7.13)$$

where μ_L is the viscosity of the suspending liquid and ψ is the volume fraction of solids. Eq. (7.13) has been found to hold for yeast and spore suspensions up to 14 vol% solids [21]. Many other cell suspensions do not obey Eq. (7.13); cell concentration can have a much stronger influence on rheological properties than is predicted by the Vand equation. As an example, Figure 7.11 shows how cell concentration affects the apparent viscosity of various pseudoplastic plant-cell suspensions [22]; a doubling in cell concentration causes the apparent viscosity to increase by a factor of up to 90. Similar results have been found for mould pellets in liquid culture [23]. When viscosity is so strongly dependent on cell concentration, a steep drop in viscosity can be achieved by diluting the broth with water or medium. Periodic removal of part of the culture and refilling with fresh medium reduces the viscosity and improves fluid flow in viscous fermentations.

7.8.2 Cell Morphology

Morphological characteristics exert a profound influence on broth rheology. Disperse filamentous growth produces 'structure' in the broth, resulting in pseudoplasticity, yield-stress behaviour, or both. On the other hand, broths containing pelleted cells tends to be more Newtonian, depending on how

Figure 7.11 Relationship between apparent viscosity and cell concentration for plant-cell suspensions forming aggregates of various size. (○) *Cudrania tricuspidata* 44–149 μm ; (●) *C. tricuspidata* 149–297 μm ; (□) *Vinca rosea* 44–149 μm ; (■) *V. rosea* 149–297 μm ; (△) *Nicotiana tabacum* 150–800 μm . (From H. Tanaka, 1982, Oxygen transfer in broths of plant cells at high density. *Biotechnol. Bioeng.* 24, 425–442.)



readily the pellets are deformed during flow. The extent of branching of hyphal cells can also affect rheology; cells with a high branching frequency are generally less flexible than non-branching cells and produce higher viscosities.

Sample rheological data for pseudoplastic mycelial broths are shown in Figure 7.12 [9]. Pelleted mycelia are more closely Newtonian in behaviour than filamentous cells; the flow behaviour index n for pellets is closer to unity. As indicated in Figure 7.12(b), the consistency index, and therefore the apparent viscosity, can differ by several orders of magnitude depending on cell morphology.

7.8.3 Osmotic Pressure

Osmotic pressure of the culture medium affects cell turgor pressure. This in turn affects the hyphal flexibility of filamentous cells; increased osmotic pressure gives a lower turgor pressure making the hyphae more flexible. Improved hyphal flexibility reduces broth viscosity, and can also have a marked effect on yield stress.

7.8.4 Product and Substrate Concentrations

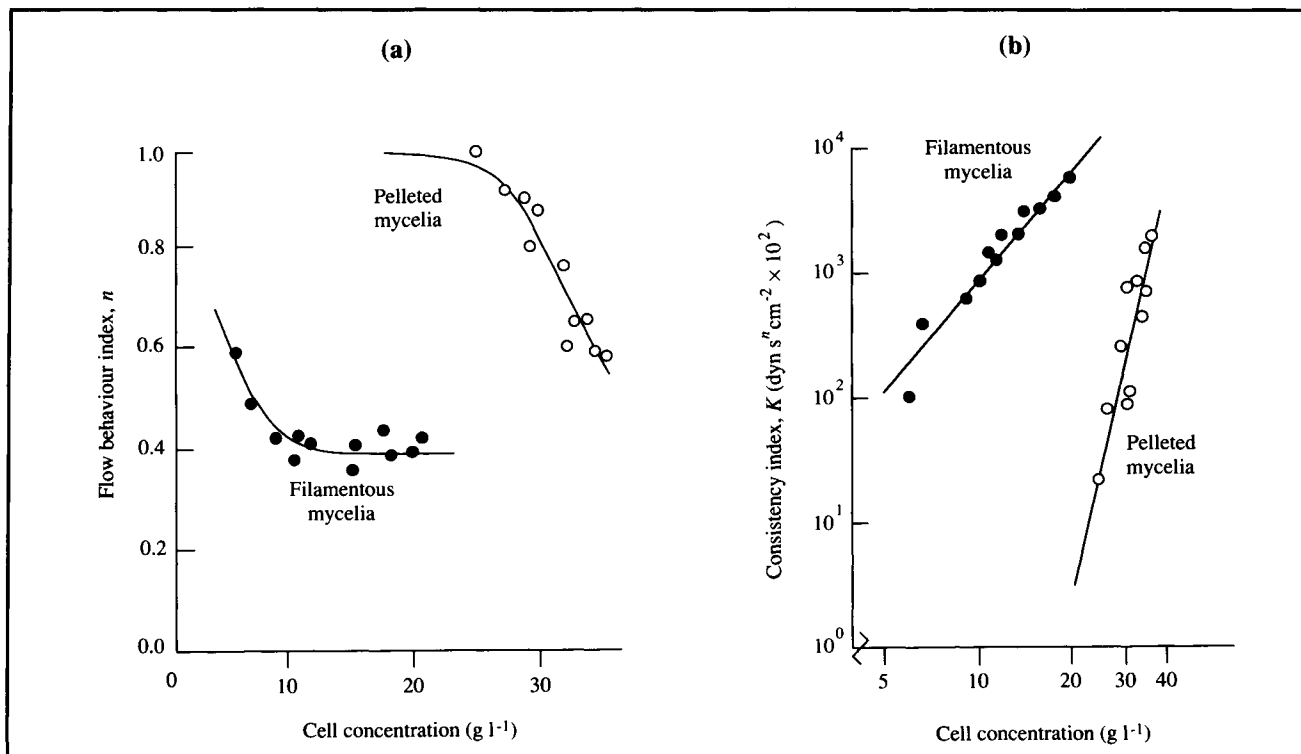
When the product of fermentation is a polymer, continued excretion in batch culture raises the broth viscosity. For example, during production of exopolysaccharide by *Aureobasidium pullulans*, apparent viscosity measured at a shear rate of 1 s^{-1} can reach as high as 24 000 cP [18]. Cell concentration usually has a negligible effect on overall viscosity in these fermentations; the rheological properties of the fluid are dominated by the dissolved polymer. Other products having a similar effect on culture rheology include dextran, alginate and xanthan gum.

In contrast, when the fermentation medium contains polymeric substrate such as starch, apparent viscosity will decrease as the fermentation progresses and the polymer is broken down. There could also be a progressive change from non-Newtonian to Newtonian behaviour. In mycelial fermentations this change is usually short lived; as the cells grow and develop a structured filamentous network, the broth becomes increasingly pseudoplastic and viscous.

7.9 Mixing

Mixing is a physical operation which reduces non-uniformities in fluid by eliminating gradients of concentration, temperature and other properties. Mixing is accomplished by interchanging material between different locations to produce a mingling of components. If a system is perfectly

Figure 7.12 Effect of morphology on the rheology of mycelial broths. (From J.H. Kim, J.M. Lebeault and M. Reuss, 1983, Comparative study on rheological properties of mycelial broth in filamentous and pelleted forms. *Eur. J. Appl. Microbiol. Biotechnol.* 18, 11–16.)



mixed, there is a random homogeneous distribution of system properties. Mixing involves:

- (i) blending soluble components of the medium such as sugars;
- (ii) dispersing gases such as air through the liquid in the form of small bubbles;
- (iii) maintaining suspension of solid particles such as cells;
- (iv) where necessary, dispersing immiscible liquids to form an emulsion or suspension of fine drops; and
- (v) promoting heat transfer to or from the liquid.

Mixing is one of the most important operations in bioprocessing. To create the optimal environment for fermentation, bioreactors must provide the cells access to all substrates, including oxygen in aerobic culture. It is not enough to just fill the fermenter with nutrient-rich medium; unless the culture is mixed, zones of nutrient depletion will develop as the cells rapidly consume the materials they need. This problem is heightened if mixing does not maintain a uniform suspension of biomass; substrate concentrations can quickly drop to zero in areas where cells settle out of suspension. Another important

function of mixing is heat transfer. Bioreactors must be capable of transferring heat to or from the broth rapidly enough so that the desired temperature is maintained. Cooling water is used to take up excess heat from fermentations; the rate of heat transfer from the broth through the walls of the vessel to the cooling water depends on mixing conditions in the vessel. The effectiveness of mixing depends in turn on the rheological properties of the culture fluid.

Mixing can be achieved in many different ways. In this chapter we will concentrate on the most common mixing technique in bioprocessing: mechanical agitation using an impeller.

7.9.1 Mixing Equipment

Mixing is usually carried out in a stirred tank, such as that shown in Figure 7.13. Stirred tanks are usually cylindrical in shape. If possible, the base of the tank is rounded at the edges rather than angled; this eliminates sharp corners and pockets into which fluid currents may not penetrate and discourages formation of stagnant regions. Mixing is achieved using an

impeller mounted in the tank; for use with Newtonian fluids, the ratio of tank diameter to impeller diameter is normally about 3:1. The impeller is usually positioned overhead on a centrally-located *stirrer shaft*. Sometimes, stirrer shafts are designed to enter at the bottom of the vessel; the disadvantage of this arrangement is that leaks can develop if the seal between the shaft and the tank floor is not perfect. The stirrer shaft is driven rapidly by the stirrer motor; the effect of the rotating impeller is to pump the liquid and create a regular flow pattern. Liquid is forced away from the impeller, circulates through the vessel, and periodically returns to the impeller region. For efficient mixing with a single impeller, the depth of liquid in the tank should be no more than 1.0–1.25 times the tank diameter.

Baffles, which are vertical strips of metal mounted against the wall of the tank, are installed to reduce vortexing and swirling of the liquid. Baffles are attached to the tank by means of welded brackets; four equally-spaced baffles are usually sufficient to prevent vortex formation. The optimum baffle width depends on the impeller design and fluid viscosity but is of the order $1/10$ – $1/12$ the tank diameter. For low-viscosity liquids, baffles are usually attached perpendicular to the wall as illustrated in Figure 7.14(a). Alternatively, as shown in Figures 7.14(b) and (c), baffles can be mounted away from the wall with a clearance of about $1/50$ the tank diameter, or set at an angle. These arrangements prevent sedimentation and development of stagnant zones at the inner edge of the baffle during mixing of viscous cell suspensions.

Many impeller designs are available for mixing applications; a small selection is illustrated in Figure 7.15. Further details and descriptions of impellers can be found elsewhere [24, 25]. Some impellers have flat blades; in others such as the

propeller and helical screw, the slope of the individual blades varies continuously. Specification of the *pitch* of a propeller blade refers to its properties as a segment of a screw; pitch is the advance per revolution. Choice of impeller depends on several factors, including viscosity of the liquid to be mixed and sensitivity of the system to mechanical shear. The recommended viscosity ranges for a number of common impellers are indicated in Figure 7.16. For low-to-medium-viscosity liquids, propellers and flat-blade turbines are recommended. The most frequently-used impeller in the fermentation industry is the 6-flat-blade disc-mounted turbine shown in Figure 7.15; this impeller is also known as the *Rushton turbine*.

Figure 7.14 Baffle arrangements. (a) Baffles attached to the wall for low-viscosity liquids. (b) Baffles set away from the wall for moderate-viscosity liquids. (c) Baffles set away from the wall and at an angle for high-viscosity liquids. (From F.A. Holland and F.S. Chapman, 1966, *Liquid Mixing and Processing in Stirred Tanks*, Reinhold, New York.)

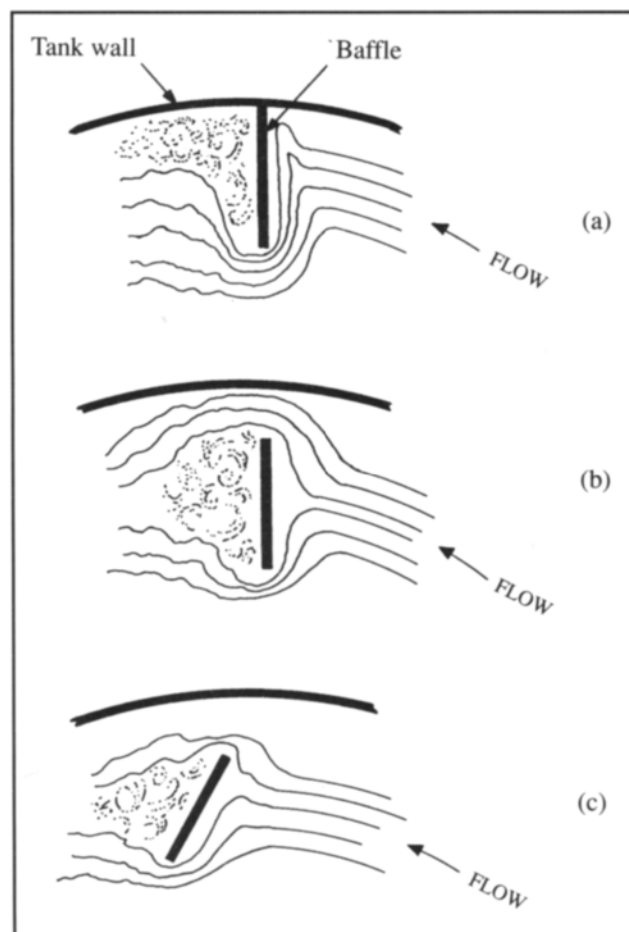


Figure 7.13 Typical configuration of a stirred tank.

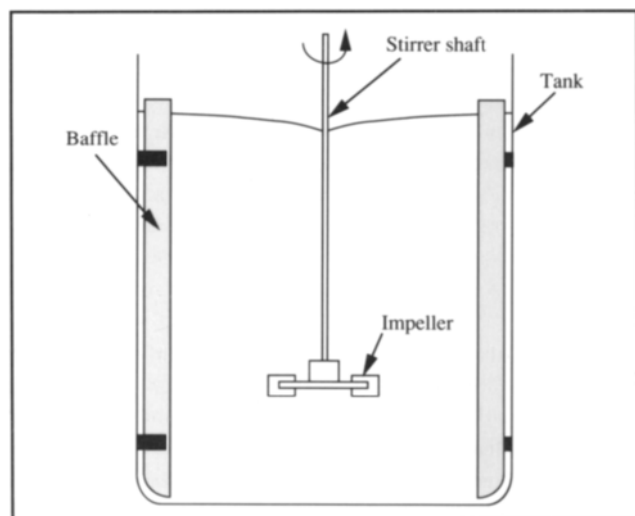


Figure 7.15 Impeller designs.

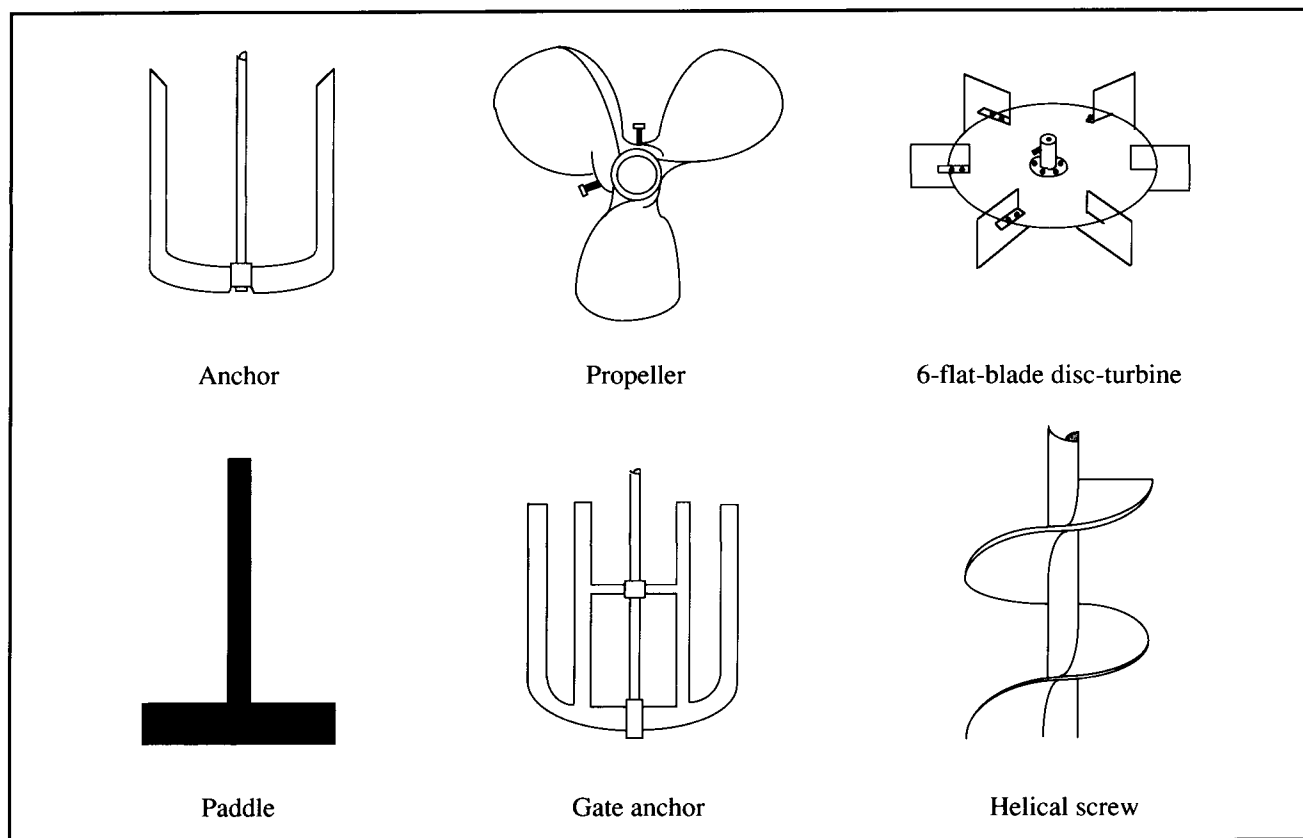
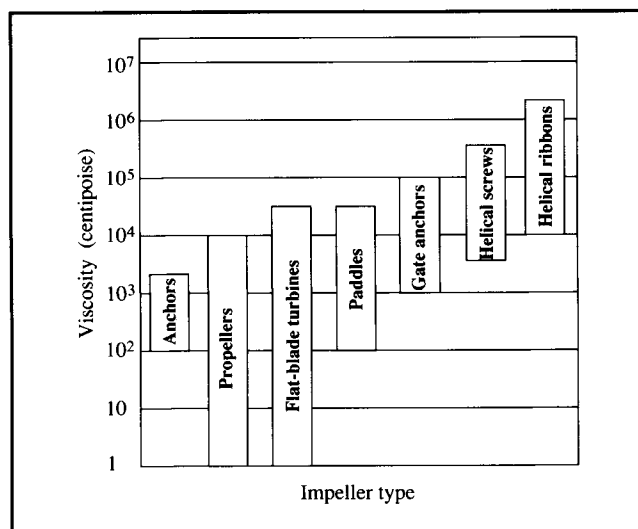


Figure 7.16 Viscosity ranges for different impellers. (From F.A. Holland and F.S. Chapman, 1966, *Liquid Mixing and Processing in Stirred Tanks*, Reinhold, New York.)



7.9.2 Flow Patterns in Agitated Tanks

The flow pattern in an agitated tank depends on the impeller design, the properties of the fluid, and the size and geometric proportions of the vessel, baffles and agitator.

Although most stirrers are rotational in action, simple circular flow of liquid around the shaft such as that illustrated in Figure 7.17(a) is generally disadvantageous and should be avoided. In circular flow, liquid moves in a streamline fashion and there is little mixing between fluid at different heights in the tank. Circular flow also leads to vortex development as shown in Figure 7.17(b). At high impeller speeds, the vortex may reach down to the impeller so that gas from the surrounding atmosphere is drawn into the liquid; this is generally undesirable as it produces very high mechanical stresses in the stirrer shaft, bearings and seal. Prevention of circular flow has a high priority in design of mixing systems, and is usually achieved by installing baffles which interrupt the circular flow pattern and create turbulence in the fluid.

As well as circular flow, motion of fluid occurs in the radial direction (i.e. from the stirrer out to the sides of the tank and

back again) and in the axial direction (i.e. up and down the height of the tank). Axial and radial flows are generated at the impeller, and it is these components of fluid motion which are primarily responsible for bulk mixing. Impellers are broadly classified as *axial flow* or *radial flow* depending on the direction of liquid leaving the impeller; some impellers have both axial- and radial-flow characteristics.

7.9.2.1 Radial-flow impellers

Radial-flow impellers have blades which are parallel to the vertical axis of the stirrer shaft and tank; the six-flat-blade disc turbine of Figure 7.15 is an example. The flow pattern set up by high-speed rotation of a radial-flow impeller is illustrated in

Figure 7.17 (a) Circular flow in an unbaffled stirred tank viewed from above. (b) Vortex formation during circular flow. (From J.H. Rushton, E.W. Costich and H.J. Everett, 1950, Power characteristics of mixing impellers: Part I. *Chem. Eng. Prog.* 46, 395–404.)

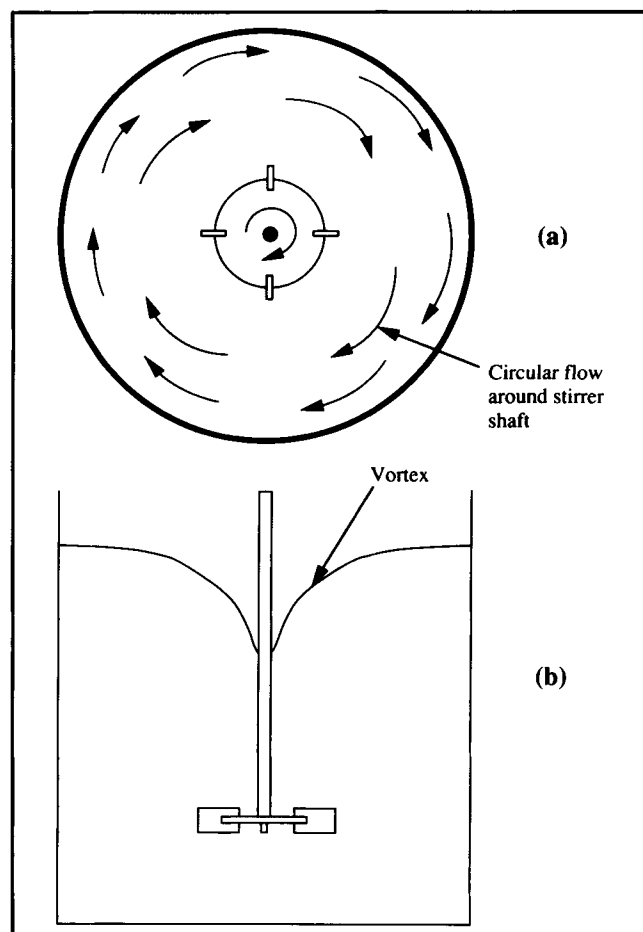


Figure 7.18. Liquid is driven radially from the impeller against the walls of the tank where it divides into two streams, one flowing up to the top of the tank and the other flowing down to the bottom. These streams eventually reach the central axis of the tank and are drawn back to the impeller. Radial-flow impellers also set up circular flow which must be reduced by baffles.

7.9.2.2 Axial-flow impellers

In general, axial-flow impellers have blades which make an angle of less than 90° to the plane of rotation and promote axial top-to-bottom motion. Propellers are axial-flow devices, as are pitched-blade turbines such as that shown in Figure 7.19. The flow pattern set up by a typical axial-flow impeller is illustrated in Figure 7.20. Fluid leaving the impeller is driven downwards until it is deflected from the floor of the vessel. It then spreads out over the floor and flows up along the wall before being drawn back to the impeller. Axial-flow impellers are particularly useful when strong vertical currents are required. For example, if the fluid contains solids, a strong axial flow of liquid leaving the impeller will discourage settling at the bottom of the tank.

7.9.3 Mechanism of Mixing

As illustrated in Figures 7.18 and 7.20, large liquid-circulation loops develop in stirred vessels. For mixing to be effective, fluid circulated by the impeller must sweep the entire vessel in a reasonable time. In addition, the velocity of fluid leaving the impeller must be sufficient to carry material into the most remote parts of the tank. Turbulence must also be developed in the fluid; mixing is certain to be poor unless flow in the tank is turbulent. All these factors are important in mixing, which can be described as a combination of three physical processes:

- (i) distribution;
- (ii) dispersion; and
- (iii) diffusion.

Distribution is sometimes called *macromixing*; diffusion is also called *micromixing*. Dispersion can be classified as either micro- or macromixing depending on the scale of fluid motion.

The pattern of bulk fluid flow in a vessel stirred by a radial-flow impeller is shown in detail in Figure 7.21. Near the impeller there is a region of high turbulence where fluid currents converge and exchange material. Away from the impeller, flow is slower and largely streamline; mixing in these regions is much less intense than near the impeller. The contents of the vessel are recirculated through the mixing zone in a very regular manner due to the periodic pumping action of the impeller.

Let us consider what happens when a small amount of liquid dye is dropped onto the top of the fluid in Figure 7.21.

Figure 7.18 Flow pattern produced by a radial-flow impeller in a baffled tank. (a) side view; (b) bottom view. (From J.H. Rushton, E.W. Costich and H.J. Everett, 1950, Power characteristics of mixing impellers. Part I. *Chem. Eng. Prog.* 46, 395–404.)

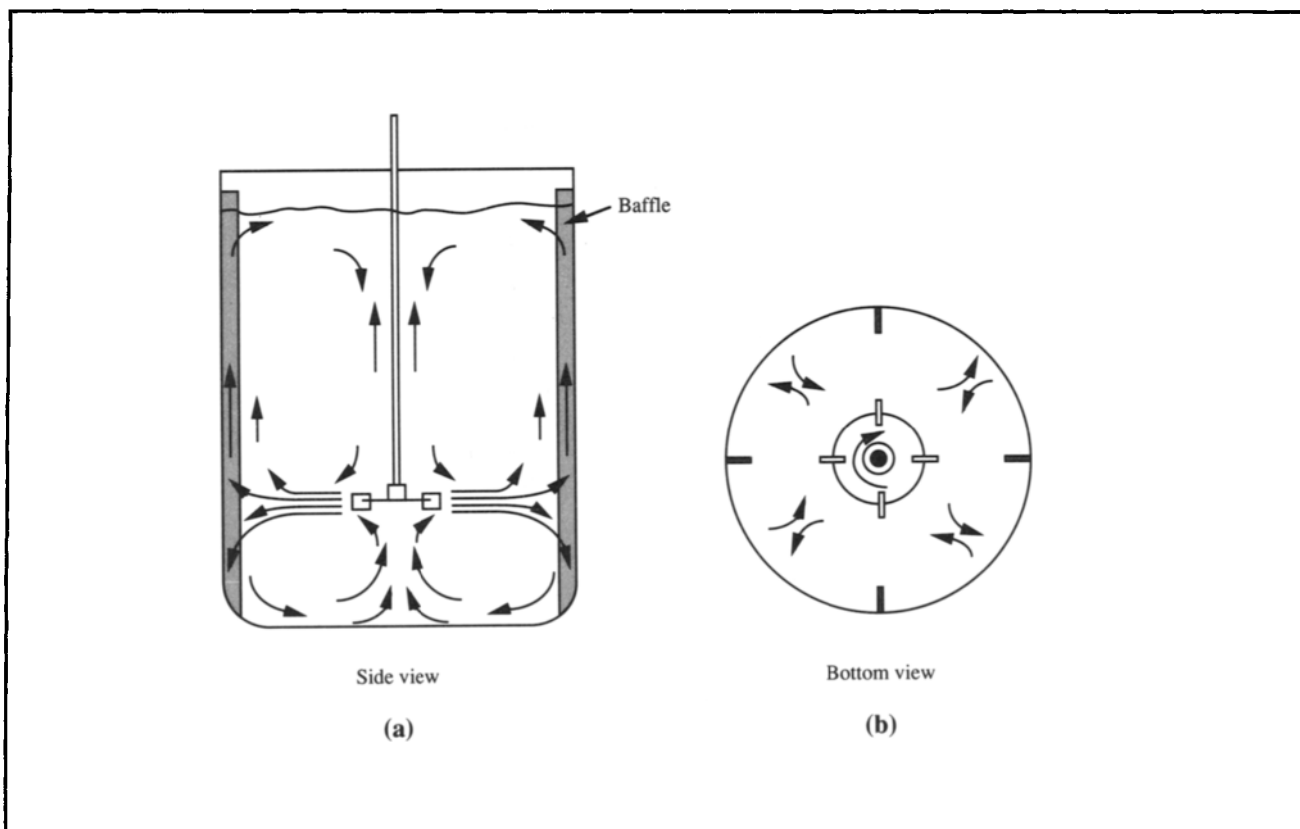
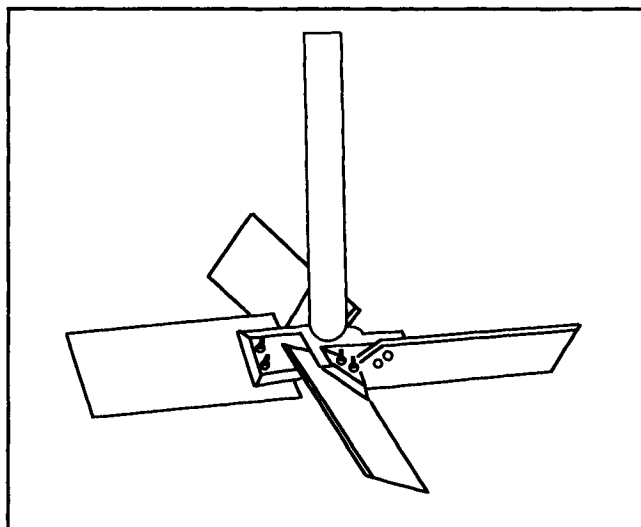


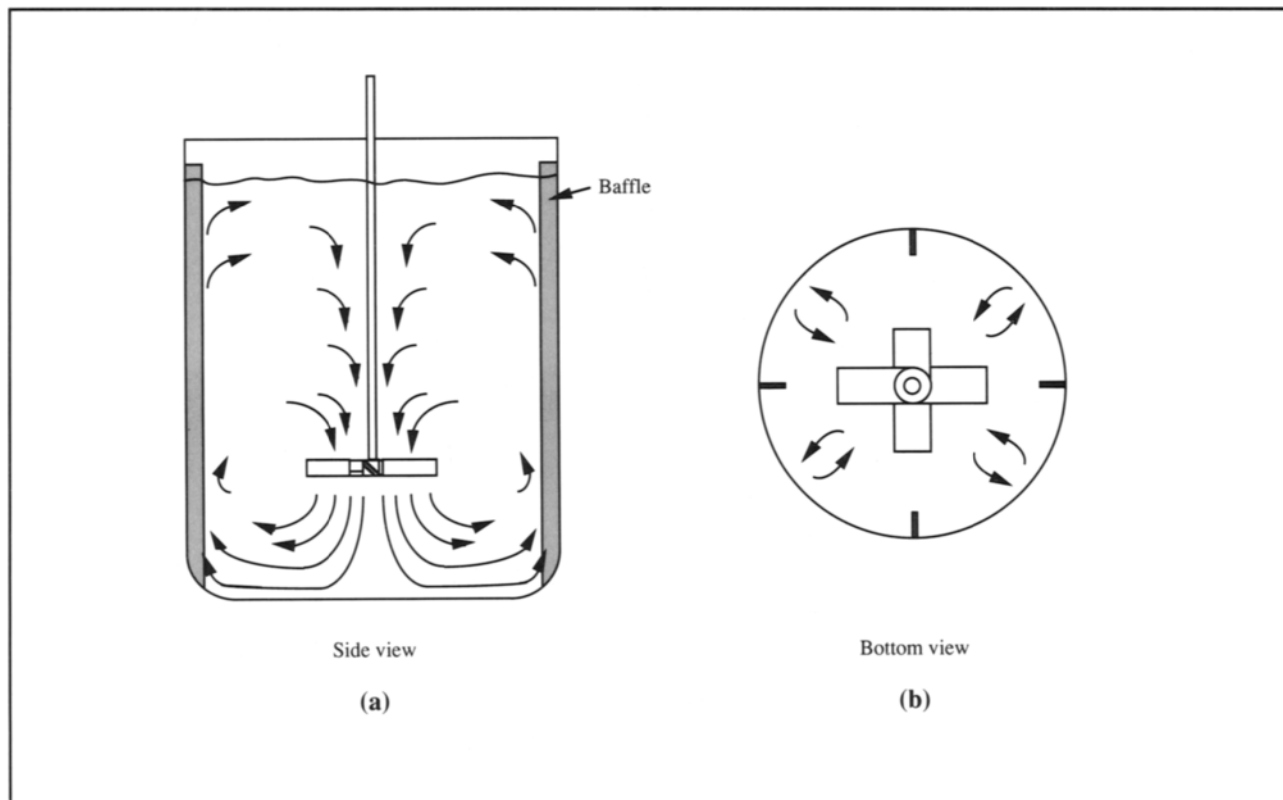
Figure 7.19 Pitched-blade turbine.



First, the dye is swept by circulating currents down to the impeller. At the impeller there is vigorous and turbulent motion of fluid; the dye is mechanically dispersed into smaller volumes and distributed between the large circulation loops. These smaller parcels of dye are then carried around the tank, dispersing all the while into those parts of the system not yet containing dye. Returning again to the impeller, the dye aliquots are broken up into even smaller volumes for further distribution. After a time, dye is homogeneously distributed throughout the tank with uniform concentration.

The process whereby dye is transported to all regions of the vessel by bulk circulation currents is called *distribution*. Distribution is an important process in mixing, but can be relatively slow. In large tanks, the size of the circulation paths is also large and the time taken to traverse them is long; this, together with the regularity of fluid pumping at the impeller, inhibits rapid mixing. Accordingly, *distribution is often the slowest step in the mixing process*. If the rotational speed of the impeller is sufficiently high, superimposed on the distribution

Figure 7.20 Flow pattern produced by an axial-flow impeller in a baffled tank. (a) side view; (b) bottom view. (From J.H. Rushton, E.W. Costich and H.J. Everett, 1950, Power characteristics of mixing impellers. Part I. *Chem. Eng. Prog.* 46, 395–404.)



process is turbulence. Turbulent flow occurs when fluid no longer travels along streamlines but moves erratically in the form of cross-currents. The kinetic energy of turbulent fluid is directed into regions of rotational flow called *eddies*; masses of eddies of various size coexist during turbulent flow. Large eddies are continuously formed by action of the stirrer; these break down into small eddies which produce even smaller eddies. Eddies, like spinning tops, possess kinetic energy which is transferred to eddies of decreasing size. When the eddies become very small they can no longer sustain rotational motion and their kinetic energy is dissipated as heat. At steady state in a mixed tank, most of the energy from the stirrer is dissipated through eddies as heat; energy lost as fluid collides with the tank walls is generally negligible.

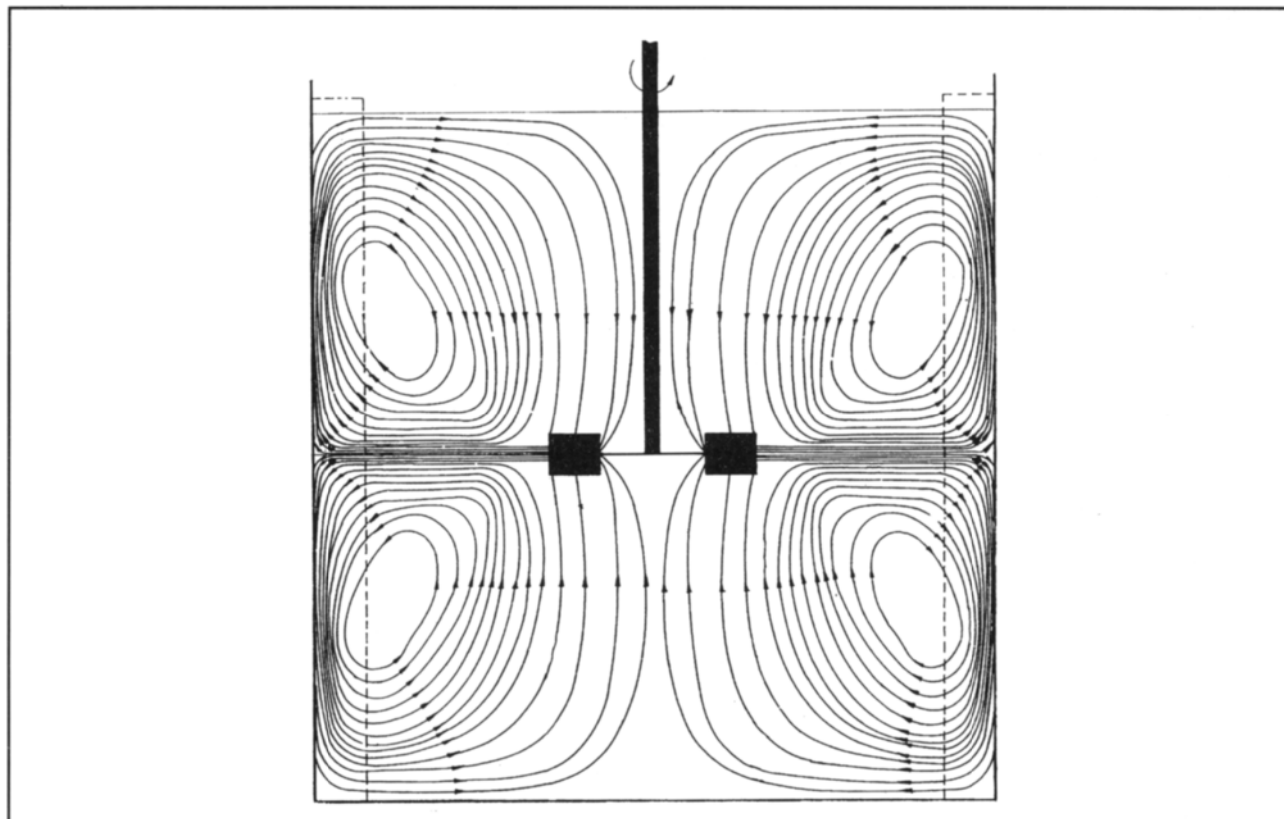
The process of breaking up bulk flow into smaller and smaller eddies is called *dispersion*; dispersion facilitates rapid transfer of material throughout the vessel. The degree of homogeneity possible as a result of dispersion is limited by the size of the smallest eddies which may be formed in a particular fluid. This size is given approximately as the *Kolmogorov scale of mixing*, or *scale of turbulence*, λ :

$$\lambda = \left(\frac{\nu^3}{\epsilon} \right)^{1/4} \quad (7.14)$$

where λ is the characteristic dimension of the smallest eddies, ν is the kinematic viscosity of the fluid, and ϵ is the local rate of turbulent energy dissipation per unit mass of liquid. At steady state, the rate of energy dissipation by turbulence is equal to the power supplied by the impeller. According to Eq. (7.14), the greater the power input to the fluid, the smaller are the eddies. λ is also dependent on viscosity; at a given power input, smaller eddies are produced in low-viscosity fluids. For low-viscosity liquids such as water, λ is usually in the range 30–100 μm . For such fluids, this is the smallest scale of mixing achievable by dispersion.

Within eddies there is little mixing because rotational flow occurs in streamlines. Therefore, to achieve mixing on a scale smaller than the Kolmogorov scale, we must rely on *diffusion*. Molecular diffusion is generally regarded as a slow process; however, over small distances it can be accomplished quite rapidly. Within eddies of 30–100 μm diameter, homogeneity is achieved in about 1 s for low-viscosity fluids. Consequently,

Figure 7.21 Flow pattern developed by a centrally-positioned radial-flow impeller. (From R.M. Voncken, J.W. Rotte and A.Th. ten Houten, Circulation model for continuous-flow, turbine-stirred, baffled tanks. In: *Mixing—Theory Related to Practice*, Proc. Symp. 10, AIChE–IChE Joint Meeting, London, June 1965.)



if power input to a stirred vessel produces eddies of this dimension, mixing on a molecular scale is accomplished virtually simultaneously.

7.9.4 Assessing Mixing Effectiveness

As explained in the above section, to achieve rapid mixing in a stirred tank the agitator must provide good bulk circulation or macromixing. Micromixing at or near the molecular scale is also important, but usually occurs relatively quickly compared with macromixing. Assessment of mixing effectiveness can therefore be reduced in most cases to measuring the rate of bulk flow.

Mixing time is a useful parameter for assessing mixing efficiency and is applied to characterise bulk flow in fermenters and reactors. The mixing time t_m is the time required to achieve a given degree of homogeneity starting from the completely segregated state. It can be measured by injecting a tracer into the vessel and following its concentration at a fixed point in the tank. Tracers in common use include acids, bases

and concentrated salt solutions; corresponding detectors are pH probes and conductivity cells. Mixing time can also be determined by measuring the temperature response after addition of a small quantity of heated liquid.

Let us assume that a small pulse of tracer is added to fluid in a stirred tank already containing tracer material at concentration C_i . When flow in the system is circulatory, the tracer concentration measured at some fixed point in the tank will follow a pattern similar to that shown in Figure 7.22. Before mixing is complete, a relatively high concentration will be detected every time the bulk flow brings tracer to the measurement point. The peaks in concentration will be separated by a period approximately equal to the average time taken for fluid to traverse one bulk circulation loop. In stirred vessels this period is called the *circulation time*, t_c . After several circulations the desired degree of homogeneity is reached.

Definition of the mixing time t_m depends on the degree of homogeneity required. Usually, mixing time is defined as the time after which the concentration of tracer differs from the

Figure 7.22 Concentration response after dye is injected into a stirred tank.

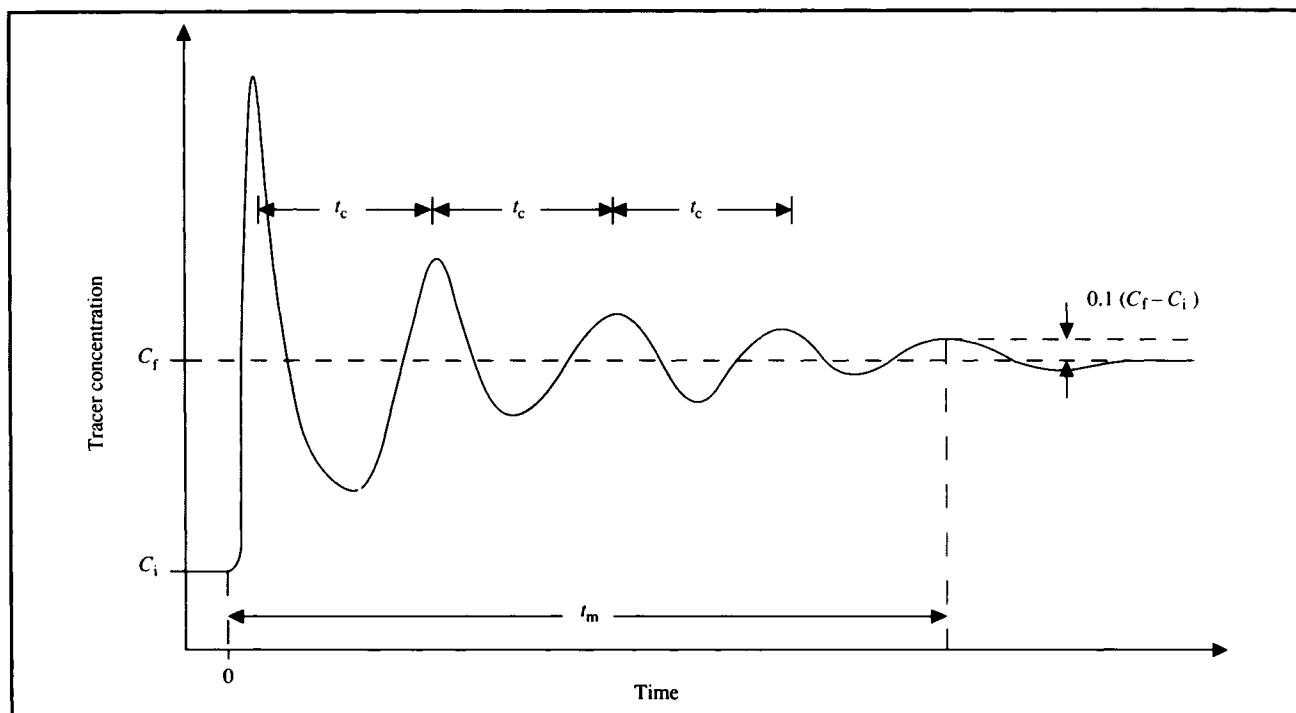
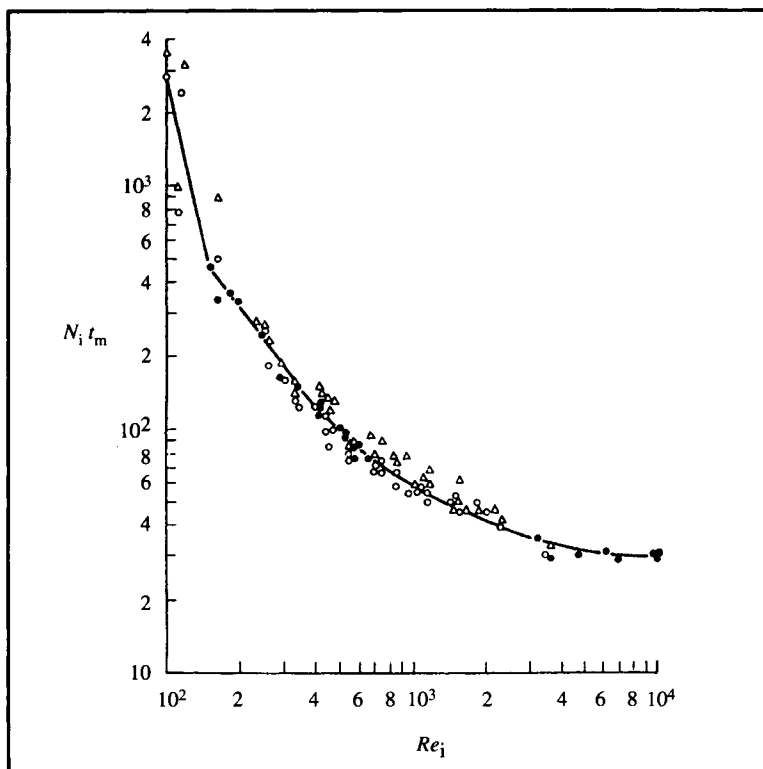


Figure 7.23 Variation of mixing time with Reynolds number for a six-blade Rushton turbine in a baffled tank. The impeller is located one-third the tank diameter off the floor of the vessel; the impeller diameter is one-third the tank diameter. The liquid height is equal to the tank diameter; the tank has four baffles of width one-tenth the tank diameter. Several measurement techniques and tank sizes were used: (●) thermal method, 1.8-m diameter vessel; (○) thermal method, 0.24-m vessel; (△) decoloration method, 0.24-m vessel. (Reprinted from C.J. Hoogendoorn and A.P. den Hartog, Model studies on mixers in the viscous flow region, *Chem. Eng. Sci.* 22, 1689–1699. Copyright 1967, with permission from Pergamon Press Ltd, Oxford.)



final concentration C_f by less than 10% of the total concentration difference ($C_f - C_i$). However, there is no single, universally-applied definition of mixing time; sometimes deviations greater or less than 10% are specified. Nevertheless, at t_m the tracer concentration is relatively steady and the fluid composition approaches uniformity. For a single-phase liquid in a stirred tank with several baffles and small impeller, there is an approximate relationship between mixing time and circulation time [26]:

$$t_m = 4t_c. \quad (7.15)$$

Industrial-scale stirred vessels with working volumes between 1 and 100 m³ have mixing times between about 30 and 120 s, depending on conditions.

Intuitively, we can predict that mixing time in stirred tanks will depend on variables such as the size of the tank and impeller, fluid properties such as viscosity, and stirrer speed. The relationship between mixing time and several of these variables has been determined experimentally for different impellers [27]; results for a Rushton turbine in a baffled tank are shown in Figure 7.23. The dimensionless product $N_i t_m$ is plotted as a function of the

impeller Reynolds number Re_i defined in Eq. (7.2); t_m is the mixing time based on a 10% deviation from final conditions, and N_i is rotational speed of the stirrer. $N_i t_m$ represents the number of stirrer rotations required to homogenise the liquid. At low Reynolds number, $N_i t_m$ increases significantly with decreasing Re_i . However, as Reynolds number is increased above about 5×10^3 , $N_i t_m$ approaches a constant value which persists at high Re_i . For Rushton turbines, this constant value can be estimated using the following relationship [28]:

$$N_i t_m = \frac{1.54 V}{D_i^3} \quad \text{at high } Re_i \quad (7.16)$$

where V is liquid volume and D_i is impeller diameter. Thus, $N_i t_m$ at high Reynolds number depends only on the size of the tank and stirrer. The relationship between $N_i t_m$ and Re_i for most impellers is similar to that shown in Figure 7.23 for Rushton turbines. In some systems a slight increase in $N_i t_m$ with increasing Re_i has been reported; however, in practice we can assume $N_i t_m$ reaches a constant value [29]. With $N_i t_m$ constant, mixing time reduces in direct proportion to increase in stirrer speed.

Example 7.1 Estimation of mixing time

A fermentation broth with viscosity 10^{-2} Pa s and density 1000 kg m^{-3} is agitated in a 2.7 m^3 baffled tank using a Rushton turbine with diameter 0.5 m and stirrer speed 1 s^{-1} . Estimate the mixing time.

Solution:

From Eq. (7.2):

$$Re_i = \frac{1 \text{ s}^{-1} (0.5 \text{ m})^2 1000 \text{ kg m}^{-3}}{10^{-2} \text{ kg m}^{-1} \text{ s}^{-1}} = 2.5 \times 10^4.$$

$Re_i > 5 \times 10^3$; therefore $N_i t_m$ is constant and can be calculated from Eq. (7.16):

$$N_i t_m = \frac{1.54 (2.7 \text{ m}^3)}{(0.5 \text{ m})^3} = 33.3.$$

Therefore:

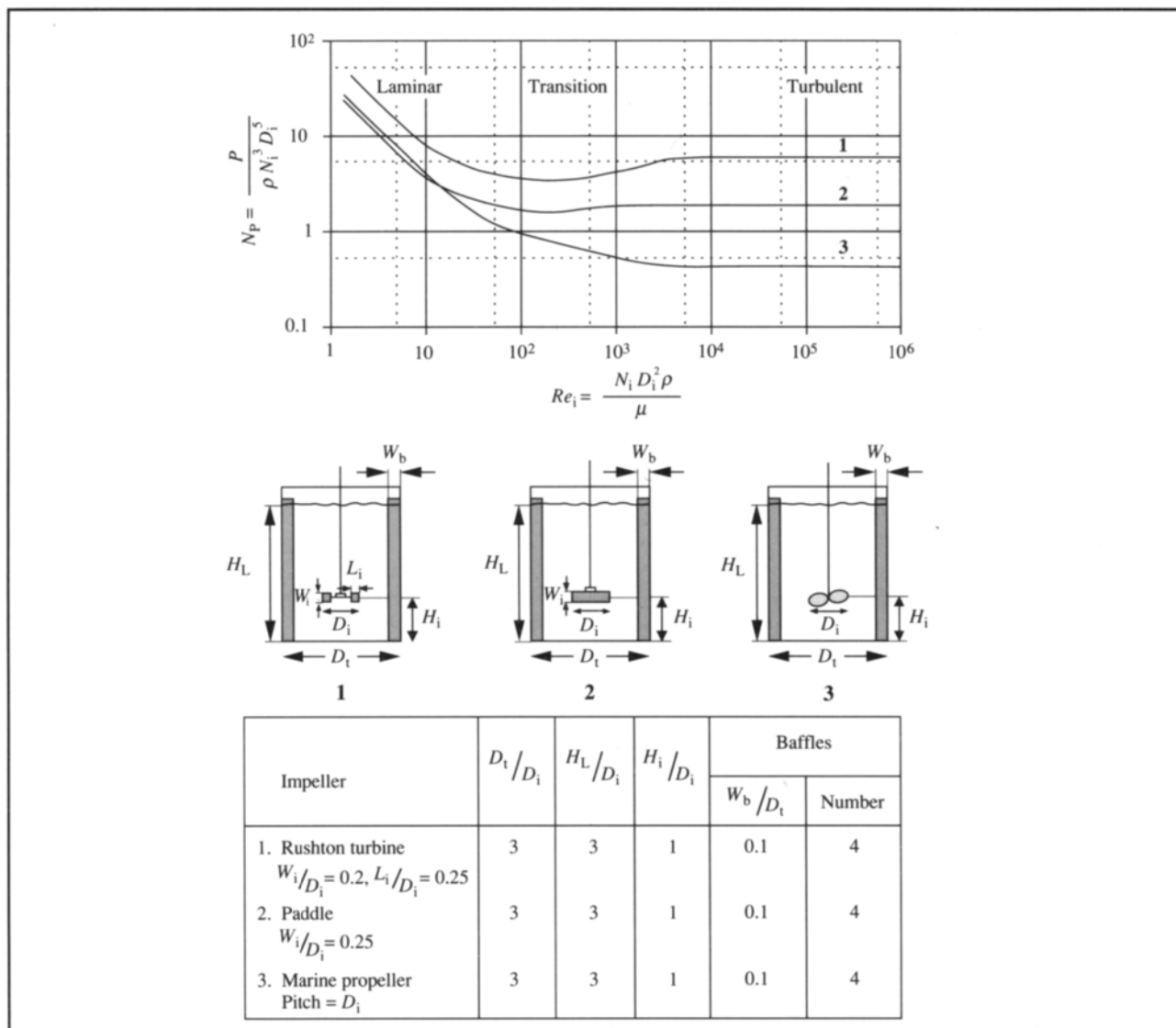
$$t_m = \frac{33.3}{1 \text{ s}^{-1}} = 33.3 \text{ s}.$$

The mixing time is about 33 s.

For rapid and effective mixing, t_m should be as small as possible. From Eq. (7.16) we can conclude that, in a tank of fixed volume, mixing time is reduced if we use a large impeller and high stirring speed. However, as the power requirements for mixing are also dependent on impeller diameter and stirrer

speed, it is not always possible to achieve small mixing times without consuming enormous amounts of energy, especially in large vessels. Relationships between power requirements, mixing time, tank size, fluid properties and other operating variables are explored further in the following section.

Figure 7.24 Correlation between power number and Reynolds number for Rushton turbine, paddle and marine propeller without sparging. (From J.H. Rushton, E.W. Costich and H.J. Everett, 1950, Power characteristics of mixing impellers. Parts I and II. *Chem. Eng. Prog.* 46, 395–404, 467–476.)



7.10 Power Requirements for Mixing

Usually, electrical power is used to drive impellers in stirred vessels. For a given stirrer speed, the power required depends on the resistance offered by the fluid to rotation of the impeller. Average power consumption per unit volume for industrial bioreactors ranges from 10 kW m⁻³ for small vessels (ca. 0.1 m³), to 1–2 kW m⁻³ for large vessels (ca. 100 m³). Friction in the stirrer motor gearbox and seals reduces the energy transmitted to the fluid; therefore, the electrical power

consumed by stirrer motors is always greater than the mixing power by an amount depending on the efficiency of the drive. Energy costs for operation of stirrers in bioreactors are an important consideration in process economics. General guidelines for calculating power requirements are discussed below.

7.10.1 Ungassed Newtonian Fluids

Mixing power for non-aerated fluids depends on the stirrer speed, the impeller diameter and geometry, and properties of

the fluid such as density and viscosity. The relationship between these variables is usually expressed in terms of dimensionless numbers such as the impeller Reynolds number Re_i and the *power number* N_p . N_p is defined as:

$$N_p = \frac{P}{\rho N_i^3 D_i^5} \quad (7.17)$$

where P is power, ρ is fluid density, N_i is stirrer speed and D_i is impeller diameter. The relationship between Re_i and N_p has been determined experimentally for a range of impeller and tank configurations. The results for five impeller designs: Rushton turbine, paddle, marine propeller, anchor and helical ribbon, are shown in Figures 7.24 and 7.25 [29–31]. Once the value of N_p is known, the power required is calculated from Eq. (7.17) as:

$$P = N_p \rho N_i^3 D_i^5 \quad (7.18)$$

For a given impeller, the general relationship between power number and Reynolds number depends on the flow regime in the tank. Three flow regimes can be identified in Figures 7.24 and 7.25:

- (i) *Laminar regime*. The laminar regime corresponds to $Re_i < 10$ for many impellers; for stirrers with very small wall-clearance such as the anchor and helical-ribbon mixer, laminar flow persists until $Re_i = 100$ or greater. In the laminar regime:

$$N_p \propto \frac{1}{Re_i} \quad \text{or} \quad P = k_1 \mu N_i^2 D_i^3 \quad (7.19)$$

where k_1 is a proportionality constant. Values of k_1 for the impellers illustrated in Figures 7.24 and 7.25 are listed in Table 7.3 [29]. Power required for laminar flow is independent of the density of the fluid but directly proportional to fluid viscosity.

- (ii) *Turbulent regime*. Power number is independent of Reynolds number in turbulent flow. Therefore:

$$P = N'_p \rho N_i^3 D_i^5 \quad (7.20)$$

where N'_p is the constant value of the power number in the turbulent regime. Approximate values of N'_p for the impellers of Figures 7.24 and 7.25 are listed in Table 7.3 [29]. N'_p for turbines is significantly higher than for most other impellers, indicating that turbines transmit more

power to the fluid than other designs. Power required for turbulent flow is independent of the viscosity of the fluid but proportional to fluid density. The turbulent regime is fully developed at $Re_i > 10^3$ or 10^4 for most small impellers in baffled vessels. For the same impellers in vessels without baffles, the power curves are somewhat different from those shown in Figure 7.24. Without baffles, turbulence is not fully developed until $Re_i > 10^5$; even then the value of N'_p is reduced to between $1/2$ and $1/10$ that with baffles [29–31].

- (iii) *Transition regime*. Between laminar and turbulent flow lies the transition regime. Both density and viscosity affect power requirements in this regime. There is usually a gradual transition from laminar to fully-developed turbulent flow in stirred tanks; the flow pattern and Reynolds-number range for transition depend on system geometry.

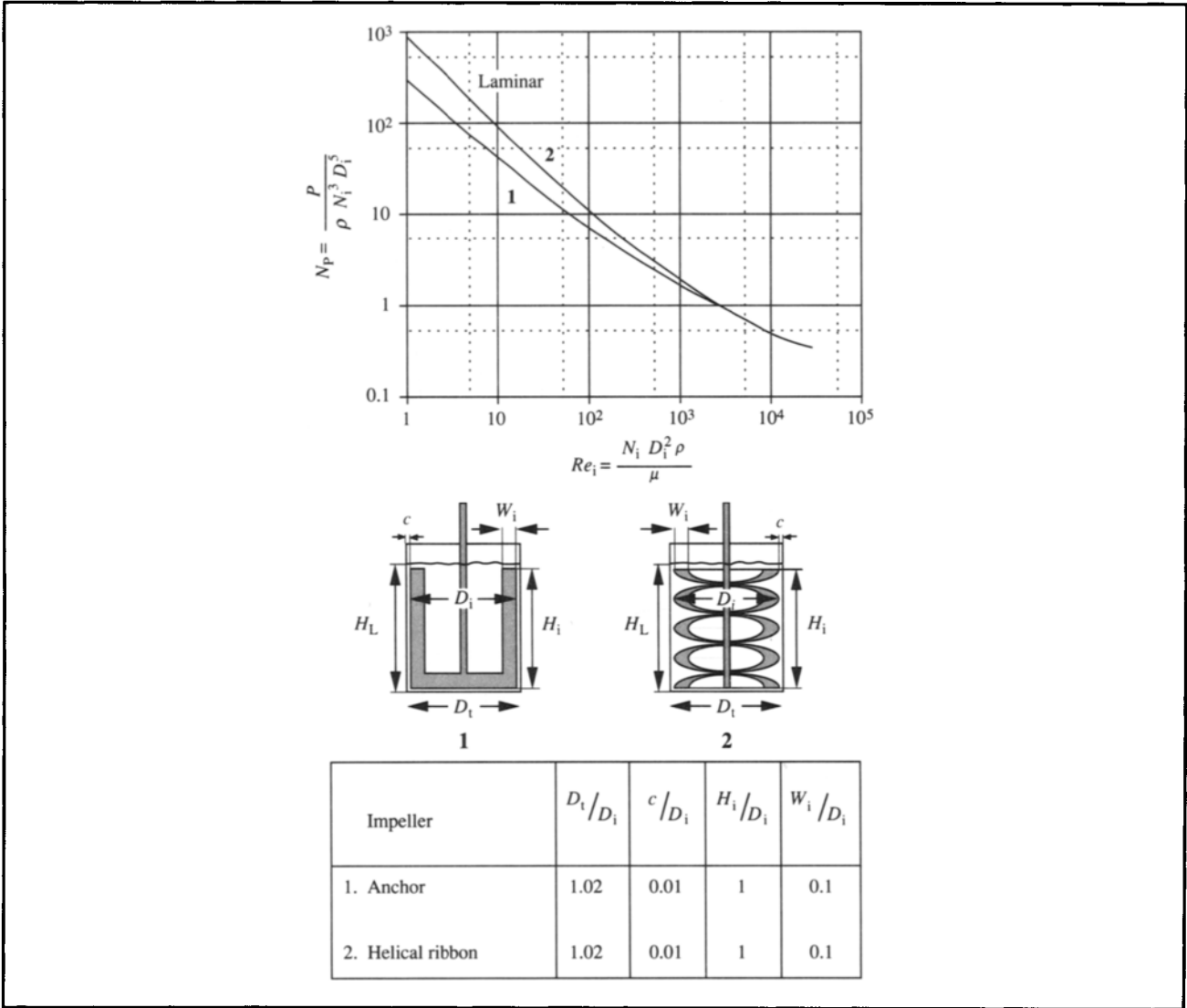
Eqs (7.19) and (7.20) express the strong dependence of power consumption on stirrer diameter and, to a lesser extent, stirrer speed. Small changes in impeller size have a large effect on power requirements, as would be expected from dependency on impeller diameter raised to the third or fifth power. In the turbulent regime, a 10% increase in impeller diameter increases the power required by more than 60%; a 10% increase in stirrer speed raises the power required by over 30%.

Frictional drag, and therefore the power required for stirring, depend on the geometry of the impeller and configuration of the tank. The curves of Figures 7.24 and 7.25 refer to the particular geometries specified and will change if the number or size of baffles, the number, length, width, pitch or angle of blades on the impeller, the height of impeller from the bottom of the tank, etc. are changed. For a Rushton turbine in a baffled tank under fully turbulent conditions ($Re_i > 10^4$), the power number lies between about 2 and 10 depending on these parameters [25, 31]. For propellers, impeller pitch has a significant effect on power number in the turbulent regime [25].

Table 7.3 Constants in Eqs (7.19) and (7.20)

Impeller type	k_1 ($Re_i = 1$)	N'_p ($Re_i = 10^5$)
Rushton turbine	70	5–6
Paddle	35	2
Marine propeller	40	0.35
Anchor	420	0.35
Helical ribbon	1000	0.35

Figure 7.25 Correlation between power number and Reynolds number for anchor and helical-ribbon impellers without sparging. (From M. Zlokarnik and H. Judat, 1988, *Stirring*, In: W. Gerhartz, Ed, *Ullmann's Encyclopedia of Industrial Chemistry*, vol. B2, pp. 25-1-25-33, VCH, Weinheim.)



Example 7.2 Calculation of power requirements

A fermentation broth with viscosity 10^{-2} Pa s and density 1000 kg m^{-3} is agitated in a 50 m^3 baffled tank using a marine propeller 1.3 m in diameter. The tank geometry is as specified in Figure 7.24. Calculate the power required for a stirrer speed of 4 s^{-1} .

Solution:

From Eq. (7.2):

$$Re_i = \frac{4 \text{ s}^{-1} (1.3 \text{ m})^2 1000 \text{ kg m}^{-3}}{10^{-2} \text{ kg m}^{-1} \text{ s}^{-1}} = 6.76 \times 10^5.$$

From Figure 7.24, flow at this Re_i is fully turbulent. From Table 7.3, N_p' is 0.35; therefore:

$$P = (0.35) 1000 \text{ kg m}^{-3} (4 \text{ s}^{-1})^3 (1.3 \text{ m})^5 = 8.3 \times 10^4 \text{ kg m}^2 \text{ s}^{-3}$$

$$P = 83 \text{ kW.}$$

7.10.2 Ungassed Non-Newtonian Fluids

Estimation of power requirements for non-Newtonian fluids is more difficult. It may be impossible with highly viscous fluids to achieve fully-developed turbulence so that N_p is always dependent on Re_i . In addition, because the viscosity of non-Newtonian liquids varies with shear conditions, the impeller Reynolds number used to correlate power requirements must be re-defined. Some power correlations have been developed using an impeller Reynolds number based on the apparent viscosity μ_a :

$$Re_i = \frac{N_i D_i^2 \rho}{\mu_a} \quad (7.21)$$

so that, from Eq. (7.8) for power-law fluids:

$$Re_i = \frac{N_i D_i^2 \rho}{K \dot{\gamma}^{n-1}} \quad (7.22)$$

where n is the flow behaviour index and K is the consistency index. A problem with application of Eq. (7.22) is evaluation of $\dot{\gamma}$. For stirred tanks, an approximate relation for pseudoplastic fluids is often used:

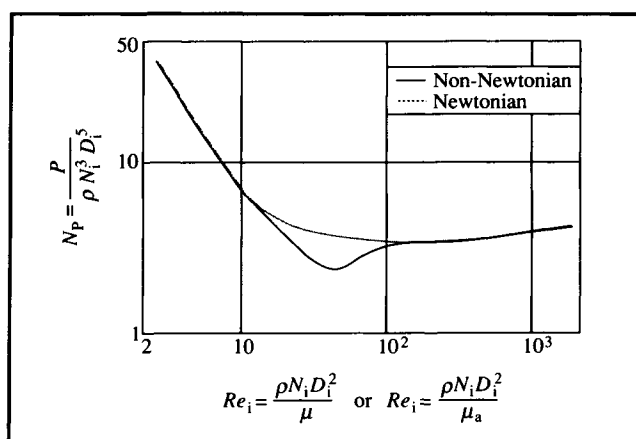
$$\dot{\gamma} = k N_i \quad (7.11)$$

where k is a constant with magnitude dependent on the geometry of the impeller. The relationship of Eq. (7.11) is discussed further in Section 7.13; for turbine impellers k is about 10. Substituting Eq. (7.11) into Eq. (7.22) gives an appropriate Reynolds number for pseudoplastic fluids:

$$Re_i = \frac{N_i^{2-n} D_i^2 \rho}{k^{n-1} K} \quad (7.23)$$

The relationship between power number and Reynolds number for a Rushton turbine in a baffled tank containing pseudoplastic non-Newtonian fluid is shown in Figure 7.26 [32, 35]. The upper line was measured using Newtonian fluid for which Re_i is defined by Eq. (7.2); this line corresponds to part of the curve already shown in Figure 7.24. The lower line gives the N_p - Re_i relationship for pseudoplastic fluid with Re_i

Figure 7.26 Correlation between power number and Reynolds number for a Rushton turbine in unaerated non-Newtonian fluid in a baffled tank. (From A.B. Metzner, R.H. Feehs, H. Lopez Ramos, R.E. Otto and J.D. Tuthill, 1961, *Agitation of viscous Newtonian and non-Newtonian fluids. AIChE J.* 7, 3-9.)



defined by Eq. (7.23). The laminar region extends to higher Reynolds numbers in pseudoplastic fluids than in Newtonian systems. At Re_i below 10 and above 200 the results for Newtonian and non-Newtonian fluids are essentially the same; in the intermediate range, pseudoplastic liquids consume less power than Newtonian fluids.

There are several practical difficulties with application of Figure 7.26 for design of bioreactors. As discussed further in Section 7.13, flow patterns in pseudoplastic and Newtonian fluids differ significantly. Even when there is high turbulence near the impeller in pseudoplastic systems, the bulk liquid may be moving very slowly and consuming relatively little power. Another problem is that, as illustrated in Figure 7.10, the non-Newtonian parameters K and n , and therefore μ_a , can vary substantially during fermentation.

7.10.3 Gassed Fluids

Liquids into which gas is sparged have reduced power requirements. Gas bubbles decrease the density of the fluid; however, the influence of density on power requirements as expressed by Eq. (7.20) does not adequately explain all the power characteristics of gas-liquid systems. The presence of bubbles also

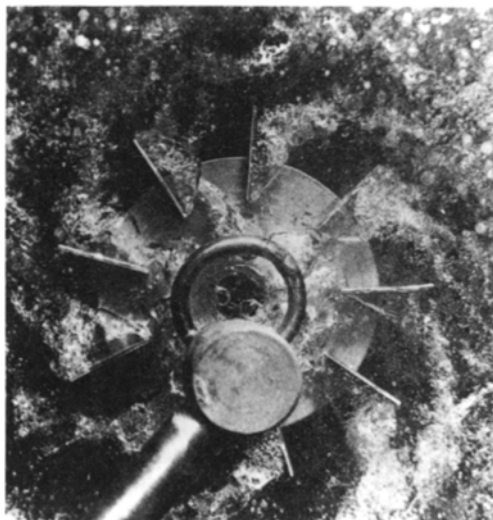
affects the hydrodynamic behaviour of fluid around the impeller. Large gas-filled cavities develop behind the stirrer blades in aerated liquids; these cavities reduce the resistance to fluid flow and decrease the drag coefficient of the impeller. Typical gas cavities are shown in Figure 7.27; this photograph taken through the base of a baffled tank shows a nine-blade disc turbine with sparger positioned just below the impeller [33].

All of the changes in hydrodynamic behaviour due to gassing are not completely understood. Power consumption is strongly controlled by gas-cavity formation; because this process is discontinuous and appears somewhat random, reduction in power consumption is typically non-uniform. The random nature of gas dispersion in agitated tanks means that it is difficult to obtain an accurate prediction of power requirements. However, an expression for the ratio of gassed to ungassed power as a function of operating conditions has been obtained [34]:

$$\frac{P_g}{P_0} = 0.10 \left(\frac{F_g}{N_i V} \right)^{-0.25} \left(\frac{N_i^2 D_i^4}{g W_i V^{2/3}} \right)^{-0.20} \quad (7.24)$$

where P_g is power consumption with sparging, P_0 is power consumption without sparging, F_g is volumetric gas flow rate, N_i is stirrer speed, V is liquid volume, D_i is impeller diameter, g is gravitational acceleration, and W_i is impeller blade width. The average deviation of experimental values from Eq. (7.24)

Figure 7.27 Gas cavities formed behind the blades of a 7.6-cm nine-blade flat-disc turbine in water sparged with air. The stirrer speed was 720 rpm. (From W. Bruijn, K. van't Riet and J.M. Smith, 1974, Power consumption with aerated Rushton turbines. *Trans. IChE* 52, 88–104.)



is about 12%. With sparging, the power consumed could be reduced to as little as half the ungassed value, depending on gas flow rate [33].

7.11 Scale-Up of Mixing Systems

Design of industrial-scale bioprocesses is usually based on the performance of small-scale prototypes. Determining optimum operating conditions at production scale is expensive and time-consuming; accordingly, it is always better to know whether a particular process will work properly before it is constructed in full size. Ideally, scale-up should be carried out so that conditions in the large vessel are as close as possible to those producing good results in the small vessel. As mixing is an important function of bioreactors, it would seem desirable to keep the mixing time constant on scale-up. Unfortunately, as explained below, the relationship between mixing time and power consumption makes this rarely possible in practice.

As the volume of mixing vessels is increased, so too are the lengths of the flow paths for bulk circulation. To keep the mixing time constant, the velocity of fluid in the tank must be increased in proportion to the size. As a rough guide, under turbulent conditions the power per unit volume is proportional to the fluid velocity squared:

$$P/V \propto v^2 \quad (7.25)$$

where P is power, V is liquid volume, and v is fluid linear velocity. The effect of this relationship on power requirements is illustrated in the following example. Suppose a cylindrical 1 m³ pilot-scale stirred tank is scaled up to 100 m³. If the tanks are geometrically similar, the length of the flow path in the large tank is about 4.5 times that in the small tank. Therefore, to keep the same mixing time, fluid velocity in the large tank must be approximately 4.5 times faster. From Eq. (7.25) this would entail a (4.5)² or 20-fold increase in power per unit volume. So, if the power input to the 1 m³ pilot-scale vessel is P , the power required for the same mixing time in the 100 m³ tank is about 2000 P . This represents an extremely large increase in power, much greater than is economically or technically feasible with most equipment used for stirring. Because the criterion of constant mixing time can hardly ever be applied for scale-up, it is inevitable that mixing times increase with scale. If instead of mixing time, P/V is kept constant during scale-up, mixing time can be expected to increase in proportion to vessel diameter raised to the power 0.67 [29].

Reduced productivity and performance often accompany scale-up of bioreactors as a result of lower mixing efficiency and subsequent alteration of the physical environment. One way of improving the design procedure is to use *scale-down methods*.

The general idea behind scale-down is that small-scale experiments to determine operating parameters are carried out under conditions that can actually be realised, physically and economically, at production scale. For example, if we decide that power input to a large-scale vessel cannot exceed a certain limit, we can calculate the corresponding mixing time and use an appropriate power input to a small-scale reactor to simulate mixing conditions in the large-scale system. Using this approach, as long as the flow regime is the same in the small- and large-scale fermenters, there is a better chance that results achieved in the small-scale unit will be reproducible in the larger system.

7.12 Improving Mixing in Fermenters

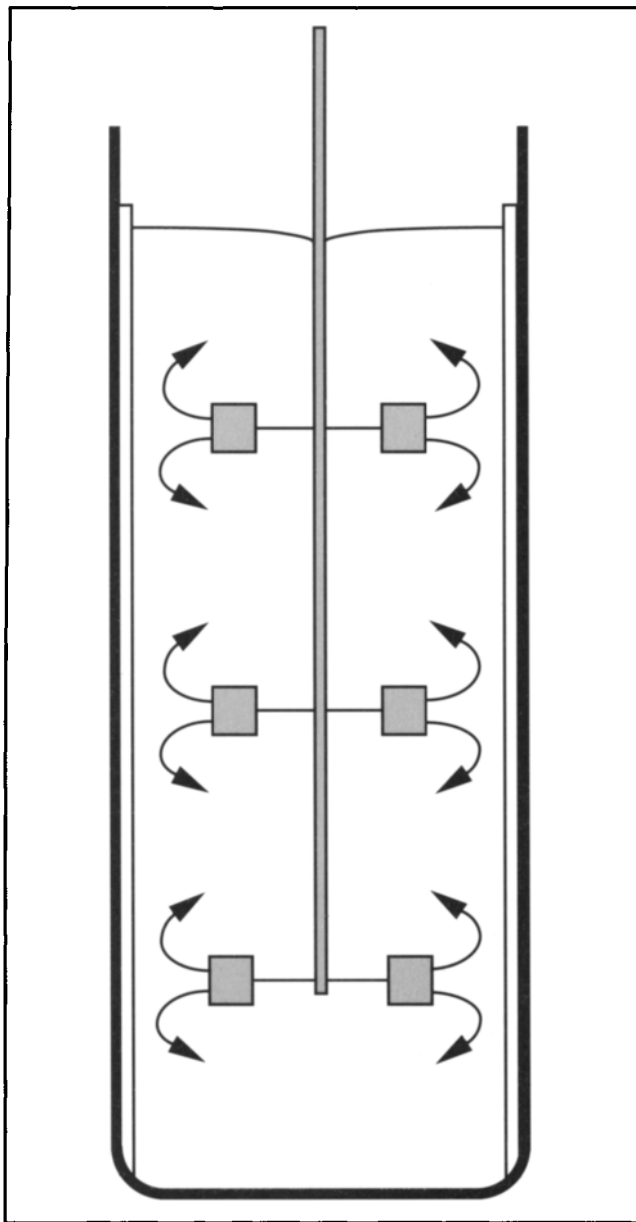
Sometimes, for the reasons outlined in the previous section, it is not possible to reduce mixing times by simply raising the power input to the stirrer. So, while increasing the stirrer speed is an obvious way of improving fluid circulation, other techniques may be required.

Mixing can sometimes be improved by changing the configuration of the system. Baffles should be installed; this is routine for stirred fermenters and produces greater turbulence. For efficient mixing the impeller should be mounted below the geometric centre of the vessel. In standard designs the impeller is located about one impeller diameter, or one-third the tank diameter, above the bottom of the tank. Mixing is facilitated when circulation currents below the impeller are smaller than those above; fluid particles leaving the impeller at the same instant then take different periods of time to return and exchange material. Rate of distribution throughout the vessel is increased when upper and lower circulation loops are asynchronous.

Another device for improving mixing is multiple impellers, although this requires an increase in power input. Typical bioreactors used for aerobic culture are tall cylindrical vessels with liquid depths significantly greater than the tank diameter. This design produces a higher hydrostatic pressure at the bottom of the vessel, and gives rising air bubbles a longer contact time with the liquid. Effective mixing in tall fermenters requires more than one impeller, as illustrated in Figure 7.28. Each impeller generates its own circulation currents. The distance between impellers should be 1.0 to 1.5 impeller diameters. If the impellers are spaced too far apart, unagitated zones develop between them; conversely, impellers located too close together produce flow streams which interfere with each other and disrupt circulation to the far reaches of the vessel. In ungassed systems with spacing between impellers of at least one impeller diameter, the power dissipated by multiple impellers is approximated by the following relationship:

$$(P)_n = n(P)_1 \quad (7.26)$$

Figure 7.28 Multiple impellers in a tall fermenter.



where $(P)_n$ is the power required by n impellers and $(P)_1$ is the power required by a single impeller. Two turbines spaced less than one impeller diameter apart can draw as much as 2.4 times the power of a single turbine. When the vessel is sparged with gas the power relationship may not be so simple. As described in Section 7.10.3, the presence of gas bubbles reduces the power required for the lowest impeller. However the quantity of gas passing through the upper impellers is often much

smaller; when this occurs the power drawn by each additional impeller is large compared with a single-impeller system [36].

Additional mixing problems can occur in fermenters when material is fed into the system during operation. If bulk distribution is slow, fermenters operated continuously or in fed-batch mode develop high localised concentrations of substrate near the feed point. This has been observed particularly in large-scale processes for production of single-cell-protein from methanol. Because high levels of methanol are toxic to cells, biomass yields decrease significantly when mixing of feed material into the broth is slow. Problems like this can be alleviated by using multiple injection points to aid distribution of substrate. It is much less expensive to do this than to increase the fluid velocity and power input.

7.13 Effect of Rheological Properties on Mixing

For effective mixing there must be turbulent conditions in the mixing vessel. Intensity of turbulence is represented by the impeller Reynolds number Re_i . As shown in Figure 7.23 for a baffled tank with turbine impeller, once Re_i falls below about 5×10^3 turbulence is damped and mixing time increases significantly. Re_i as defined in Eq. (7.2) decreases in direct proportion to increase in viscosity. Accordingly, non-turbulent conditions and poor mixing are likely to occur during agitation of highly viscous fluids. Increasing the impeller speed is an obvious solution but, as discussed in Section 7.11, this requires considerable increase in power consumption and therefore may not be feasible.

Most non-Newtonian fluids in bioprocessing are pseudoplastic. Because the apparent viscosity of these fluids depends on the shear rate, the rheological behaviour of many culture broths depends on shear conditions in the fermenter. Metzner and Otto [32] have proposed that the average shear rate in a stirred vessel is a linear function of stirrer speed:

$$\dot{\gamma}_{av} = kN_i \quad (7.27)$$

where $\dot{\gamma}_{av}$ is the average shear rate, k is a constant dependent on impeller design and N_i is stirrer speed. Experimental values of k are summarised in Table 7.4. The validity of Eq. (7.27) was established in studies by Metzner *et al.* [35]. However, shear rate in stirred vessels is far from uniform, being strongly dependent on distance from the impeller. Figure 7.29 indicates the rapid decline in shear rate in pseudoplastic fluid with increasing radial distance from the tip of a flat-blade turbine impeller [37]. The maximum shear rate close to the impeller is much higher than the average calculated from Eq. (7.27).

Pseudoplastic fluids are shear thinning, i.e. their apparent viscosity decreases with increasing shear. Accordingly, in stirred

vessels, pseudoplastic fluids have relatively low apparent viscosity in the high-shear zone near the impeller, and relatively high apparent viscosity when the fluid is away from the impeller. As a result, flow patterns similar to that illustrated in Figure 7.30 can develop; a small circulating pool of highly sheared fluid surrounds the impeller while the bulk liquid scarcely moves at all. In bioreactors containing non-Newtonian broths, this can lead to development of stagnant zones away from the impeller.

The effects of local fluid thinning in pseudoplastic fluids can be countered by modifying the geometry of the system or impeller design. Stirrers of larger diameter are recommended. For turbine impellers, instead of the conventional tank-to-impeller diameter ratio of 3:1 used with low viscosity fluids, this ratio is reduced to between 1.6 and 2. Different impeller designs which sweep the entire volume of the vessel are also recommended. The most common types used for viscous mixing are helical impellers and gate- and paddle-anchors mounted with small clearance between the impeller and tank wall. Mixing with these stirrers is accomplished at low speed without high-velocity streams. Helical agitators have been used to reduce shear damage and improve mixing in viscous cell suspensions [38].

Alternative impeller designs such as the helical ribbon and anchor improve mixing in viscous fluids; however their application in fermenters is only possible when oxygen demand in the culture is relatively low. Although large-diameter impellers operating at relatively slow speed give superior bulk mixing, high-shear systems with small, high-speed impellers are preferable for breaking up gas bubbles and promoting oxygen transfer to the liquid. In design of fermenters for viscous cultures, a compromise is usually required between mixing effectiveness and adequate mass transfer.

7.14 Role of Shear in Stirred Fermenters

Mixing in bioreactors must provide the shear conditions necessary to disperse bubbles, droplets and cell flocs. Dispersion of

Table 7.4 Observed values of k in Eq. (7.27)

(From S. Nagata, 1975, *Mixing: Principles and Applications*, Kodansha, Tokyo)

Impeller type	k
Rushton turbine	10–13
Paddle	10–13
Curved-blade paddle	7.1
Propeller	10
Anchor	20–25
Helical ribbon	30

Figure 7.29 Shear rate in pseudoplastic fluid as a function of stirrer speed and radial distance from the impeller: (○) impeller tip; (▲) 0.10 in; (■) 0.20 in; (▽) 0.34 in; (●) 0.50 in; (□) 1.00 in. The impeller diameter is 4 in. (From A.B. Metzner and J.S. Taylor, 1960, Flow patterns in agitated vessels. *AIChE J.* 6, 109–114.)

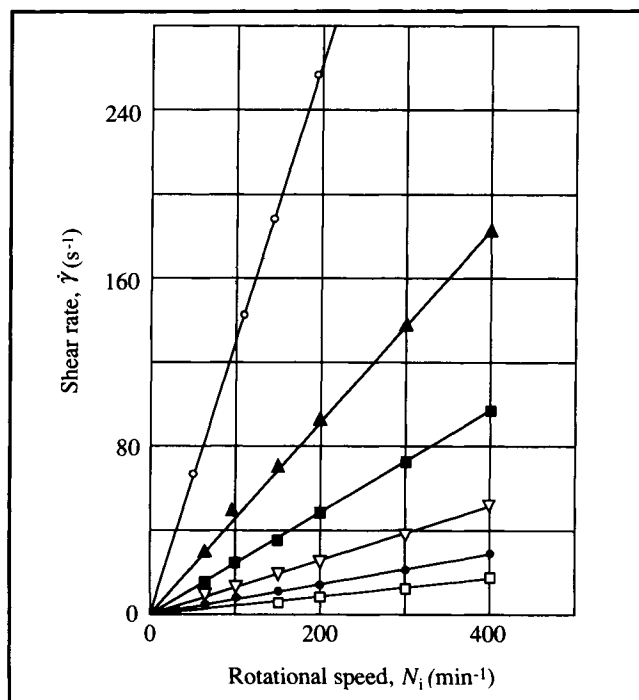
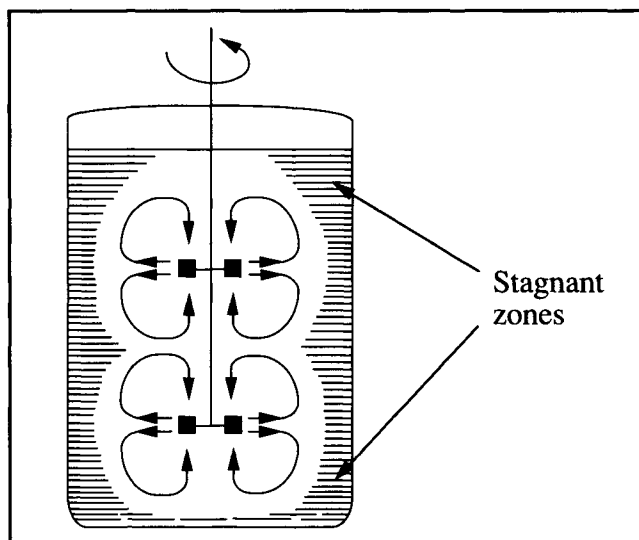


Figure 7.30 Mixing pattern for pseudoplastic liquid in a stirred fermenter.



gas bubbles by agitation involves a balance between opposing forces. Shear forces in turbulent eddies stretch and distort the bubbles and break them into smaller sizes; at the same time, surface tension at the gas–liquid interface tends to restore the bubbles to their spherical shape. In the case of solid material such as cell flocs or aggregates, shear forces in turbulent flow are resisted by the mechanical strength of the particles.

While bubble break-up is required in fermenters to facilitate oxygen transfer, disruption of cells is undesirable. Different cell types display different levels of *shear sensitivity*; insect, mammalian and plant cells are known to be particularly sensitive to mechanical forces. Bioreactors used for culture of these cells must limit the intensity of shear while still providing adequate mixing and mass transfer. At the present time, the effects of shear on cells are not well understood. Cell disruption is an obvious outcome of high shear forces; however more subtle changes such as retardation of growth and product synthesis, denaturation of extracellular proteins, change in morphology, and thickening of the cell wall, may also occur.

Because there is significant spatial variation in shear intensity in stirred vessels, the precise shear conditions experienced by cells are poorly defined. There have been many publications in recent years addressing the problem of shear damage, especially in insect- and mammalian-cell cultures. Several mechanisms have been considered in terms of their contribution to cell damage:

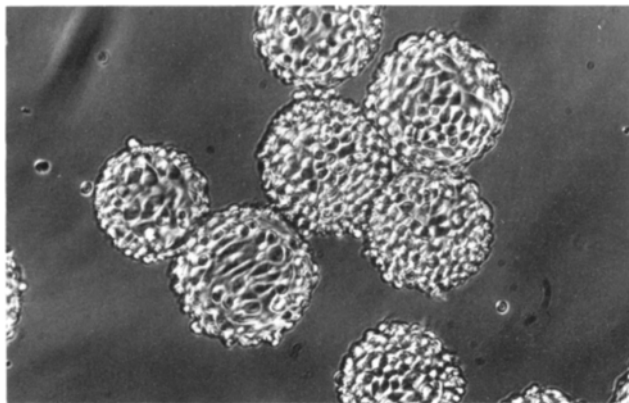
- (i) interaction between cells and turbulent eddies;
- (ii) collisions between cells, collision of cells with the impeller, and collision of cells with stationary surfaces in the vessel;
- (iii) generation of shear forces in the boundary layers and wakes near solid objects in the reactor, especially the impeller;
- (iv) generation of shear forces as bubbles rise through liquid; and
- (v) bursting of bubbles at the liquid surface.

Detailed discussion of these effects can be found elsewhere [39–50]. In general, when gas bubbles are not present in the liquid, interactions between cells and turbulent eddies are considered most likely to damage cells. However, if the vessel is sparged with air, shear damage can occur at much lower impeller speeds due to shear effects associated with bubbles [44].

7.14.1 Interaction Between Cells and Turbulent Eddies

Hydrodynamic effects have been studied mainly with animal cells because shear damage is a significant problem in large-scale culture. Many animal cells used in bioprocessing are *anchorage-dependent*; this means that the cells must be attached to a solid

Figure 7.31 Chinese hamster ovary (CHO) cells attached to microcarrier beads; magnification $\times 85$. (Photograph courtesy of J. Crowley.)



surface for survival. In bioreactors, the surface area required for cell attachment is provided very effectively by *microcarrier beads*, which range in diameter from 80 to 200 μm . As shown in Figure 7.31, cells cover the surface of the beads which are then suspended in nutrient medium. There are many benefits associated with use of microcarriers; however, a disadvantage is that cells attached to microcarriers cannot easily change position or rotate in response to shear forces in the fluid. This, coupled with the lack of a protective cell wall, make animal cells on microcarriers especially susceptible to shear damage.

Interactions between microcarriers and eddies in turbulent flow have the potential to cause mechanical damage to cells.

The intensity of shear associated with these interactions is dependent on the relative sizes of the eddies and microcarrier particles. If the particles are small relative to the eddies, they tend to be captured or entrained in the eddies as shown in Figure 7.32(a). As fluid motion within eddies is laminar, if the density of the microcarriers is about the same as the suspending fluid, there is little relative motion of the particles. Accordingly, the velocity difference between the fluid streamlines and the microcarriers is small, except for brief periods of acceleration when the bead enters a new eddy. On average, therefore, if the particles are smaller than the eddies, the shear effects of eddy–cell interactions are minimal.

If the stirrer speed is increased and the average eddy size reduced, interactions between eddies and microcarriers can occur in two possible ways. A single eddy that cannot fully engulf the particle will act on part of its surface and cause the particle to rotate in the fluid; this will result in a relatively low level of shear at the surface of the bead. However, much higher shear stresses result when several eddies with opposing rotation interact with the particle simultaneously, as illustrated in Figure 7.32(b). It has been found experimentally that detrimental effects start to occur when the Kolmogorov scale (Eq. 7.14) for eddy size drops below $2/3-1/2$ the diameter of the microcarrier beads [41, 42, 49]. Excessive agitation leads to formation of eddies with size small enough and of sufficient energy to cause damage to the cells. These findings for cells on microcarriers apply also to freely suspended cells; however, because cells are smaller than microcarriers, eddy sizes causing shear damage are also smaller.

Example 7.3 Operating conditions for turbulent shear damage

Microcarrier beads 120 μm in diameter are used to culture recombinant CHO cells for production of growth hormone. It is proposed to use a 6-cm turbine impeller to mix the culture in a 3.5-litre stirred tank. Air and carbon dioxide are supplied by flow through the reactor headspace. The microcarrier suspension has a density of approximately 1010 kg m^{-3} and a viscosity of $1.3 \times 10^{-3} \text{ Pa s}$. Estimate the maximum allowable stirrer speed which avoids turbulent shear damage of the cells.

Solution:

Damage due to eddies is avoided if the Kolmogorov scale remains greater than $2/3-1/2$ the diameter of the beads. Let us determine the stirrer speed required to create eddies with size $\lambda = 2/3$ (120 μm) = 80 μm = $8 \times 10^{-5} \text{ m}$. The stirrer power producing eddies of this dimension can be estimated using Eq. (7.14) and the properties of the fluid.

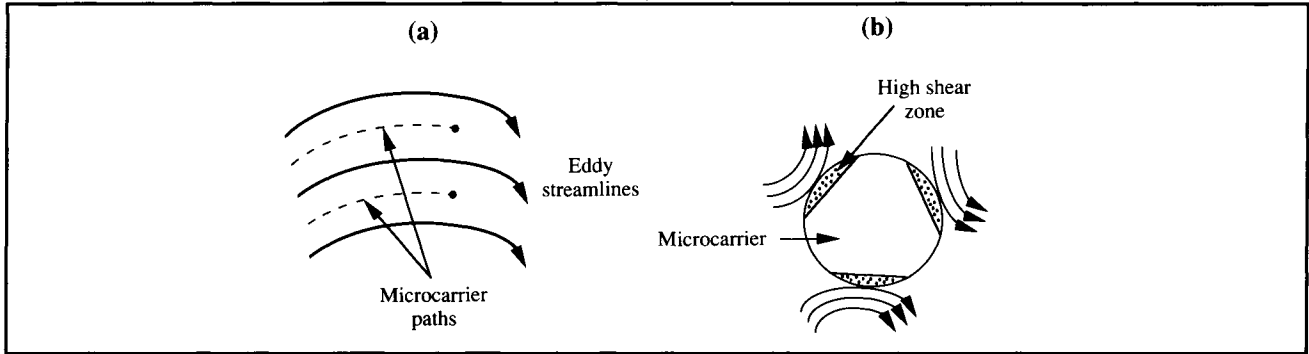
$$\text{Kinematic viscosity, } \nu = \frac{\mu}{\rho} = \frac{1.3 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}}{1010 \text{ kg m}^{-3}} = 1.29 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}.$$

Raising both sides of Eq. (7.14) to the fourth power:

$$\lambda^4 = \frac{\nu^3}{\epsilon}$$

so that

Figure 7.32 Eddy–microcarrier interactions. (a) Microcarriers are captured in large eddies and move within the streamline flow. (b) When several eddies with opposing rotation interact with the microcarrier simultaneously, high levels of shear develop on the bead surface. (From R.S. Cherry and E.T. Papoutsakis, 1986, Hydrodynamic effects on cells in agitated tissue culture reactors. *Bioprocess Eng.* 1, 29–41.)



$$\varepsilon = \frac{v^3}{\lambda^4}.$$

Therefore:

$$\varepsilon = \frac{(1.29 \times 10^{-6})^3 \text{ m}^6 \text{ s}^{-3}}{(8 \times 10^{-5})^4 \text{ m}^4} = 0.052 \text{ m}^2 \text{ s}^{-3}.$$

ε represents the power dissipated per unit mass of fluid. The mass of fluid in which turbulent power is dissipated may be taken as the entire contents of the tank; however power dissipation occurs unevenly throughout the vessel with highest levels in the vicinity of the impeller. ε based on the mass of fluid near the impeller is considered the more appropriate value in these calculations [50]; fluid mass in the impeller zone is roughly equal to ρD_i^3 where ρ is fluid density and D_i is impeller diameter. Therefore, the stirrer power P is equal to ε multiplied by ρD_i^3 :

$$P = (0.052 \text{ m}^2 \text{ s}^{-3}) (1010 \text{ kg m}^{-3}) (6 \times 10^{-2} \text{ m})^3$$

$$P = 1.13 \times 10^{-2} \text{ kg m}^2 \text{ s}^{-3} = 1.13 \times 10^{-2} \text{ W}.$$

From Figure 7.24 and Table 7.3, N_p' is about 5 for a turbine impeller operating in the turbulent regime, depending on tank geometry. The stirrer speed corresponding to these conditions can be calculated from Eq. (7.20):

$$N_i^3 = \frac{P}{N_p' \rho D_i^5}$$

$$N_i^3 = \frac{1.13 \times 10^{-2} \text{ kg m}^2 \text{ s}^{-3}}{(5) (1010 \text{ kg m}^{-3}) (6 \times 10^{-2} \text{ m})^5} = 2.89 \text{ s}^{-3}.$$

Taking the cube root:

$$N_i = 1.42 \text{ s}^{-1} = 85.5 \text{ rpm}.$$

Flow is just turbulent with $Re_i = 4 \times 10^3$. This analysis indicates that shear damage from turbulent eddies is not expected until the stirrer speed exceeds about 85 rpm. If the culture were sparged with gas, it is possible that shear damage would occur due to other mechanisms, e.g. bursting bubbles.

As indicated by Eq. (7.14), if the viscosity of the fluid is increased, the size of the smallest eddies also increases. Increasing the fluid viscosity should, therefore, reduce shear damage in bioreactors. This effect has been demonstrated by addition of thickening agents to animal-cell growth medium; moderate increases in viscosity have been shown to significantly reduce turbulent cell death [42].

7.14.2 Bubble Shear

When liquid containing shear-sensitive cells is sparged with air, other damaging mechanisms come into play. From experiments conducted so far, these appear to be associated primarily with bubbles bursting at the surface of the liquid. Breakage of the thin bubble film and rapid flow from the bubble rim back into the liquid generate high shear forces capable of damaging certain types of cell. Further discussion of these effects can be found in the literature [43–47].

7.15 Summary of Chapter 7

Chapter 7 covers a wide range of topics in fluid dynamics, rheology and mixing. At the end of Chapter 7 you should:

- (i) understand the difference between *laminar* and *turbulent* flow;
- (ii) be able to describe how fluid *boundary layers* develop in terms of *viscous drag*;
- (iii) be able to define *viscosity* in terms of *Newton's law*;
- (iv) know what *Newtonian* and *non-Newtonian* fluids are, and the difference between viscosity for Newtonian fluids and *apparent viscosity* for non-Newtonian fluids;
- (v) be familiar with equipment used for mixing in stirred vessels;
- (vi) be able to describe the mechanisms of mixing and their effect on *mixing time*;
- (vii) understand the effects of scale-up on mixing;
- (viii) know how liquid properties, gas sparging, impeller size and stirrer speed affect power consumption in stirred vessels; and
- (ix) understand how cells can be damaged by shear in stirred fermenters.

Problems

7.1 Rheology of fermentation broth

The fungus *Aureobasidium pullulans* is used to produce an extracellular polysaccharide by fermentation of sucrose. After 120 h fermentation, the following measurements of shear stress and shear rate were made with a rotating-cylinder viscometer.

<i>Shear stress</i> (dyn cm ⁻²)	<i>Shear rate</i> (s ⁻¹)
44.1	10.2
235.3	170
357.1	340
457.1	510
636.8	1020

- (a) Plot the rheogram for this fluid.
- (b) Determine the appropriate non-Newtonian parameters.
- (c) What is the apparent viscosity at shear rates of:
 - (i) 15 s⁻¹; and
 - (ii) 200 s⁻¹?

7.2 Rheology of yeast suspensions

Apparent viscosities for pseudoplastic cell suspensions at varying cell concentrations are measured using a coaxial-cylinder rotary viscometer. The results are:

<i>Cell concentration</i> (%)	<i>Shear rate</i> (s ⁻¹)	<i>Apparent viscosity</i> (cP)
1.5	10	1.5
	100	1.5
3	10	2.0
	100	2.0
6	20	2.5
	45	2.4
10.5	10	4.7
	20	4.0
	50	4.1
	100	3.8
12	1.8	40
	4.0	30
	7.0	22
	20	15
18	40	12
	1.8	140
	7.0	85
	20	62
21	40	55
	1.8	710
	4.0	630
	7.0	480
	40	330
	70	290

Show on an appropriate plot how *K* and *n* vary with cell concentration.

7.3 Impeller viscometer

The rheology of a *Penicillium chrysogenum* broth is examined using an impeller viscometer. The density of the cell suspension is approximately 1000 kg m^{-3} . Samples of broth are poured into a glass beaker of diameter 15 cm and stirred slowly using a Rushton turbine of diameter 4 cm. The average shear rate generated by this impeller is greater than the stirrer speed by a factor of about 10.2. When the stirrer shaft is attached to a device for measuring torque and rotational speed, the following results are recorded.

Stirrer speed (s^{-1})	Torque (N m)
0.185	3.57×10^{-6}
0.163	3.45×10^{-6}
0.126	3.31×10^{-6}
0.111	3.20×10^{-6}

- Can the rheology be described using a power-law model? If so, evaluate K and n .
- Viscosity measurements using impeller viscometers must be carried out under laminar flow conditions. Check that flow in this experiment is laminar.
- Use of turbines for impeller viscometry restricts the range of shear rates that can be tested. How is the situation improved with a helical-ribbon impeller?

7.4 Particle suspension and gas dispersion

Cells in fermenters must be kept from settling out of suspension. The minimum stirrer speed required to keep the bottom of the tank free of cells can be estimated roughly using a relation given by Zwietering [51]:

$$N_i^* = C v_L^{0.1} D_p^{0.2} \left(\frac{g(\rho_p - \rho_L)}{\rho_L} \right)^{0.45} D_i^{-0.85} x^{0.13}$$

where:

- N_i^* = minimum stirrer speed for suspension of solids, s^{-1} ;
 C = a constant (= 7.7 for a turbine impeller with diameter one-third that of the tank);
 v_L = liquid kinematic viscosity, $\text{m}^2 \text{s}^{-1}$;
 D_p = mean cell diameter, m;
 g = gravitational acceleration, m s^{-2} ;
 ρ_p = density of the cells, kg m^{-3} ;
 ρ_L = density of the suspending liquid, kg m^{-3} ;
 D_i = impeller diameter, m; and
 x = cell concentration, wt%.

A certain minimum stirrer speed is also required in aerobic

systems for proper dispersion of air bubbles. From the data of Westerterp *et al.* [52], the minimum stirrer tip speed (tip speed = $\pi N_i D_i$) for this purpose can be assumed to lie between 1.5 and 2.5 m s^{-1} , depending on the surface tension between the gas and liquid, the fluid density, and the tank-to-impeller diameter ratio.

A fermentation broth contains 40 wt% cells of average dimension $10 \mu\text{m}$ and density 1.04 g cm^{-3} . The diameter of the impeller in the fermenter is 30 cm. Assuming that the density and viscosity of the suspending medium are the same as water, determine which takes more power to achieve, cell suspension or bubble dispersion.

7.5 Scale-up of mixing system

To ensure turbulent conditions and minimum mixing time during agitation with a turbine impeller, the Reynolds number must be at least 10^4 .

- A stirred laboratory-scale fermenter with a turbine impeller 5 cm in diameter is operated at 800 rpm. If the density of broth being stirred is close to that of water, what is the upper limit for viscosity of the suspension if adequate mixing is to be maintained?
- The mixing system is scaled up so the tank and impeller are 15 times the diameter of the laboratory equipment. The stirrer in the large vessel is operated so that the stirrer tip speed (tip speed = $\pi N_i D_i$) is the same as in the laboratory apparatus. How does scale-up affect the maximum viscosity allowable for maintenance of turbulent mixing conditions?

7.6 Effect of viscosity on power requirements

A cylindrical bioreactor of diameter 3 m has four baffles. A Rushton turbine mounted in the reactor has a diameter one-third the tank diameter and is operated at a speed of 90 rpm. The density of the fluid is approximately 1 g cm^{-3} . The reactor is used to culture an anaerobic organism that does not require gas sparging. The broth can be assumed Newtonian. As the cells grow, the viscosity of the broth increases.

- Compare power requirements when the viscosity is:
 - approximately that of water;
 - 100 times greater than water; and
 - 10^4 times greater than water.
- When the viscosity is 1000 times greater than water, estimate the power required to achieve turbulence.

7.7 Electrical power required for mixing

Laboratory-scale fermenters are usually mixed using small stirrers with electric motors rated between 100 and 500 W. One

such motor is used to drive a 7-cm turbine impeller in a small reactor containing fluid with the properties of water. The stirrer speed is 900 rpm. Estimate the power requirements for this process. How do you explain the difference between the amount of electrical power consumed by the motor and the power dissipated by the stirrer?

7.8 Mixing time with aeration

A cylindrical stirred bioreactor of diameter and height 2 m has a Rushton turbine one-third the tank diameter in size. The bioreactor contains Newtonian culture broth with the same density as water and with viscosity 4 cP.

- (a) If the specific power consumption must not exceed 1.5 kW m^{-3} , determine the maximum allowable stirrer speed. What is the mixing time under these conditions?
- (b) The tank is now aerated. In the presence of gas bubbles, the approximate relationship between ungassed power number $(N_p)_0$ and gassed power number $(N_p)_g$ is: $(N_p)_g = 0.5 (N_p)_0$. What maximum stirrer speed is now possible in the sparged reactor? Estimate the mixing time.

References

1. Skelland, A.H.P. (1967) *Non-Newtonian Flow and Heat Transfer*, John Wiley, New York.
2. van Wazer, J.R., J.W. Lyons, K.Y. Kim and R.E. Colwell (1963) *Viscosity and Flow Measurement*, Interscience, New York.
3. Whorlow, R.W. (1980) *Rheological Techniques*, Ellis Horwood, Chichester.
4. Charles, M. (1978) Technical aspects of the rheological properties of microbial cultures. *Adv. Biochem. Eng.* **8**, 1–62.
5. Metz, B., N.W.F. Kossen and J.C. van Suijdam (1979) The rheology of mould suspensions. *Adv. Biochem. Eng.* **11**, 103–156.
6. Sherman, P. (1970) *Industrial Rheology*, Academic Press, London.
7. Bongenaar, J.J.T.M., N.W.F. Kossen, B. Metz and F.W. Meijboom (1973) A method for characterizing the rheological properties of viscous fermentation broths. *Biotechnol. Bioeng.* **15**, 201–206.
8. Roels, J.A., J. van den Berg and R.M. Voncken (1974) The rheology of mycelial broths. *Biotechnol. Bioeng.* **16**, 181–208.
9. Kim, J.H., J.M. Lebeault and M. Reuss (1983) Comparative study on rheological properties of mycelial broth in filamentous and pelleted forms. *Eur. J. Appl. Microbiol. Biotechnol.* **18**, 11–16.
10. Rapp, P., H. Reng, D.-C. Hempel and F. Wagner (1984) Cellulose degradation and monitoring of viscosity decrease in cultures of *Cellulomonas uda* grown on printed newspaper. *Biotechnol. Bioeng.* **26**, 1167–1175.
11. Labuza, T.P., D. Barrera Santos and R.N. Roop (1970) Engineering factors in single-cell protein production. I. Fluid properties and concentration of yeast by evaporation. *Biotechnol. Bioeng.* **12**, 123–134.
12. Berkman-Dik, T., M. Özilgen and T.F. Bozoğlu (1992) Salt, EDTA, and pH effects on rheological behavior of mold suspensions. *Enzyme Microbiol. Technol.* **14**, 944–948.
13. Deindoerfer, F.H. and E.L. Gaden (1955) Effects of liquid physical properties on oxygen transfer in penicillin fermentation. *Appl. Microbiol.* **3**, 253–257.
14. Ju, L.-K., C.S. Ho and J.F. Shanahan (1991) Effects of carbon dioxide on the rheological behavior and oxygen transfer in submerged penicillin fermentations. *Biotechnol. Bioeng.* **38**, 1223–1232.
15. Taguchi, H. and S. Miyamoto (1966) Power requirement in non-Newtonian fermentation broth. *Biotechnol. Bioeng.* **8**, 43–54.
16. Deindoerfer, F.H. and J.M. West (1960) Rheological examination of some fermentation broths. *J. Biochem. Microbiol. Technol. Eng.* **2**, 165–175.
17. Tuffile, C.M. and F. Pinho (1970) Determination of oxygen-transfer coefficients in viscous streptomycete fermentations. *Biotechnol. Bioeng.* **12**, 849–871.
18. LeDuy, A., A.A. Marsan and B. Coupal (1974) A study of the rheological properties of a non-Newtonian fermentation broth. *Biotechnol. Bioeng.* **16**, 61–76.
19. Kato, A., S. Kawazoe and Y. Soh (1978) Viscosity of the broth of tobacco cells in suspension culture. *J. Ferment. Technol.* **56**, 224–228.
20. Ballica, R., D.D.Y. Ryu, R.L. Powell and D. Owen (1992) Rheological properties of plant cell suspensions. *Biotechnol. Prog.* **8**, 413–420.
21. Deindoerfer, F.H. and J.M. West (1960) Rheological properties of fermentation broths. *Adv. Appl. Microbiol.* **2**, 265–273.
22. Tanaka, H. (1982) Oxygen transfer in broths of plant cells at high density. *Biotechnol. Bioeng.* **24**, 425–442.
23. Laine, J. and R. Kuoppamäki (1979) Development of the design of large-scale fermentors. *Ind. Eng. Chem. Process Des. Dev.* **18**, 501–506.
24. Holland, F.A. and F.S. Chapman (1966) *Liquid Mixing and Processing in Stirred Tanks*, Reinhold, New York.

25. Bates, R.L., P.L. Fondy and J.G. Fenic (1966) Impeller characteristics and power. In: V.W. Uhl and J.B. Gray (Eds), *Mixing: Theory and Practice*, vol. 1, pp. 111–178, Academic Press, New York.
26. Nagata, S. (1975) *Mixing: Principles and Applications*, Kodansha, Tokyo.
27. Hoogendoorn, C.J. and A.P. den Hartog (1967) Model studies on mixers in the viscous flow region. *Chem. Eng. Sci.* **22**, 1689–1699.
28. Kossen, N.W.F. and N.M.G. Oosterhuis (1985) Modelling and scaling-up of bioreactors. In: H.-J. Rehm and G. Reed (Eds), *Biotechnology*, vol. 2, pp. 571–605, VCH, Weinheim.
29. Zlokarnik, M. and H. Judat (1988) Stirring. In: W. Gerhartz (Ed), *Ullmann's Encyclopedia of Industrial Chemistry*, vol. B2, pp. 25-1–25-33, VCH, Weinheim.
30. Rushton, J.H., E.W. Costich and H.J. Everett (1950) Power characteristics of mixing impellers. Part I. *Chem. Eng. Prog.* **46**, 395–404.
31. Rushton, J.H., E.W. Costich and H.J. Everett (1950) Power characteristics of mixing impellers. Part II. *Chem. Eng. Prog.* **46**, 467–476.
32. Metzner, A.B. and R.E. Otto (1957) Agitation of non-Newtonian fluids. *AIChE J.* **3**, 3–10.
33. Bruijn, W., K. van't Riet and J.M. Smith (1974) Power consumption with aerated Rushton turbines. *Trans. IChE* **52**, 88–104.
34. Hughmark, G.A. (1980) Power requirements and interfacial area in gas–liquid turbine agitated systems. *Ind. Eng. Chem. Process Des. Dev.* **19**, 638–641.
35. Metzner, A.B., R.H. Feehs, H. Lopez Ramos, R.E. Otto and J.D. Tuthill (1961) Agitation of viscous Newtonian and non-Newtonian fluids. *AIChE J.* **7**, 3–9.
36. Nienow, A.W. and M.D. Lilly (1979) Power drawn by multiple impellers in sparged agitated vessels. *Biotechnol. Bioeng.* **21**, 2341–2345.
37. Metzner, A.B. and J.S. Taylor (1960) Flow patterns in agitated vessels. *AIChE J.* **6**, 109–114.
38. Jolicoeur, M., C. Chavarie, P.J. Carreau and J. Archambault (1992) Development of a helical-ribbon impeller bioreactor for high-density plant cell suspension culture. *Biotechnol. Bioeng.* **39**, 511–521.
39. Cherry, R.S. and E.T. Papoutsakis (1986) Hydrodynamic effects on cells in agitated tissue culture reactors. *Bioprocess Eng.* **1**, 29–41.
40. Tramper, J., D. Joustra and J.M. Vlak (1987) Bioreactor design for growth of shear-sensitive insect cells. In: C. Webb and F. Mavituna (Eds), *Plant and Animal Cells: Process Possibilities*, pp. 125–136, Ellis Horwood, Chichester.
41. Croughan, M.S., J.-F. Hamel and D.I.C. Wang (1987) Hydrodynamic effects on animal cells grown in microcarrier cultures. *Biotechnol. Bioeng.* **29**, 130–141.
42. Croughan, M.S., E.S. Sayre and D.I.C. Wang (1989) Viscous reduction of turbulent damage in animal cell culture. *Biotechnol. Bioeng.* **33**, 862–872.
43. Handa-Corrigan, A., A.N. Emery and R.E. Spier (1989) Effect of gas–liquid interfaces on the growth of suspended mammalian cells: mechanisms of cell damage by bubbles. *Enzyme Microb. Technol.* **11**, 230–235.
44. Kunas, K.T. and E.T. Papoutsakis (1990) Damage mechanisms of suspended animal cells in agitated bioreactors with and without bubble entrainment. *Biotechnol. Bioeng.* **36**, 476–483.
45. Jöbses, I., D. Martens and J. Tramper (1991) Lethal events during gas sparging in animal cell culture. *Biotechnol. Bioeng.* **37**, 484–490.
46. Chalmers, J.J. and Bavarian, F. (1991) Microscopic visualization of insect cell-bubble interactions. II. The bubble film and bubble rupture. *Biotechnol. Prog.* **7**, 151–158.
47. Cherry, R.S. and C.T. Hulle (1992) Cell death in the thin films of bursting bubbles. *Biotechnol. Prog.* **8**, 11–18.
48. van't Riet, K. and J. Tramper (1991) *Basic Bioreactor Design*, Chapter 8, Marcel Dekker, New York.
49. McQueen, A., E. Meilhoc and J.E. Bailey (1987) Flow effects on the viability and lysis of suspended mammalian cells. *Biotechnol. Lett.* **9**, 831–836.
50. Cherry, R.S. and E.T. Papoutsakis (1988) Physical mechanisms of cell damage in microcarrier cell culture bioreactors. *Biotechnol. Bioeng.* **32**, 1001–1014.
51. Zwietering, Th.N. (1958) Suspending of solid particles in liquid by agitators. *Chem. Eng. Sci.* **8**, 244–253.
52. Westerterp, K.R., L.L. van Dierendonck and J.A. de Kraa (1963) Interfacial areas in agitated gas–liquid contactors. *Chem. Eng. Sci.* **18**, 157–176.

Suggestions for Further Reading

1. Atkinson, B. and F. Mavituna (1991) *Biochemical Engineering and Biotechnology Handbook*, 2nd edn, Chapter 11, Macmillan, Basingstoke.
2. Oldshue, J.Y. (1983) *Fluid Mixing Technology*, McGraw-Hill, New York.
3. Thomas, C.R. (1990) Problems of shear in biotechnology. In: M.A. Winkler (Ed), *Chemical Engineering Problems in Biotechnology*, pp. 23–93, Elsevier Applied Science, Barking.

Heat Transfer

In this chapter we are concerned with the process of heat flow between hot and cold systems. The rate at which heat is transferred depends directly on two variables: the temperature difference between the hot and cold bodies, and the surface area available for heat exchange. It is also influenced by many other factors, such as the geometry and physical properties of the system and, if fluid is present, the flow conditions. Fluids are often heated or cooled in bioprocessing. Typical examples are removal of heat during fermenter operation using cooling water, and heating of raw medium to sterilisation temperature by steam.

As shown in Chapters 5 and 6, energy balances allow us to determine the heating and cooling requirements of fermenters and enzyme reactors. Once the rate of heat transfer for a particular purpose is known, the surface area required to achieve this rate can be calculated using design equations. Estimating the heat-transfer surface area is a central objective in design as this parameter determines the size of heat-exchange equipment. In this chapter the principles governing heat transfer are outlined with applications in bioprocess design.

First, let us look at the types of equipment used for industrial heat-exchange.

8.1 Heat-Transfer Equipment

In bioprocessing, heat exchange occurs most frequently between fluids. Equipment is provided to allow transfer of heat while preventing the fluids from actually coming into contact with each other. In most heat exchangers, heat is transferred through a solid metal wall which separates the fluid streams. Sufficient surface area is provided so that the desired rate of heat transfer can be achieved. Heat transfer is facilitated by agitation and turbulent flow of the fluids.

8.1.1 Bioreactors

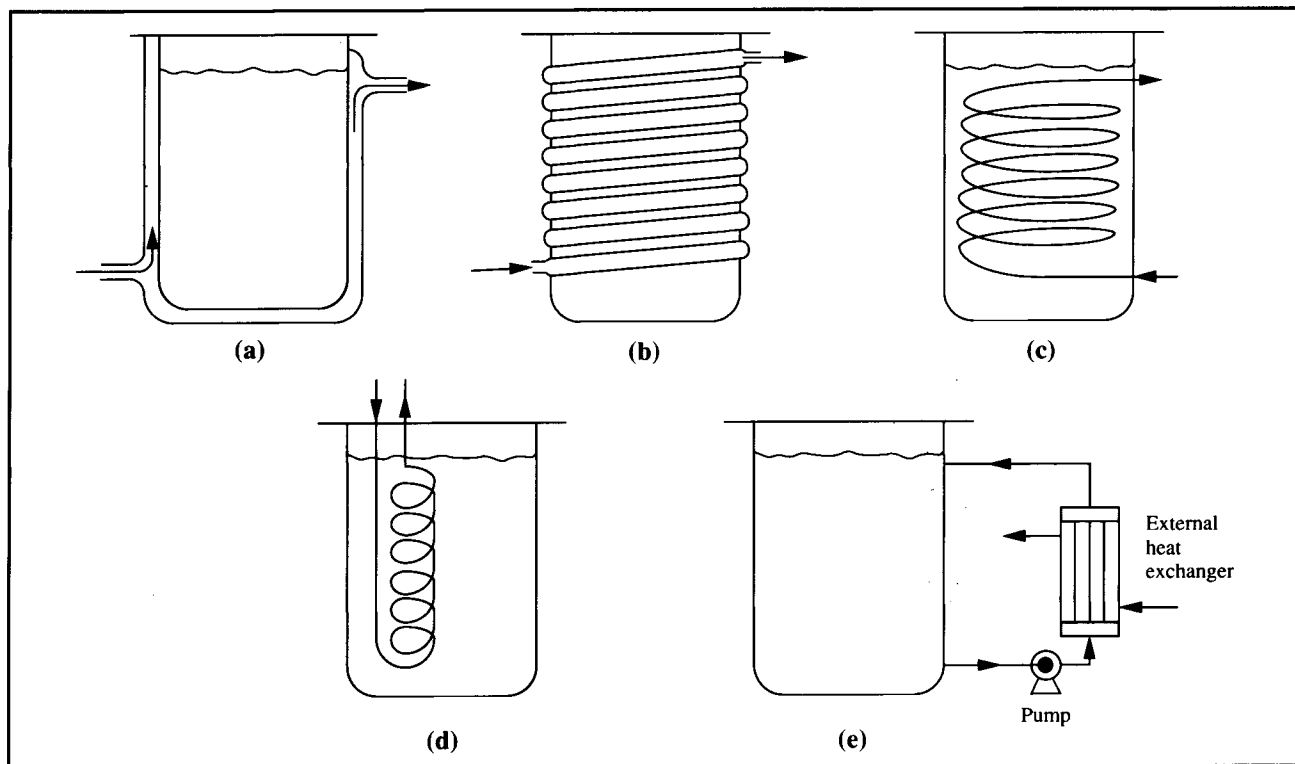
Two applications of heat transfer are common in bioreactor operation. The first is *in situ* batch sterilisation of liquid medium. In this process, the fermenter vessel containing medium is heated using steam and held at the sterilisation temperature for a period of time; cooling water is then used to bring the temperature back to normal operating conditions. Sterilisation is discussed in more detail in Chapter 13. The

other application of heat transfer is for temperature control during reactor operation. Metabolic activity of cells generates a substantial amount of heat in fermenters; this heat must be removed to avoid temperature increases. Most fermentations take place in the range 30–37°C; in large-scale operations, cooling water is used to maintain the temperature usually to within 1°C. Small-scale fermenters have different heat-exchange requirements; because the external surface area to volume ratio is much greater and heat losses through the wall of the vessel more significant, laboratory-scale units often require heating rather than cooling. Many enzyme reactions also require heating to maintain optimum temperature.

Equipment used for heat exchange in bioreactors usually takes one of the forms illustrated in Figure 8.1. The fermenter may have an external jacket (Figure 8.1a) or coil (Figure 8.1b) through which steam or cooling water is circulated. Alternatively, helical (Figure 8.1c) or baffle (Figure 8.1d) coils may be located internally. Another method is to pump liquid from the reactor through a separate heat-exchange unit as shown in Figure 8.1(e).

The surface area available for heat transfer is lower in the external jacket and coil designs of Figures 8.1(a) and (b) than when internal coils are completely submerged in the reactor contents. External jackets on bioreactors provide sufficient heat-transfer area for laboratory and other small-scale systems; however they are likely to be inadequate for large-scale fermentations. Internal coils are frequently used in production vessels; the coil can be operated with high liquid velocity and the entire tube surface is exposed to the reactor contents providing a relatively large heat-transfer area. There are some disadvantages with internal structures: they interfere with mixing in the vessel and make cleaning of the reactor difficult;

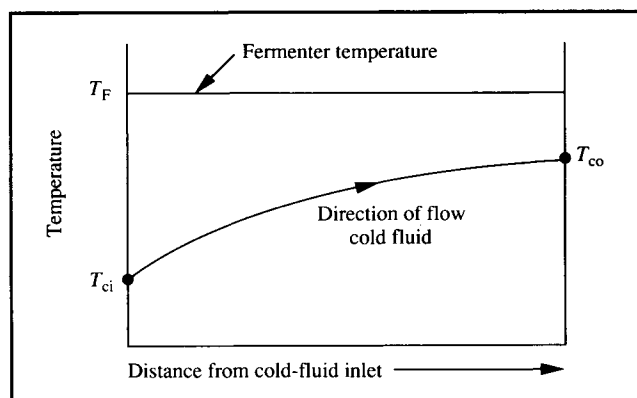
Figure 8.1 Heat-transfer configurations for bioreactors: (a) jacketed vessel; (b) external coil; (c) internal helical coil; (d) internal baffle-type coil; (e) external heat exchanger.



another problem is film growth of cells on the heat-transfer surface. In contrast, the external heat exchange unit shown in Figure 8.1(e) is independent of the reactor, easy to scale-up, and can provide better heat-transfer capabilities than any of the other configurations. However, conditions of sterility must be met, the cells must be able to withstand the shear forces imposed during pumping, and, in aerobic fermentations, the residence time in the heat exchanger must be small enough to ensure the medium does not become depleted of oxygen.

When internal coils such as those in Figures 8.1(c) and (d) are used to carry cooling water for removal of heat from a fermenter, the variation of water temperature with distance through the coil is as shown in Figure 8.2. The temperature of the cooling water rises as it flows through the tube and takes up heat from the fermenter contents. The water temperature increases steadily from its inlet temperature T_{ci} to the outlet temperature T_{co} . On the other hand, if the fermenter contents are well mixed, temperature gradients in the bulk fluid are negligible and the temperature is uniform at T_F .

Figure 8.2 Temperature changes for control of fermentation temperature using cooling water.



8.1.2 General Equipment For Heat Transfer

Many types of general-purpose equipment are used industrially for heat-exchange operations. The simplest form of

heat-transfer equipment is the double-pipe heat exchanger; for larger capacities, more elaborate shell-and-tube units containing hundreds of square metres of heat-exchange area are required. These devices are described below.

8.1.2.1 Double-pipe heat exchanger

A *double-pipe heat exchanger* consists of two metal pipes, one inside the other as shown in Figure 8.3. One fluid flows through the inner tube while the other fluid flows in the annular space between the pipe walls. When one of the fluids is hotter than the other, heat flows from it through the wall of the inner tube into the other fluid. As a result, the hot fluid becomes cooler and the cold fluid becomes warmer.

Double-pipe heat exchangers can be operated with counter-current or cocurrent flow of fluid. If, as indicated in Figure 8.3, the two fluids enter at opposite ends of the device and pass in

opposite directions through the pipes, the flow is *counter-current*. Cold fluid entering the device meets the hot fluid just as it is leaving, i.e. cold fluid at its lowest temperature is placed in thermal contact with hot fluid also at its lowest temperature. Changes in temperature of the two fluids as they flow counter-currently through the length of the pipe are shown in Figure 8.4. The four terminal temperatures are as follows: T_{hi} is the inlet temperature of the hot fluid, T_{ho} is the outlet temperature of the hot fluid, T_{ci} is the inlet temperature of the cold fluid, and T_{co} is the outlet temperature of the cold fluid leaving the system. A sign of efficient operation is T_{co} close to T_{hi} , or T_{ho} close to T_{ci} .

The alternative to countercurrent flow is *cocurrent* or *parallel* flow. In this mode of operation, both fluids enter their respective tubes at the same end of the exchanger and flow in the same direction to the other end. The temperature curves for cocurrent flow are given in Figure 8.5. Cocurrent operation

Figure 8.3 Double-pipe heat exchanger. (From A.S. Foust, L.A. Wenzel, C.W. Clump, L. Maus and L.B. Andersen, 1960, *Principles of Unit Operations*, John Wiley, New York.)

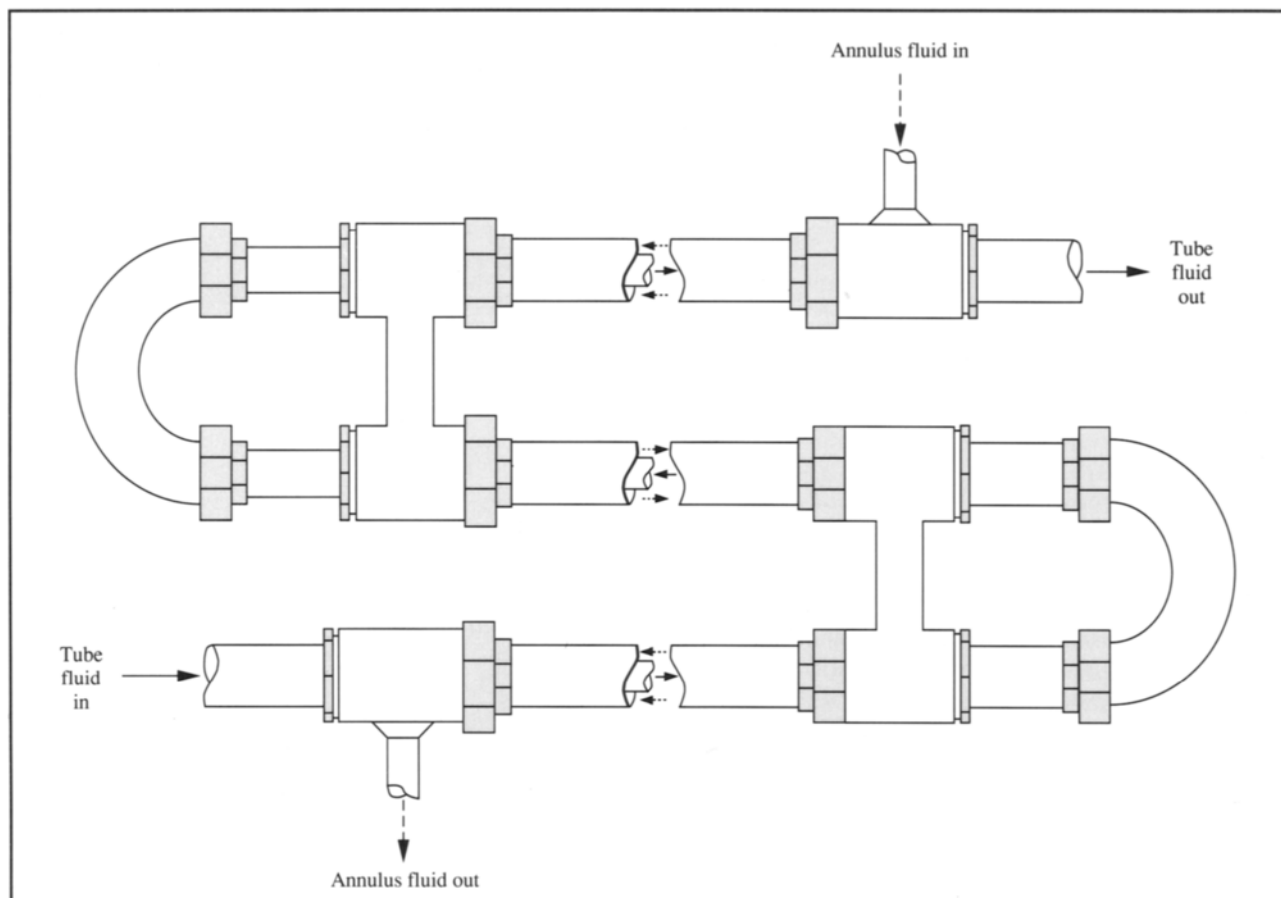
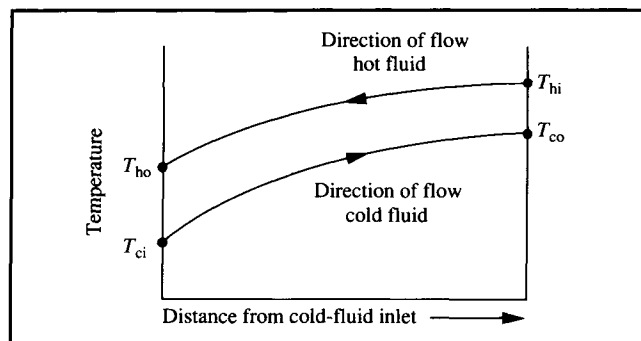


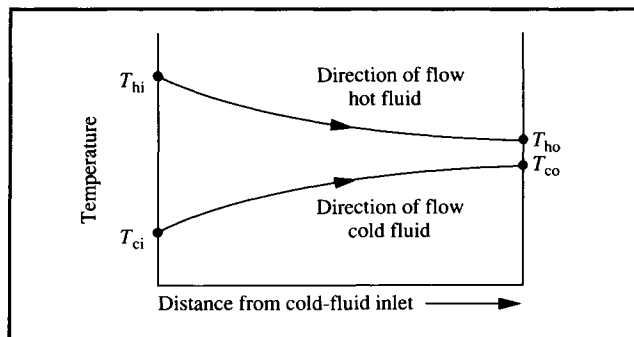
Figure 8.4 Temperature changes for countercurrent flow in a double-pipe heat exchanger.



is not as effective as countercurrent; it is not possible using cocurrent flow to bring the exit temperature of one fluid close to the entrance temperature of the other. Instead, the exit temperatures of both streams lie between the two entrance temperatures. Less heat can be transferred in parallel flow than in countercurrent flow; consequently, parallel flow is applied less frequently.

Double-pipe heat exchangers can be extended to several passes arranged in a vertical stack, as illustrated in Figure 8.3. However, when large surface areas are needed to achieve the desired rate of heat transfer, the weight of the outer pipe becomes so great that an alternative design, the shell-and-tube

Figure 8.5 Temperature changes for cocurrent flow in a double-pipe heat exchanger.



heat exchanger, is a better and more economical choice. As a rule of thumb, if the heat-transfer area between the fluids must be more than 10–15 m², a shell-and-tube exchanger is required.

8.1.2.2 Shell-and-tube heat exchangers

Shell-and-tube heat exchangers are used for heating and cooling all types of fluid. They have the advantage of containing very large surface areas in a relatively small volume. The simplest form, called a *single-pass shell-and-tube heat exchanger*, is shown in Figure 8.6.

Figure 8.6 Single-pass shell-and-tube heat exchanger. (From A.S. Foust, L.A. Wenzel, C.W. Clump, L. Maus and L.B. Andersen, 1960, *Principles of Unit Operations*, John Wiley, New York.)

